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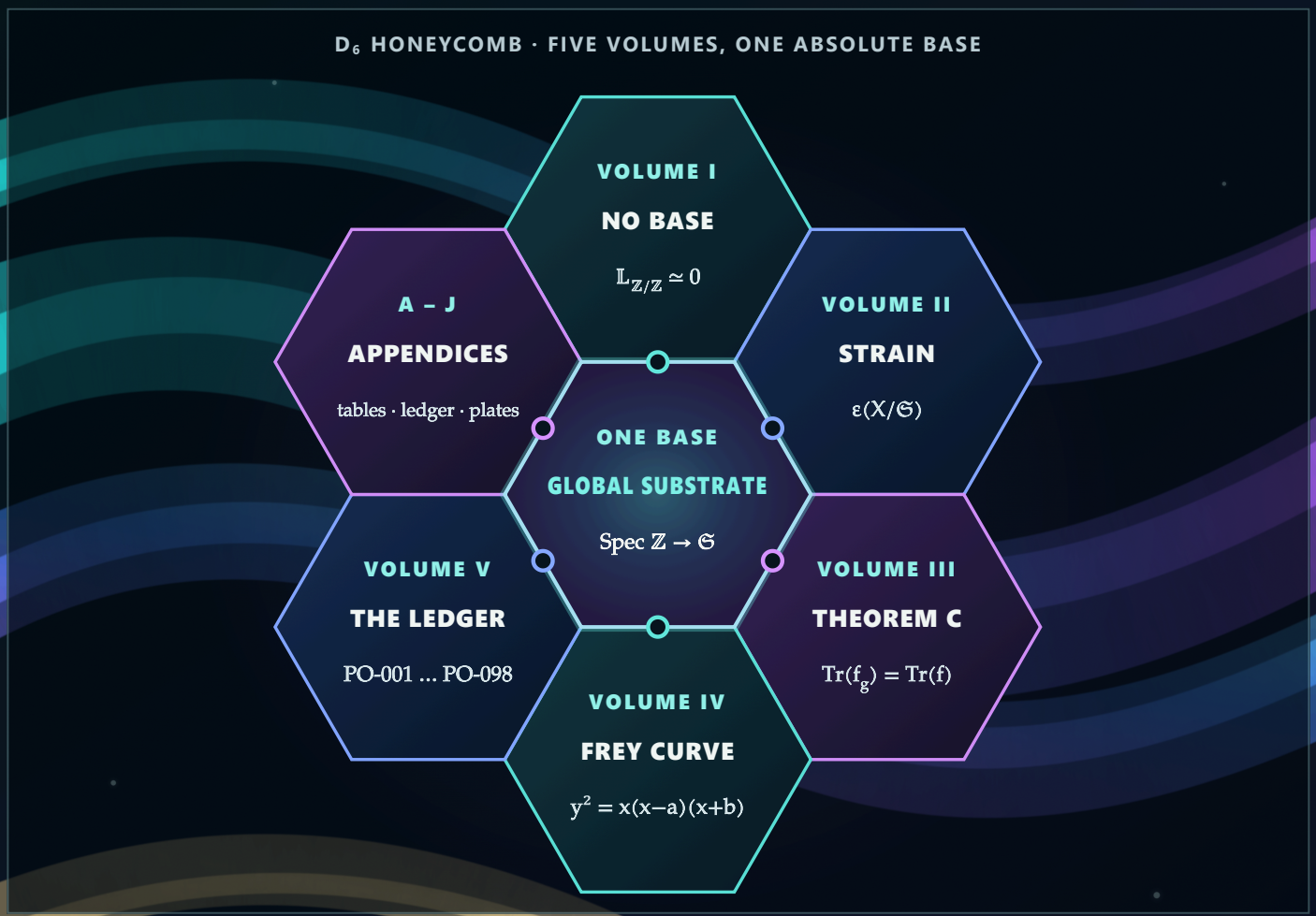
COMPLETE FIVE-VOLUME MONOGRAPH · VOLUMES I–V

Strained Prismatic Geometry

Deformation over the Absolute Base and the abc Conjecture

RESEARCH ARCHITECTURE EDITION 1



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Strained Prismatic Geometry

The Complete Monograph

Strained Prismatic Geometry

Deformation over the Absolute Base and the abc Conjecture

Vincent Liu

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The abc conjecture remains open. *The abc conjecture is an open problem, and this book does not prove it.* What this book contains is: a proposed foundational framework in which Spec$\mathbb{Z}$ would acquire a base direction, a derivation, a relative cotangent complex, and an arithmetic Riemann–Hurwitz formula; an unconditional core consisting of established ingredients, formal constructions, and classical deductions; a conditional derivation of *abc* from a named finite list of proof obligations; and a falsifiable programme for discharging or refuting them. Every theorem-like assertion carries one of six epistemic tags: [KNOWN], [REPROVED], [NEW], [CONDITIONAL], [OPEN], or [DISPUTED]. The conventions governing these tags are set out in *Epistemic Status of This Book*, which the reader is asked to read before Chapter 1.

Machine assistance. The research programme recorded in this volume was conducted with substantial machine assistance. The architecture of that assistance, the division of labour between author and machine, and the provenance discipline used for retained claims are documented in Chapter 56 and in the Preface. Responsibility for every statement in this book, and for every error in it, is the author's alone.

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To everyone who read a long proof carefully.

Series Note

Strained Prismatic Geometry is the sixth work in the WiseMotif Monograph of Mathematical Intelligence and belongs to the Grothendieckian Programme. The first two works of the series are bound closely together. *The Grothendieckian Machine* set out how mathematical worlds are constructed, compared and carried from one setting to another, and how a machine may hold them as typed, versioned and verifiable objects. *GM-Omega-Infinity: Topos-Native Intelligence* stated what an intelligence must be, internally, if it is to reason inside such worlds. The third, fourth and fifth works, *Arithmetic Spectral Descent*, *Fractured Adelic Geometry* and *The Archimedean Prism*, took the Riemann Hypothesis, the conjecture of Birch and Swinnerton-Dyer and the Hodge Conjecture as research subjects of the Programme. The present work is a self-contained system in the same sense, and it is the Programme's fourth research subject.

The Programme has a single research aim: machine intelligence that discovers mathematical structure on its own account. That aim asks for reasoning architectures held to the standard of proof, and for a discipline of recursive mathematical discovery, in which a proposed structure is tested, what survives the test is abstracted into a new setting, and the question is posed again in that setting. Its model is Grothendieck's rising sea. A hard problem is not attacked where it stands; a theory is raised around it until the problem becomes natural and gives way. A research subject of the Programme is a problem on which this method is carried through in full, with every step recorded in a form that a mathematician can check and a machine can execute.

This work carries it through on the *abc* conjecture of Masser and Oesterlé. It begins from a contrast. Over a function field the polynomial *abc* theorem of Mason and Stothers is easy, because the proof runs on a base: one differentiates along it, and Riemann–Hurwitz counts what the derivative sees. Over $\mathbb{Z}$ there is no base: $\mathbb{Z}$ is initial among rings, $\mathrm{Spec}\mathbb{Z}$ is terminal among schemes, and its cotangent complex is zero. The recursion therefore changes the object rather than the inequality. It asks for a substrate $\mathfrak{S}$ — prismatic geometry at every finite prime, a q-deformation over the Habiro ring at all primes at once, and condensed analytic geometry at the real place, glued along the product formula — over which $\mathrm{Spec}\mathbb{Z}$ acquires a relative cotangent direction, and it reads bad reduction as strain in that direction. Each step is then tested against what is known: the function-field limit, the Tate curve, the Frey curve, the extremal triples of the abc records. What fails is recorded as a proof obligation. The recursion ends in a master inequality, Theorem A, an *abc* inequality over $\mathbb{Q}$, Theorem B, both conditional on a named ledger of ninety-eight obligations; an unconditional categorical theorem of trace invariance, Theorem C, as the answer to the indeterminacy of identifications; an obstruction atlas of fourteen entries; twenty-two critical experiments with kill criteria; and a companion program, `abctk`, which recomputes the book's records of *abc* triples and its certified benchmarks, and which keeps the open comparisons open.

The work keeps one editorial principle throughout. No result is presented at a strength greater than its proof. Every substantive assertion carries a label where it is made — **[KNOWN]**, **[REPROVED]**, **[CONDITIONAL]**, **[NEW]**, **[OPEN]** or **[DISPUTED]** — and every conditional statement names the obligations it rests on. The four labels **[KNOWN]**, **[CONDITIONAL]**, **[NEW]** and **[OPEN]** are those

of the whole series, in the same sense; this work adds [REPROVED] for a known result proved again here, with its source, and [DISPUTED] for a published disagreement it records without adjudicating. Computation is reported as computation, at the level of certification it has, and is never promoted to a theorem. An obstruction the work fails to remove is stated as an obstruction, in the place where it stops the argument.

The *abc* conjecture remains open. This work does not prove it. It is not an exposition, verification, refutation or reformulation of inter-universal Teichmüller theory, and it takes no side in the published disagreement about that theory. What it offers is the state of the question after the method has been applied without concession: what the substrate would deliver, what it would need, and where each route to it stops.

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Abstract

This monograph asks why the *abc* conjecture is a theorem over function fields, with a proof of half a page, and open over $\mathbb{Q}$; and what object would allow the function-field proof to be imitated in arithmetic rather than merely admired.

[NEW] The diagnosis is negative and exact. The Mason–Stothers theorem consumes four inputs: a base, a derivation along it, a non-zero sheaf of relative differentials, and a degree–ramification formula. Over $\mathbb{Z}$ all four fail for one reason: $\mathbb{Z}$ is initial among commutative rings, so the cotangent complex of $\mathrm{Spec}\mathbb{Z}$ over its only base is zero. A no-go theorem (Chapter 3) shows that no 1-categorical repair which keeps an initial lift of $\mathbb{Z}$ can change this, and the obstruction is unfolded into eight requirements against which six decades of manufactured derivations, and this book's own proposal, are scored (Chapter 4).

[CONDITIONAL] The proposal changes the base. A global substrate $\mathfrak{S}$ is assembled from prismatic geometry and the Cartier–Witt stack at every finite prime, a q-deformation over the Habiro ring valid at all primes at once, and condensed analytic geometry at the real place, glued along the product formula, so that $\mathrm{Spec}\mathbb{Z} \to \mathfrak{S}$ carries a relative cotangent direction. Bad reduction becomes strain in that direction: a strain complex, a strain class and an arithmetic Kodaira–Spencer class $\kappa(X)$, whose function-field limit is the classical Kodaira–Spencer map and the geometric Szpiro inequality. A Θ-dilation Θ_λ acts on the strain by $q \mapsto q^\lambda$ on Tate parameters and $\tau \mapsto \lambda\tau$ on periods; the comparison of its finite-place and archimedean computations has an unconditional analytic core with explicit error $24(\lambda+1)|q|(1-|q|)^{-2}$ (Chapter 30).

[NEW] Indeterminacy of identifications is answered structurally. Theorem C (Chapter 36) proves that the categorical trace of an endomorphism of a dualisable object, equivariant for a connected ∞-groupoid of gluing choices, is a canonical point of $\mathrm{End}(\mathbf{1})^{hG}$ and is independent of every choice. Its instance over the substrate is conditional, and the chapter states which part of the objection of Scholze and Stix the theorem addresses and which parts it leaves standing.

[CONDITIONAL] The master inequality, Theorem A (Chapter 39), bounds the strain trace $\vartheta(X)$ by $(1+\varepsilon)$ times the conductor and different terms plus a constant independent of X, by an eight-step argument whose arithmetic weight rests on one obligation, the archimedean ramification estimate PO-079. Through the height form of Szpiro and the Frey–Hellegouarch curve, with its complete 2-adic classification, Theorem B (Chapter 43) gives $\max(|a|,|b|,|c|) \le K_B(\varepsilon)\,\mathrm{rad}(abc)^{1+\varepsilon}$ — conditional on sixty-one named obligations of the ledger of ninety-eight and on three named hypotheses, with every constant tracked (Chapter 40). The Diophantine endgame is unconditional: [NEW] the height form of Szpiro with exponent $6+\delta$ implies *abc* with exponent $1+\delta/6$ and explicit constants (Chapter 42).

The computational part is executable, and it is reported as computation. The companion package `abctk` performs exact radical sieving, Frey-curve construction, Tate local analysis and certified interval enclosures, with sixty-one tests; it recomputes the book's records of *abc* triples and keeps open the comparison that would identify its benchmarks with the strain trace (PO-087–PO-089). A Lean 4 blueprint and a generated proof-obligation ledger (Appendix H) fix what a formal and a human audit must check.

[OPEN] What remains is stated as a programme. An obstruction atlas of fourteen entries, twenty-two critical experiments with three decisive computations against named rival accounts and three kill criteria, an audit of what has and has not been proved (Chapter 61), and a programme for the next decade (Chapter 62). The book is not an exposition, verification, refutation or reformulation of inter-universal Teichmüller theory; the published disagreement about that theory is recorded, [DISPUTED], without being adjudicated.

The *abc* conjecture remains open. This monograph does not prove it.

Preface

This book does not prove the *abc* conjecture. It develops a framework for approaching the conjecture, constructs an architecture in which its proposed mechanism can be stated, and derives *abc* conditionally from explicitly named proof obligations. Some parts of the book are established mathematics, some are new arguments proved here, and some are conjectural comparison or descent statements on which the main conditional deduction rests. The distinction is not a qualification added after the event. It is part of the mathematical content of the book, and it is recorded statement by statement in the epistemic tags, Chapter 55, Chapter 61, and Appendix H.

The project began from a contrast that is elementary to state and difficult to ignore. Over a function field $k(t)$, the polynomial *abc* theorem is the Mason–Stothers theorem. One differentiates, forms a Wronskian, and counts zeros; geometrically, Riemann–Hurwitz supplies the governing identity. Over $\mathbb{Z}$, this proof has no literal analogue. The obstruction is not that a sufficiently ingenious arithmetic derivative has merely failed to be found. The integers are initial among commutative unital rings, and $\mathrm{Spec}\mathbb{Z}$ is terminal among schemes. Ordinary arithmetic therefore has no intrinsic base direction beneath $\mathrm{Spec}\mathbb{Z}$, no non-zero relative cotangent complex in that direction, and no corresponding Riemann–Hurwitz calculation. Chapter 3 makes this negative statement precise. The rest of the volume asks whether the missing direction can be manufactured without pretending that ordinary algebraic geometry already contains it.

That question, rather than the prestige of the conjecture, is the reason for introducing a new substrate. A direct attack on an inequality of heights may obscure the fact that the function-field proof is powered by relative geometry. If arithmetic lacks the relevant relative structure, then one may either accept that the analogy ends there or seek a category in which the arithmetic object becomes relative. This book takes the second course. It combines three kinds of input: absolute prismatic geometry at the finite primes, a q-deformation organised through the Habiro ring, and condensed and analytic geometry at the infinite place. Their proposed gluing is the global substrate $\mathfrak{S}$. The map

$$\mathrm{Spec}\mathbb{Z} \longrightarrow \mathfrak{S}$$

is intended to carry a non-zero relative cotangent object. Pulling that direction through arithmetic families gives the strain complex, the strain class, and the arithmetic Kodaira–Spencer class used later in the volume.

The word *intended* in the preceding paragraph is exact. Prisms, Cartier–Witt geometry in its established range, condensed mathematics, Arakelov degree, cotangent complexes, and categorical traces are not invented here. The proposed all-place assembly, its effective descent, the non-vanishing and arithmetic interpretation of its cotangent direction, and several coherence statements are open. The main theorem depends on them. It would be misleading to infer the existence of the global object merely from the strength of its local ingredients. One of the recurring lessons of the book is that a collection of good local theories is not yet a

global theory: effectivity, comparison, and higher coherence are mathematical theorems, not punctuation.

This work belongs to the Grothendieckian Programme of the WiseMotif Monograph of Mathematical Intelligence, and it is the Programme's fourth research subject, after the Riemann Hypothesis, the conjecture of Birch and Swinnerton-Dyer and the Hodge Conjecture. The Programme has a single aim: machine intelligence that discovers mathematical structure on its own account, under reasoning architectures held to the standard of proof. A research subject is a problem on which that aim is tested in full. The test is not whether a machine can produce text about a famous problem; it is whether the method can raise a structure around the problem, check that structure against everything already known, and say precisely where it fails.

The method is recursive mathematical discovery. A structure is proposed; it is tested against the cases in which the answer is known; what survives is abstracted into a new setting, and the question is posed again there. What does not survive is not discarded. It becomes a proof obligation or an obstruction, with a statement, a status and an owner, and it is carried forward as part of the result. This book is the record of one such recursion on the *abc* conjecture: the substrate $\mathfrak{S}$ is the structure proposed; the function-field limit, the Tate curve, the Frey–Hellegouarch curve and the extremal triples of the *abc* records are the cases it is tested against; and the ledger of Appendix H, the atlas of Chapter 58 and the experiments of Chapter 59 are what the recursion leaves behind.

Grothendieck described two ways of opening a hard nut: to strike it until it breaks, or to immerse it in a rising sea until it softens and opens of itself. The results that bear on *abc* over $\mathbb{Q}$ are, almost without exception, of the first kind. Linear forms in logarithms give the exponential bounds of Stewart, Tijdeman and Yu; the Frey–Hellegouarch curve converts *abc* into Szpiro's inequality; each is a precise blow at one point. This book tries the second way. It does not look for a sharper estimate of a linear form. It asks what the base would have to be over which $\operatorname{Spec}\mathbb{Z}$ is a relative object, so that the half-page proof over a function field has somewhere to run, writes down the requirements such a base must meet, and follows their consequences. The sea has not risen far enough: the substrate is specified, its local pillars are established mathematics, and its global assembly rests on obligations of the first rank of difficulty. But the specification is exact, it reproduces the function-field theorem in the limit, and it locates the remaining difficulty in a finite, named set of statements. That is the state the book reports.

The new material of the volume lies in several distinct layers. They should not be conflated.

(i) The first layer is the *design diagnosis*. Chapters 2–5 isolate the exact uses of a base direction in the Mason–Stothers proof, prove the absolute obstruction in ordinary algebraic geometry, and turn that obstruction into a list of requirements for any proposed arithmetic replacement.

(ii) The second layer is the *substrate architecture*. The finite prismatic legs, the all-primes q-deformation, and the archimedean gauge are placed in a common gluing problem governed by the product formula. This is a proposal with established local components and open global interfaces; the book's contribution is to state those interfaces sufficiently precisely that they may be proved or refuted.

(iii) The third layer is the *strain formalism*. The relative cotangent complex over $\mathfrak{S}$, its strain sheaf $\mathfrak{s}$, the class $\varepsilon(X/\mathfrak{S})$, and the arithmetic Kodaira–Spencer class $\kappa(X)$ organise bad reduction as deformation in the proposed absolute direction. The familiar function-field differential is the compulsory limit, not a decorative analogy.

(iv) The fourth layer is the Θ-*dilation*. Instead of treating a Θ-link as an indeterminate identification between worlds, the book asks for an honest endomorphism Θ_λ of the substrate data, together with its higher equivariance. The Θ-line, its theta-value ledger, the dilation cocycle, and the semigroup law separate what is classical on Tate curves from what remains to be descended globally.

(v) The fifth layer is the *invariance mechanism*. Gluing choices are organised as an ∞-groupoid $\mathcal{G}$. The object sought downstream is not a preferred identification but a class function: the categorical trace of a $\mathcal{G}$-equivariant endomorphism. This is the book's architectural answer to indeterminacy. It is meaningful only if dualisability, equivariance, and descent are proved in the stated category.

(vi) The sixth layer is the *audited Diophantine passage*. The volume separates the proposed trace inequality from the classical deductions Theorem A ⇒ Szpiro ⇒ *abc*, distinguishes the discriminant and height forms of Szpiro, retains the 2-adic correction, and displays the dependence of every constant. The Frey-curve calculations and exponent conversions are unconditional once the appropriate height-form inequality is granted.

(vii) The final layer is methodological. The proof-obligation ledger, falsification experiments, certified-computation protocol, formalisation blueprint, and provenance discipline are part of the architecture. They do not make an open assertion true. They make it harder for an open assertion to pass unnoticed as a theorem.

I regard the last layer as more important than the proposed framework itself. A mathematical programme may be fertile even if its central construction fails. It becomes dangerous when the boundary between theorem, analogy, expectation, and computation is allowed to drift. This is particularly acute in a long work, where a conditional sentence in an early chapter may be silently consumed as unconditional hundreds of pages later. The tags [KNOWN], [REPROVED], [NEW], [CONDITIONAL], [OPEN], and [DISPUTED] are intended to prevent that drift. Appendix H extracts the 98 open obligations, PO-001–PO-098; Chapter 59 gives twenty-two tests with predetermined refuting outcomes, together with three computations specified sharply enough that a third party can run them and compare the answer this framework predicts against the answers three rival accounts predict; Chapter 61 states what survives if the new architecture is removed. The pre-consolidation ledger contained duplicate identifiers, stale generated counts, and dangling references. Those editorial defects have been repaired by a whole-book renumbering and regeneration; they are recorded here rather than suppressed because a ledger that cannot identify its own nodes cannot certify a dependency chain.

The ten parts are bound as one monograph in five volumes, two parts to each, with the ten appendices as the sixth cell of the series' hexagonal cover; where the text says *this volume*, it means the whole monograph. Each volume ends at a result its reader can hold without the others, and the Leitfaden that follows this preface gives routes that cross them.

Volume I: The Problem and the Absolute Base (Parts 0–I, Chapters 1–12). [KNOWN] Part 0 states the Masser–Oesterlé conjecture in its constant, finiteness and quality forms, with its elliptic and height-theoretic equivalents and every ε-conversion visible, and proves the Mason–Stothers theorem three times to isolate the four inputs it consumes. [NEW] It then proves the absolute obstruction and the no-go theorem for initial lifts, extracts eight design requirements, and scores the surrogates of six decades — p-derivations, $\mathbb{F}_1$-geometries, the Fargues–Fontaine curve, prismatic and q-de Rham cohomology, inter-universal Teichmüller theory — and this book's own proposal against them. Chapter 5 states Theorems A, B and C with their obligations in full. [KNOWN] Part I builds the finite-place pillar from first principles: δ-rings and Witt vectors, prisms and prismatic cohomology, the Cartier–Witt stack,

the syntomic stack and F-gauges, and the Rees dictionary that makes a filtration a degeneration. [CONDITIONAL] It ends with the absolute base $\mathfrak{A}$ of Chapter 11, whose universal property is unconditional and whose geometricity and cotangent descent are the first obligations of the ledger.

Volume II: The Global Substrate and Its Deformation Theory (Parts II–III, Chapters 13–26). [KNOWN] Part II supplies the two remaining pillars: the q-de Rham complex and the Habiro ring, which sees every root of unity at once, and condensed mathematics with analytic stacks, which put the real place into the same homological category as the primes. [CONDITIONAL] Chapter 18 glues the pillars along the product formula into the global substrate $\mathfrak{S}$ and shows its structure morphism non-degenerate modulo two named obligations. [CONDITIONAL] Part III places $\mathrm{Spec}\mathbb{Z}$ over $\mathfrak{S}$: the strain complex, the strain class and the arithmetic Kodaira–Spencer class, specialised to semistable elliptic curves, logarithmic boundary data and the Θ-line. [NEW] It closes by replacing chosen identifications with a gluing ∞-groupoid $\mathcal{G}$, provably not contractible in the model case, and an indeterminacy calculus of orbits and stabilisers. The volume ends with the object on which the rest of the book acts.

Volume III: Dilation, Traces and the Master Inequality (Parts IV–V, Chapters 27–40). [NEW] Part IV makes dilation categorical on gauge stacks and proves its semigroup law. [CONDITIONAL] It constructs the Θ-dilation Θ_λ, proves the dilation comparison theorem with an unconditional analytic core, recasts multiradiality as descent along $\mathcal{G}$ without any claim about inter-universal Teichmüller theory, and organises the dilation cocycle and the obstruction package that must vanish before a trace can be taken. [NEW] Part V installs six-functor formalisms and categorical traces and proves Theorem C, trace invariance under groupoids of identifications. [CONDITIONAL] It identifies the height with the first variation of the trace of Θ_λ, restores Riemann–Hurwitz over the substrate, and states and proves the master inequality, Theorem A, conditional on fifty-nine obligations, with its constants audited in Chapter 40. The volume ends where the book's new mathematics ends: at Theorem A and the one obligation, PO-079, that carries its arithmetic weight.

Volume IV: The Descent to *abc* and Its Computation (Parts VI–VII, Chapters 41–52). [CONDITIONAL] Part VI carries Theorem A through Szpiro's inequality to *abc*. [NEW] The Frey–Hellegouarch calculation, the complete 2-adic classification and both implications between the height form of Szpiro and *abc* are proved with every constant tracked. [CONDITIONAL] Theorem B over $\mathbb{Q}$, *abc* over number fields, and the classical consequences — asymptotic Fermat, Fermat–Catalan, Roth, Erdős–Woods — are each labelled by the form of *abc* they need, and none of them is available unconditionally. Chapter 47 confronts the framework with the extremal triples. Part VII specifies the computation: certified arithmetic in the Habiro ring, algorithms for prismatic and syntomic cohomology, interval-certified trace benchmarks, the *abc*-verification engine and the reproducibility contract, implemented in `abctk`. It records, as a result of its own, that no program yet evaluates the strain trace.

Volume V: Formalisation, Audit and Prospects (Parts VIII–IX, Chapters 53–62). Part VIII gives the Lean 4 blueprints for the substrate and for the trace theorem, treats the proof-obligation ledger as a formal artefact with a soundness condition that is stated and left open, and gives an engineering account of the machine-assisted method by which the book was made and of the history of verification in which it asks to be judged. [OPEN] Part IX is the critical retrospective: the obstruction atlas, the critical experiments and kill criteria, the test of the substrate against Vojta, Szpiro and uniformity, the audit of what has been proved, and a programme for the next decade.

The appendices (A–J). The ten appendices collect the conventions on higher categories, condensed mathematics, prisms and Arakelov geometry that the text uses; the tables of δ-rings and of the Habiro ring; the glossary; the proof-obligation ledger, generated from the sources; the atlas of the 186 plates; and the computational resources.

The intellectual debts of the book are extensive. Grothendieck's insistence on relative geometry, functoriality, and the organisation of mathematics by sites and descent is the background against which the absolute obstruction becomes visible. Deligne's work provides models of how arithmetic, cohomology, weights, and categorical structure can remain exact under great conceptual pressure. Faltings, Vojta, and Bombieri shaped the modern language of heights and the Diophantine inequalities to which the final parts return. The arithmetic intersection and adelic-vector-bundle theories used here owe much to the work of Chen and Moriwaki.

At the finite primes, the decisive modern input is the prismatic theory of Bhatt and Scholze, together with the Cartier–Witt and syntomic developments that surround it. At the infinite place, the possibility of putting topological and analytic coefficients into a workable homological category comes from condensed mathematics and the work of Clausen and Scholze. The book also draws on Buium's arithmetic differential geometry, Borger's Frobenius-lift viewpoint, Habiro's cyclotomic completion, the q-de Rham tradition, Deninger's dynamical analogies, and the long development of Arakelov geometry. None of these theories was created for the specific gluing attempted here, and citing them does not transfer their established status to that gluing.

Mochizuki's inter-universal Teichmüller theory deserves a separate word. It is a major and highly original attempt to reorganise arithmetic information in order to reach *abc*-type inequalities. This book is not an exposition, verification, refutation, or reformulation of IUT. Its substrate, Θ-dilation, and trace mechanism are independently defined. Where the vocabulary or motivating difficulties meet, I have tried to state the comparison without claiming an identity of methods. The criticisms associated with the Scholze–Stix discussion are treated here as a demand for a precise invariance mechanism: if a quantity is transported through non-canonical choices, what categorical structure makes the final scalar independent of them? My proposed answer is equivariance and categorical trace. That answer must be proved on its own terms. Agreement of intended conclusions is not evidence that either theory verifies the other.

This book was made with substantial machine assistance. Concealing that fact would contradict its own account of provenance. Language models operating in multiple roles assisted with literature triage, search across the manuscript, LaTeX construction, consistency checks, adversarial criticism, computational experiments, and the maintenance of notation and dependency records. The process was deliberately divided among construction, criticism, and verification roles because fluent generation is not verification. A model which has just produced an argument is poorly placed to be its only critic; a model which recalls a plausible citation is not a bibliographic authority; agreement among generated chapters can be false coherence produced by shared phrasing rather than shared definitions.

Chapters 56 and 57 describe the method and its controls. Citation keys were restricted to a whitelist. Definitions and notation were fixed globally. Proof obligations were extracted mechanically. Chapters were compiled and audited in isolation. Numerical claims were separated from the unproved comparison which would identify them with the substrate trace. These controls reduce familiar failure modes—fabricated references, silent changes of notation, omitted hypotheses, and plausible but unsupported transitions—but they do not

establish mathematical truth. I read, selected, revised, and accepted the text, and I remain solely responsible for every claim, omission, error, attribution, and judgement in the volume. Machine assistance is part of the provenance, not a co-author and not a substitute for proof.

The intended reader is an arithmetic geometer willing to move between classical Diophantine geometry and newer categorical methods. The opening part assumes familiarity with algebraic number theory, algebraic curves, elliptic curves, heights, and the basic language of schemes. Parts I and II use derived algebraic geometry, cotangent complexes, p-adic Hodge theory, prisms, stacks, and condensed mathematics. Parts III–V require comfort with deformation theory, logarithmic geometry, stable ∞-categories, duality, and categorical traces. Part VI returns to elliptic curves, conductors, discriminants, Frey curves, and Vojta-style height inequalities. The computational and formal parts assume no single software system, but they do assume the reader distinguishes exact arithmetic, interval certification, reproducibility, and formal proof.

No reader is expected to follow the book linearly on a first encounter. The Leitfaden gives routes for the Diophantine reader, the foundations reader, the computational reader, and the sceptical reader. The sceptical route is not a concession. It is the correct route for anyone evaluating a new framework: read the status declaration, the main conditional interfaces, the ledger, and the kill criteria before investing in the internal vocabulary. A reader who finds a false obligation, a circular dependency, or a certified failure of a locked comparison has advanced the subject more than one who merely finds the architecture suggestive.

What the Programme takes from this subject is a method rather than a result. A conditional architecture of this size is useful only if its conditions are finite, named and attackable, and if the boundary between what is proved and what is proposed survives the length of the argument. The ledger, the six labels, the atlas, the kill criteria and the companion program are the instruments by which this book tries to keep that boundary fixed. They are offered to the reader in the same spirit as the mathematics: to be checked, and where they fail, to be corrected.

I thank the authors whose work made the local components of this programme possible, and the expositors whose precision made those components usable outside their original settings. I thank the maintainers of open mathematical software, bibliographic databases, proof assistants, and typesetting systems on which the reproducibility apparatus depends. I thank readers who insist that a beautiful diagram is not a descent theorem, that a numerical match is not an identification, and that a conditional proof must name its conditions. I am especially grateful for criticism that forced the book to distinguish original Szpiro from modified Szpiro, local constructions from global effectivity, and a generated ledger from a verified dependency certificate.

The ambition of the book is therefore limited but serious. It offers a specific account of what arithmetic would need in order to imitate the geometric proof over function fields; it proposes objects and morphisms which could supply that account; it proves the parts that can presently be proved; and it exposes the remaining claims to proof, computation, formalisation, and refutation. Whether the substrate survives those tests is an open mathematical question. The *abc* conjecture is open as well.

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2026

Epistemic Status of This Book

Read this before Chapter 1

The abc conjecture is an open problem, and this book does not prove it.

What this book contains is: a proposed foundational framework in which Spec$\mathbb{Z}$ would acquire a base direction, a derivation, a relative cotangent complex, and an arithmetic Riemann–Hurwitz formula; an unconditional core consisting of established ingredients, formal constructions, and classical deductions; a conditional derivation of *abc* from a named finite list of proof obligations; and a falsifiable programme for discharging or refuting them.

The book uses six epistemic tags on theorem-like assertions. [KNOWN] marks a result of the existing literature, cited at the point of use. [REPROVED] marks a known result reproved here in the framework's language. [NEW] marks a result first proved in this volume, unconditionally. [CONDITIONAL] marks a result proved conditionally, and names the proof obligations it depends on. [OPEN] marks a statement that is not proved here at all. [DISPUTED] records a documented disagreement without adjudicating it.

The cardinal rule. No statement tagged [NEW] may cite, transitively, a statement tagged [OPEN] or a [CONDITIONAL] statement whose obligations it does not itself declare. This rule is audited mechanically: the source audit checks the tags, while Chapter 55 and Appendix H expose the declared obligation graph. Mechanical auditing does not establish the truth of a statement tagged as proved.

The monograph comprises ten Parts, bound in five volumes, 62 chapters, ten appendices (A–J), and 186 plates. The consolidated proof-obligation ledger contains 98 open records, numbered consecutively PO-001–PO-098, with no duplicate or dangling identifier.

The global substrate, its non-zero arithmetic cotangent direction, the all-place dilation and trace comparisons, the master inequality, and the resulting *abc* implication remain conditional. The classical Mason–Stothers theorem, standard Frey-curve calculations, the formal category-theoretic trace identities, and deductions from an assumed height-form Szpiro inequality belong to the unconditional core. Chapter 61 gives the full separation; Appendix H gives the exact open interfaces.

What the tags entitle a reader to do

A tag is not a rhetorical hedge. It is an instruction about how the statement may be used, and the six instructions differ sharply.

- [KNOWN] The statement is due to others and is cited at the point of use. A reader may use it exactly as they would use it in its source. If the citation is wrong, that is an error in this book and not a weakening of the statement.
- [REPROVED] The statement is known, and is proved here again because the framework's language changes what the proof looks like or because self-containedness demands it. A reader may use it freely. The reproof is offered for inspection, not for priority.
- [NEW] The statement is proved here for the first time, and *unconditionally*. A reader may cite it as a theorem. If it is wrong, the book is wrong, and no appeal to the programme's incompleteness excuses it. This is the only tag that carries the author's full assertion.
- [CONDITIONAL] The statement is proved, but the proof consumes hypotheses that are not established. The hypotheses are named by identifier, not gestured at. A reader may use the statement only after adopting those identifiers as assumptions, and must carry them forward. A [CONDITIONAL] statement is a genuine mathematical result about an implication; it is not a claim about its own conclusion.
- [OPEN] The statement is not proved here at all. It is recorded because the framework needs it, because it is the natural next question, or because naming it precisely is itself useful. A reader may not use it for anything except as a target.
- [DISPUTED] There is a documented disagreement in the literature, and this book records it without adjudicating. A reader is being told that competent people differ and that this volume is not the place the matter is settled.

The distinction that matters most, and the one most often blurred in long arguments, is between [NEW] and [CONDITIONAL]. A [CONDITIONAL] theorem whose hypotheses are all eventually discharged becomes a theorem; a [CONDITIONAL] theorem presented as though it were already a theorem is a false claim, whatever happens later.

One set of labels across the series

The labels [KNOWN], [CONDITIONAL], [NEW] and [OPEN] are those of the whole WiseMotif Monograph of Mathematical Intelligence. *Arithmetic Spectral Descent*, *Fractured Adelic Geometry* and *The Archimedean Prism*, the earlier research subjects of the Grothendieckian Programme, use them in the same sense, so that a statement marked [NEW] or [OPEN] means the same thing in every work of the series. This work needs two more and defines them above: [REPROVED], because much of its finite-place foundation is known mathematics proved again in the language of the substrate, with its source named; and [DISPUTED], because one published disagreement bears on its subject and is recorded here without being adjudicated. The claim boundary is printed in the same form in every work of the series. For this work it reads: the *abc* conjecture remains open; this book does not prove it; and it neither verifies nor refutes inter-universal Teichmüller theory. The labels are applied by the same rule throughout: a statement receives the weakest label its proof supports, a conditional statement names its obligations where it is made, and a computation is reported at the level of certification it has and never as a theorem.

The census

The tagging is not decorative, and its extent can be measured. The volume carries 1,420 tagged statements, distributed thus.

TAG	COUNT	MEANING
[KNOWN]	223	established elsewhere, cited
[REPROVED]	322	known, reproved here
[NEW]	646	first proved here, unconditionally
[CONDITIONAL]	177	proved from named obligations
[OPEN]	48	not proved here
[DISPUTED]	4	disagreement recorded, not adjudicated
	1,420	

Of the 62 chapters, 19 contain no conditional and no open statement whatsoever; they are listed in the Leitfaden. Everything asserted there is either established mathematics, a reproof, or a new unconditional result. A reader who rejects the substrate entirely may still read those chapters, and whatever value the volume retains under that rejection is concentrated in them.

These counts are produced mechanically from the sources and can be reproduced by any reader who has them. They measure how many statements carry each tag. They do not measure importance: a single [CONDITIONAL] theorem may carry more of the argument than forty [KNOWN] lemmas, and in this book one of them does. Appendix H, not the census, is the authority on what rests on what.

Four things this discipline does not do

It is as important to say what the apparatus fails to achieve as to describe it, because an audit that is trusted beyond its warrant is worse than no audit.

(i) **It does not make a proof correct.** A statement tagged [NEW] may still be false, because the tag records the author's assertion and the mechanical checks test the shape of the dependency graph, not the validity of any argument. Every proof in this volume is offered for refutation on the ordinary terms.

(ii) **It does not detect a false lemma with true-looking dependencies.** The tag-closure audit finds a [NEW] claim that leans on a [CONDITIONAL] one. It is structurally incapable of finding a [NEW] claim whose proof is simply wrong. Chapter 55 constructs an explicit example of the second kind that passes every check.

(iii) **It does not certify the ledger itself.** The ledger's soundness condition is stated in Chapter 55, and it is *not known to hold*. The ledger certifies declared dependence, and declaration is an act of the author.

(iv) **A chapter free of [CONDITIONAL] and [OPEN] tags is not thereby independent of the programme.** It may prove only correct statements about an object whose existence is open. Chapter 27 is the sharpest instance: its dilation theory is proved, but on a fixed prism the operator it constructs is inner, so the theory acquires content only through the prism motion of Chapter 28. The text says so where it arises rather than leaving the reader to discover it.

The shape of the conditional result

The main conditional deduction is not a single leap. Theorem A, the master inequality of Chapter 39, is an eight-link argument, and its dependency closure comprises 59 of the 98 obligations. From Theorem A the passage to Szpiro and thence to *abc* is largely classical and is set out in Chapters 41–43 with its constants displayed.

The reader should therefore locate the risk correctly. It is concentrated upstream, in the construction of the substrate and in the comparison theorems that would identify an abstract trace with an arithmetic height—not in the Diophantine endgame, which is the best-understood part of the chain. The chapters carrying the most conditional statements are 45, 33 and 39, followed by 37, 29 and 41. Chapter 45 deserves particular caution: it derives Fermat, Catalan, Roth-type and Erdős–Woods consequences, and *every* principal statement in it is conditional, since each descends from Theorem B. Those consequences are attractive and they are not available. Nothing in this volume entitles anyone to cite them.

What is asked of a reader

A framework that cannot be attacked is of no use. Chapter 59 therefore states predictions specific enough for a third party to compute, together with what competing accounts predict instead and the threshold at which the difference becomes decisive. Chapter 58 catalogues the unresolved mechanisms. Appendix H lists every obligation with its status and risk.

The most valuable response to this book is not agreement. It is a demonstration that one of the obligations is false, that a declared dependency is circular, that a computation in Chapter 59 comes out the other way, or that a proof tagged `[NEW]` does not close. Any of these advances the subject. A reader who concludes only that the architecture is suggestive has not tested it.

Machine assistance was used substantially throughout, in divided construction, criticism and verification roles; the method and its controls are described in the Preface and in Chapters 56 and 57. It is part of the provenance of this volume, it is not a co-author, and it is not a substitute for proof. Responsibility for every statement here, and for every error, rests with the author alone.

Leitfaden: Dependencies and Reading Paths

The book has ten Parts, counting the introductory Part 0, with 62 chapters, ten appendices (A–J), and 186 plates. Its ledger contains 98 obligations, PO-001–PO-098. The order is mathematical rather than historical. Part 0 fixes the Diophantine problem, reconstructs the function-field proof, proves the absolute obstruction, and states the design contract. Parts I and II construct the local ingredients of the proposed absolute substrate. Part III develops deformation theory relative to that substrate. Part IV constructs and studies the Θ-dilation. Part V passes from dilation to categorical traces, heights, arithmetic Riemann–Hurwitz, and the master inequality. Part VI performs the classical descent from the master inequality through Szpiro and Frey curves to *abc*, and studies consequences and uniformity. Parts VII and VIII specify computational, reproducibility, formalisation, and proof-audit infrastructure. Part IX is the critical retrospective: it catalogues obstructions, gives falsification tests, audits what is proved, and sets out the research programme.

This order should not be confused with a claim that every chapter is a proved prerequisite of the next. Several central interfaces are conditional. The table below follows the chapter titles and the dependency preludes of the chapters.

The ten Parts

Part 0 — The Problem and the Obstruction (Chapters 1–5). Chapter 1 states *abc*, its height and Szpiro variants, and the additive–multiplicative tension. Chapter 2 isolates the derivative, Wronskian, and Riemann–Hurwitz mechanism over function fields. Chapter 3 shows why this mechanism has no ordinary arithmetic base direction. Chapter 4 compares surrogate directions already present in arithmetic geometry, and Chapter 5 states the proposed architecture, its principal conditional theorems, and the reading contract. No later Part can be read responsibly without the status distinctions of Chapters 3 and 5.

Part I — The Absolute Substrate, I: Prisms, Gauges, and the Cartier–Witt Stack (Chapters 6–12). The first construction Part passes from δ-rings and Witt vectors to prisms, absolute prismatisation, the syntomic stack, F-gauges, and Rees descent. Chapter 11 assembles these data into the proposed absolute base $\mathfrak{A}$, with its unresolved geometric interfaces isolated as PO-001–PO-007. Chapter 12 compares the resulting direction with Buium's arithmetic jet spaces. The logical spine is

$$6 \longrightarrow 7 \longrightarrow 8 \longrightarrow 9 \longrightarrow 10 \longrightarrow 11,$$

with Chapter 3 supplying the obstruction that these constructions are meant to evade.

Part II — The Absolute Substrate, II: q-Deformation, Habiro Descent, and the Analytic Glue (Chapters 13–19). Chapters 13–15 develop the q-de Rham and Habiro branch and the proposed all-primes q-derivation. Chapters 16 and 17 provide the condensed and analytic branch needed at infinity. Chapter 18 is the merge point: it asks for product-formula descent of the finite and infinite legs to the global substrate $\mathfrak{S}$. Chapter 19 develops adelic curves, Arakelov degrees, slopes, Green functions, and the proposed substrate lift of adelic vector bundles. The two hard branches are therefore

$$13 \to 14 \to 15, \qquad 16 \to 17,$$

which meet, together with Chapters 9 and 11, at Chapter 18.

Part III — Deformation Theory over the Absolute Base (Chapters 20–26). Chapter 20 defines the strain complex and strain sheaf. Chapter 21 forms the arithmetic Kodaira–Spencer class. Chapters 22 and 23 specialise to semistable elliptic curves and logarithmic boundary data. Chapter 24 builds the Θ-line and separates the classical theta calculation from the proposed substrate interpretation. Chapters 25 and 26 replace non-canonical identifications by a gluing ∞-groupoid and an indeterminacy calculus. The principal chain is

$$18 \to 20 \to 21 \to 22 \to 23 \to 24 \to 25 \to 26,$$

although Chapters 22–24 also consume the gauge and Habiro material of Parts I and II.

Part IV — The Θ-Dilation (Chapters 27–33). Chapter 27 gives an abstract theory of dilation on gauges. Chapter 28 constructs the proposed Θ_λ from the q-derivation, the Θ-line, and the gluing data. Chapter 29 treats functoriality and base change; Chapter 30 performs the local-to-global comparison; Chapter 31 recasts multiradiality as descent; Chapter 32 extracts the dilation cocycle; and Chapter 33 studies its obstruction and rigidity classes. The main dependency is

$$(15, 17, 22, 24, 25, 27) \to 28 \to (29, 30) \to 31 \to 32 \to 33.$$

Part V — Heights, Traces, and the Master Inequality (Chapters 34–40). Chapter 34 supplies the abstract six-functor language, Chapter 35 the dualisability and trace formalism, and Chapter 36 the groupoid-invariance mechanism. Chapter 37 identifies the intended arithmetic height with a trace, while Chapter 38 formulates arithmetic Riemann–Hurwitz. Chapter 39 states and conditionally proves the master inequality, Theorem A, as an eight-link argument with its hypotheses enumerated. Chapter 40 audits constants and effectivity in that theorem. The conceptual chain is

$$34 \to 35 \to 36 \to 37 \to 38 \to 39 \to 40,$$

with Chapters 18, 20, 21, 23, and 28 entering at the arithmetic interfaces.

Part VI — The Descent to abc and Its Consequences (Chapters 41–47). Chapter 41 converts the master inequality to Szpiro-type bounds. Chapter 42 gives the Frey–Hellegouarch calculation and distinguishes the discriminant and height forms. Chapter 43 proves Theorem B over $\mathbb{Q}$ conditionally on its displayed obligations, tracking the constants and stating explicitly that the conclusion is not unconditional. Chapters 44–46 treat number fields, consequences, families, and the function-field analogue. Chapter 47 searches for counterexamples and extremal behaviour. Once the correct height-form input is granted, the chain $41 \to 42 \to 43$ is

largely classical; the open burden lies upstream in Part V and in the height and wild-prime comparison obligations.

Part VII — Computational Architecture (Chapters 48–52). Chapter 48 computes in the Habiro ring, Chapter 49 treats prismatic and syntomic algorithms, Chapter 50 specifies certified trace evaluation, Chapter 51 assembles the verification engine and database, and Chapter 52 sets the reproducibility and archival standards. These chapters distinguish exact arithmetic and interval certification from the open theorem identifying a computed model with the abstract strain trace.

Part VIII — Formalisation and the Multi-Agent Mathematician (Chapters 53–57). Chapters 53 and 54 give Lean blueprints for the substrate and trace formalism. Chapter 55 treats the proof-obligation ledger as an artefact. Chapters 56 and 57 describe the machine-assisted production method, provenance, and the sociology of verification. This Part depends on the mathematical architecture but does not discharge it merely by encoding interfaces.

Part IX — The Obstruction Atlas, Falsifiability, and Prospects (Chapters 58–62). Chapter 58 catalogues the unresolved mechanisms, Chapter 59 assigns critical experiments and kill criteria, and Chapter 60 tests the substrate against Vojta, Szpiro, uniformity, and higher-dimensional analogues. Chapter 61 audits what is proved and what is not; Chapter 62 gives the decade programme. This Part should be read before one relies on any theorem from Parts V or VI.

The five volumes

The ten parts are bound as one monograph in five volumes, two parts to each; the appendices complete the set. The volumes are the coarsest unit of reading: each ends at a result its reader can hold on its own, and the routes below cross them freely.

Volume	Parts	Content
I	0–I	The problem, the absolute obstruction and the design contract; the finite-place base $\mathfrak{A}$ (Chapters 1–12)
II	II–III	The q-deformation, the analytic glue and the global substrate $\mathfrak{S}$; strain, the arithmetic Kodaira–Spencer class and the gluing ∞-groupoid (Chapters 13–26)
III	IV–V	The Θ-dilation; six functors, Theorem C and the master inequality, Theorem A (Chapters 27–40)
IV	VI–VII	Szpiro, the Frey–Hellegouarch curve and Theorem B; the computational architecture and `abctk` (Chapters 41–52)
V	VIII–IX	Formalisation, the ledger and the multi-agent method; the obstruction atlas, falsifiability and the programme (Chapters 53–62)
—	Appendices	A–J: conventions, tables, glossary, the ledger, the atlas of plates, computational resources

Chapter-level dependency table

CH.	CHAPTER TITLE	DECLARED PREREQUISITES
1	The abc Conjecture: Statements, Equivalents, and the Shape of the Difficulty	none
2	Mason–Stothers and the Anatomy of an Easy Theorem	1
3	The Absolute Obstruction: Why Spec$\mathbb{Z}$ Has No Base	2
4	An Atlas of Surrogates: Six Decades of Manufactured Derivations	2, 3
5	Design of the Framework: Main Theorems, Proof Architecture, and the Reading Contract	1–4
6	δ-Rings, Witt Vectors, and the Frobenius-Lift Calculus	2–5
7	Prisms, the Prismatic Site, and Prismatic Cohomology	6
8	Absolute Prismatisation and the Stack $\mathbb{Z}^{\Delta}$	3, 7
9	The Syntomic Stack, F-Gauges, and Filtered Frobenius Structures	7, 8
10	Rees Constructions, Graded Descent, and Gauge Fixation	7–9
11	The Absolute Base	3, 6–10
12	Arithmetic Jet Spaces: Buium's δ-Geometry inside the Substrate	3, 6, 11
13	q-de Rham Cohomology and the q-Crystalline Site	3, 4, 7
14	The Habiro Ring: Arithmetic at All Primes Simultaneously	3, 4, 7, 13
15	The q-Derivation on the Absolute Base: Habiro Descent	3, 7, 8, 11, 13, 14
16	Condensed Mathematics and Analytic Rings	3, 15
17	Analytic Stacks and the Archimedean Gauge	9, 16
18	The Global Substrate $\mathfrak{S}$: Gluing the Finite and Infinite Places	9, 11, 15–17
19	Adelic Curves, Arakelov Data, and Their Substrate Lift	16–18; Appendix F
20	Cotangent Complexes and Relative Differentials over $\mathfrak{S}$	3, 8, 17, 18
21	The Arithmetic Kodaira–Spencer Class	8, 12, 17, 18, 20
22	Semistable Models, Elliptic Curves, and Their Substrate Deformations	9, 14, 18, 21, 23
23	Log Structures, Log Shells, and the Boundary Divisor	2, 9, 14, 18, 21, 22
24	The Θ-Line, Theta Functions, and the Reconstruction of the Θ-Link	17, 18, 22, 23

Ch.	Chapter title	Declared prerequisites
25	Gluing Data as an ∞-Groupoid: The Scholze–Stix Resolution	18, 23, 24
26	The Indeterminacy Calculus: Automorphisms, Not Identifications	23–25
27	Dilation Operators on Gauge Stacks	6, 9, 10, 17
28	Construction of the Θ-Dilation	15, 17, 22, 24, 25, 27
29	Functoriality, Base Change, and Compatibility with Descent	18, 20, 25, 27, 28
30	The Dilation Comparison Theorem	17, 20, 21, 27, 28
31	Multiradiality as Descent: A Reformulation	25, 26, 28, 30; uses 36 downstream
32	The Dilation Cocycle and Its Cohomology	18, 23, 28–31
33	Obstruction Classes, Vanishing Criteria, and the Rigidity Lemma	21, 28, 32
34	Abstract Six-Functor Formalisms	16–18, 20; Appendix A
35	Dualisability, Categorical Traces, and Trace Cyclicity	24, 25, 34; Appendix A
36	Trace Invariance under Groupoids of Identifications	20, 25, 28, 35
37	Heights as Traces: The Adelic Trace Formula	18–20, 28, 35, 36
38	Arithmetic Riemann–Hurwitz over the Substrate	18, 20, 21, 23, 35–37
39	The Master Inequality (Theorem A)	3, 17, 18, 20–22, 28, 30, 32, 36–38; 40 for constants; 41–44 for consequences
40	Constants, Error Terms, and Effectivity	18, 34–39
41	From the Master Inequality to Szpiro's Conjecture	19, 37–40
42	Frey–Hellegouarch Curves and the Passage to abc	2, 19, 39, 41
43	The *abc* Conjecture over $\mathbb{Q}$ (Theorem B)	3, 19, 39–42
44	abc over Number Fields	39–43
45	Consequences: Fermat, Catalan, Roth, and the Erdős–Woods Problem	41–44
46	Uniformity, Families, and the Geometric Analogue	20, 21, 37–39, 45
47	Counterexample Search, Extremal Triples, and Numerical Evidence	1, 39, 40, 46
48	Computing in the Habiro Ring and with q-Deformations	13–15, 47
49	Algorithms for Prismatic and Syntomic Cohomology	7–10, 48
50	Certified Trace Evaluation and Interval Arithmetic	17, 23, 35–40, 49
51	The abc-Verification Engine: Architecture and Databases	40–43, 49, 50
52	Reproducibility: Artefacts, Formats, and Benchmarks	48–51

CH.	CHAPTER TITLE	DECLARED PREREQUISITES
53	Formalising the Substrate: A Lean 4 Blueprint	6–20, 52
54	Formalising ∞-Categories, Six Functors, and Traces	34–36, 53
55	The Proof-Obligation Ledger as a Formal Artefact	5, 54; Appendix H
56	The Multi-Agent Mathematician: An Architecture for Machine-Assisted Discovery	3, 39, 55; repository audits
57	Epistemic Hygiene, Provenance, and the Sociology of Verification	39, 55, 56
58	The Obstruction Atlas	3–5, 15, 17, 18, 21, 25, 26, 34–40, 53–57
59	Falsifiability: Critical Experiments and Kill Criteria	2, 18, 22, 27–29, 36, 38–40, 47–58
60	The Substrate Beyond *abc*: Vojta, Szpiro, and Uniformity	1, 2, 18, 21, 37–42, 46, 55, 59
61	What Has Been Proved and What Has Not	5, 18, 21, 28, 36, 38–42, 44, 55; Appendix H
62	A Programme for the Next Decade	46, 49–55, 58–61; Appendix H

The apparent cycle between Chapters 22 and 23 is present in the declared preludes: Chapter 22 names Chapter 23, while Chapter 23 names Chapter 22. It should be read as a coupled local package—semistable reduction on one side, logarithmic boundary structure on the other—rather than as a valid topological ordering. Chapter 31 similarly states that Chapter 36 is used but not presupposed. These exceptions are useful warnings against treating a chapter list as a proof-obligation DAG.

Reading paths

The dependency preludes are cumulative. Consequently a path advertised as self-contained is necessarily much longer than an inspection itinerary. The following paths are closed under the declared chapter prerequisites; ranges are to be read in increasing order. Appendices A and F supply the higher-categorical and Arakelov background named by Chapters 19, 34, and 35. Appendix H is added whenever the route enters the ledger or proof-status audit.

(i) *The shortest dependency-closed path to Theorem B:*

1–32, 34–43, Appendices A and F.

This is the closure of Chapter 43 after the explicitly downstream references in the prelude of Chapter 39 are excluded from the prerequisite relation. Chapter 33 is not in that closure: it may be read after Chapter 32, but no later chapter on this route declares it as an input.

(ii) *The arithmetic-geometry path through families and uniformity:*

1–32, 34–46, Appendices A and F.

This is the closure of Chapter 46. It emphasises adelic and Arakelov data, semistable curves, conductors, discriminants, logarithmic geometry, Frey curves, number fields, families, and their required substrate and trace interfaces.

(iii) *The foundations reader:*

1–18, 20–25, 27–28, 34–36, Appendix A.

This is the closure of Chapter 36. It follows the proposed manufacture of an arithmetic base direction from δ-rings and prisms through Habiro and condensed geometry to descent, dilation, six functors, and categorical trace.

(iv) *The computational and reproducibility path:*

1–32, 34–52, Appendices A and F.

This is the closure of Chapter 52. Chapters 48–52 separate exact arithmetic and executable shadows from the open comparison identifying a computed model with the abstract trace.

(v) *The dependency-closed sceptical audit:*

1–32, 34–55, 61, Appendices A, F, and H.

This is the closure of Chapter 61. A preliminary inspection may begin with Chapter 61 and Appendix H, but reconstructing its audit requires the chapters listed above.

(vi) *The full provenance, falsification, and programme path:*

1–32, 34–62, Appendices A, F, and H.

This is the closure of Chapter 62. It studies both the proposed argument and the machinery by which a reader distinguishes a theorem, a conditional interface, a computation, a formal shadow, and an unresolved obligation.

For any path, the *Epistemic Status of This Book* remains the first prerequisite. The tags at theorem level override the rhetorical momentum of the surrounding exposition, and Appendix H is the generated authority for the 98 obligations PO-001–PO-098.

An itinerary for the reader with one afternoon

The reading paths above are dependency-closed and therefore long. A referee, an editor, or a mathematician deciding whether the volume repays further attention needs something else: the shortest route that permits an informed judgement. The following itinerary is not self-contained and is not intended to be. It is arranged so that a reader who stops at any point has stopped somewhere defensible.

(i) *The status declaration* (*Epistemic Status of This Book*, in the front matter) and this Leitfaden, about a dozen pages. If the reader disagrees with the tagging discipline itself, nothing later will repair that disagreement.

(ii) *Chapter 3*, the absolute obstruction. This is the one negative result on which the entire motivation rests, it is proved here, and it is short. A reader who believes Chapter 3 is wrong should stop: the book has no reason to exist.

(iii) *Chapter 5*, the design contract, followed immediately by *Chapter 61*, what has been proved. Reading the promise and the audit back to back, out of order, is the fastest way to detect the failure mode that most threatens a long conditional argument—a claim that grows in strength between its statement and its recapitulation.

(iv) *Chapter 39*, the master inequality, read only for its hypothesis list, together with *Appendix H*. Theorem A is an eight-link argument; its dependency closure is 59 obligations. The reader need not follow the links to see what is being assumed.

(v) *Chapter 59*, falsifiability. Three of its predictions are specified precisely enough to be computed by a third party, and each is accompanied by what a competing account predicts instead. This is the chapter by which the programme should be judged, because it is the chapter that could destroy it.

(vi) *Chapter 55*, the ledger, § on soundness. The book states its own certification apparatus's soundness condition and records that the condition is not known to hold. A reader who wants to know how candid the volume is will find out here faster than anywhere else.

Where the unconditional mathematics is

The volume contains 1,420 tagged statements. The distribution across the six tags is tabulated in the *Epistemic Status* declaration; the counts are mechanical, produced from the sources of this edition by the statement scan of its status-label gate, and they can be reproduced by any reader with the sources.

Of the 62 chapters, the following 19 contain *no* conditional and *no* open statement whatsoever:

2, 3, 4, 6, 7, 8, 9, 10,
14, 16, 17, 27, 35, 42, 48, 50,
51, 52, 58.

Everything asserted in those chapters is either established mathematics attributed to its authors, a proof reproduced here for self-containedness, or a new argument proved in the text. They are, in the strict sense, the part of this book that does not depend on the programme succeeding. A reader who rejects the substrate entirely may still read them.

The complement is equally worth naming. By the census, the chapters carrying the most conditional statements are 45, 33 and 39, followed by 37, 29 and 41. Chapter 45 is instructive: it derives Fermat, Catalan, Roth-type and Erdős–Woods consequences, and *every one* of its principal statements is conditional, because each descends from Theorem B. The consequences are attractive and they are not available. Nothing in this volume entitles a reader to cite them.

Two cautions about these counts. First, a chapter free of [CONDITIONAL] and [OPEN] tags is not thereby free of dependence on the programme: it may be *about* an object whose existence is open, while proving only correct statements concerning it. Chapter 27 is the sharpest instance—its dilation theory is proved, but on a fixed prism the operator it constructs is inner, so the theory acquires content only through the prism motion of Chapter 28. This is recorded in the text as a warning, not buried. Second, the tag census counts statements, not weight; one [CONDITIONAL] theorem may carry more of the argument than forty [KNOWN] lemmas. Appendix H, not the census, is the authority on what rests on what.

Notation and Conventions

This list fixes the conventions used throughout the volume and gives a forward reference to the principal notation. It is not a substitute for the definitions: several objects below exist only under hypotheses stated in the chapter cited in the right-hand column. In particular, the presence of a symbol in this list makes no assertion that every comparison map or global construction involving it has been proved.

Global conventions

Algebra and geometry. Unless a contrary convention is stated, rings are commutative and possess a unit, ring homomorphisms preserve the unit, and modules are unital. Schemes used as geometric inputs are quasi-compact and quasi-separated; morphisms are morphisms over the displayed base. Fibre products and derived tensor products are taken over that base. A completion is derived when the surrounding category is derived and classical otherwise. The word “perfect” refers to dualisability in the appropriate derived category, not merely to finite generation of cohomology.

Grading and complexes. Complexes are cohomologically graded. The cotangent complex of a regular quotient has its conormal contribution in degree -1, hence is represented by a module followed by $[1]$. Homotopy groups of spectra and derived stacks use homological subscripts. Filtrations are decreasing unless an increasing convention is explicitly displayed. The symbol gr^{i} denotes the i-th associated graded piece.

Places, logarithms, heights, and degrees. All logarithms are natural. At a finite place v, the basic weight is $\log \mathrm{N}v$; at an archimedean place the weight n_v records the real or complex multiplicity. Local heights use the sign $-\log \|s\|_v$. Thus a finite module M has $\deg M = \log \#M$, and an effective vertical divisor has non-negative degree. For a number field K, absolute heights and field invariants are normalised by $[K:\mathbb{Q}]$ unless the adjective *relative* or *unnormalised* is present. In particular,

$$d_K = [K:\mathbb{Q}], \qquad \delta_K = d_K^{-1} \log |D_K|.$$

The Petersson and Faltings-height normalisation used for elliptic curves is the one stated in Chapter 37; changing it alters heights by a displayed additive constant, not by an unrecorded convention.

Small parameters and strain. Letters $\epsilon, \varepsilon, \eta, \delta > 0$, when used without geometric arguments, denote adjustable tolerances. The *strain class* is always written with its arguments as $\varepsilon(X/\mathfrak{S})$. It is an object of a derived category, not a small positive real number. The distinction is carried by the presence of the arguments and by nothing else. In particular it is *not* typographical: the same letter ε serves as the tolerance of Theorem 39.1.6 and as the strain class of Chapter 20, and the variant glyphs ϵ and ε are used interchangeably for the tolerance across the volume — Chapter 43 writes ε and Chapter 45 writes ϵ for one and the same quantity. A reader must therefore read the argument, not the shape of the letter. The same warning applies to ϑ: with

a gauge argument $\vartheta(G)$ is the infinitesimal generator of a dilation (Chapter 27), while with a strain package $\vartheta(X)$ is the strain degree that stands on the left of the master inequality.

Asymptotic notation. The assertions $f = O_A(g)$ and $f \ll_A g$ both mean that $|f| \le C(A)|g|$ on the stated range for some constant depending only on the displayed data A. An omitted subscript means an absolute constant. Uniformity in a number field, a curve, or a deformation parameter is never inferred from an omitted argument: it must be stated. The notation $O(1)$ means bounded in the variables then tending to a limit, with every permitted dependence recorded in the surrounding sentence.

Epistemic vocabulary. The six theorem-status marks have the following exact meanings. [KNOWN] denotes a result from the existing literature, with a citation at the point of use. [REPROVED] denotes a known result for which a proof is supplied in the language of this volume. [NEW] denotes an unconditional result first proved here; it may not depend, even transitively, on an open obligation. [CONDITIONAL] denotes a proved implication whose hypotheses include the proof obligations named in the tag. [OPEN] denotes a statement not proved in this volume. [DISPUTED] reports a publicly documented disagreement about status or interpretation and does not adjudicate it. A numbered `PO-xxx` is an open interface recorded in the proof-obligation ledger. These marks attach to assertions, not to the confidence or authority of an author.

Sets, spaces, and categories

SYMBOL	MEANING	DEFINED
$\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$	natural numbers, integers, rationals, reals, and complex numbers	standard
$\mathbb{F}_p, \mathbb{Z}_p, \mathbb{Q}_p$	prime field, p-adic integers, and p-adic field	Chs. 3, 6
$\mathrm{Spec}A, \mathrm{Spf}A$	affine spectrum and formal spectrum of A	standard
Hom, End, Aut, Ext, Map	ordinary or derived mapping objects, as determined by the ambient category	standard
$\mathcal{C}, \mathcal{D}, \mathcal{E}$	generic categories in a six-functor or trace formalism	Chs. 34–35
$\mathbb{F}_1$	the conventional symbol $\mathbb{F}_1$ for "the field with one element"; it is not an ordinary field in this volume	Chs. 3–6
$\mathrm{Fun}(\mathcal{C}, \mathcal{D})$	functor category	standard
$\overline{\mathbb{Q}}$	algebraic closure of $\mathbb{Q}$	standard
$\mathbb{A}^n, \mathbb{P}^n, \mathbb{G}_a, \mathbb{G}_m$	affine and projective space, and the additive and multiplicative group schemes	standard
Gal, Pic, Div, CH, Lie	Galois group, Picard group, divisor group, Chow group, and Lie algebra	standard
$R\Gamma$, Ind, Res	derived global sections, induction, and restriction	standard
$\mathrm{Perf}(X)$	stable category of perfect complexes on X	Chs. 8, 35
$\mathrm{Nuc}(X)$	category of nuclear modules over the condensed or analytic object X	Chs. 16, 36
$\mathrm{Cond}(\mathrm{Ab})$	category of condensed abelian groups	Ch. 16
$\mathrm{Corr}(\mathcal{C}, \mathcal{E})$	category of correspondences in $\mathcal{C}$ with coefficients in $\mathcal{E}$	Ch. 34
AnRing, An	analytic rings and the analytification functor	Chs. 17, 36
$\mathrm{Pl}(\mathbb{Q})$	finite and archimedean places of $\mathbb{Q}$	Ch. 18
$\mathrm{Ar}_{\mathbb{Q}}$	arithmetic indexing category for the global leg diagram	Ch. 18

SYMBOL	MEANING	DEFINED
THH, TC, TP	topological Hochschild homology, topological cyclic homology, and topological periodic homology	Appendices A, C
dR, syn, HT	de Rham, syntomic, and Hodge–Tate decorations	Chs. 7–9
$X^\vee$	dual of a dualisable object X	Ch. 35
$\mathrm{End}(\mathbf{1})$	endomorphism ring spectrum of the monoidal unit	Ch. 35
$X_{F,E}, X_{FF}$	Fargues–Fontaine constructions attached to the displayed input	Chs. 4, 7

Rings, δ-rings, prisms, and q-deformations

SYMBOL	MEANING	DEFINED
δ, δ_p	a δ-operator and its p-typical specialisation	Chs. 4, 6
$F(x) = x^p + p\delta_p(x)$	Frobenius lift associated with a δ_p-structure	Ch. 6
$\delta_p(n)$	arithmetic Fermat quotient $(n - n^p)/p$ in the convention of Chapter 6	Ch. 6
$\Delta = (A, I)$	a prism: a δ-ring with distinguished prism ideal I, subject to the prismatic axioms	Ch. 7
Δ	enlarged or universal prismatic object, with the displayed subscript specifying the input	Ch. 5, App. C
$\Delta_{X/A}$	prismatic cohomology of X relative to A	Ch. 7
A_{inf}	Fontaine's period ring $W(\mathcal{O}_C^\flat)$ in the perfectoid setting	Chs. 6–7
B_{dR}	de Rham period ring	Ch. 4
$\mathfrak{S}$	Breuil–Kisin prism used for local semistable models	Ch. 7
θ	Fontaine's map $A_{\mathrm{inf}} \to \mathcal{O}_C$; context distinguishes it from a theta function	Ch. 7
WCart	Cartier–Witt stack	Ch. 8
$\mathbb{Z}^\Delta$	fixed-prime absolute prismatisation $\mathbb{Z}_p^\Delta \simeq$ WCart (house shorthand at the chosen p)	Chs. 8–9
$\mathbb{Z}^{\mathrm{syn}}$	fixed-prime syntomic stack $\mathbb{Z}_p^{\mathrm{syn}}$ (house shorthand at the chosen p)	Ch. 9
$J^n(X)$	n-th arithmetic jet space	Ch. 4
ψ	arithmetic Kodaira–Spencer or jet-theoretic operator in the local comparison	Ch. 12
$\mathrm{KS}_{\mathcal{E}/C}$	Kodaira–Spencer morphism of a family $\mathcal{E}/C$	Ch. 12
Λ	context-dependent: Borger's Λ-structure, the q-deformation base $\mathbb{Z}[[q-1]]$, or a displayed lattice or dilation-parameter monoid	Chs. 4, 13–14, 27, 30, 32, 39
$[n]_q = [n]_q$	q-integer $(q^n - 1)/(q - 1)$	Ch. 13
$[p]_q = [p]_q$	cyclotomic q-integer governing the p-prismatic specialisation	Chs. 7, 13
$[n]_q!$	q-factorial $\prod_{i=1}^{n} [i]_q$, with $[0]_q! = 1$	Ch. 13
q-dR	q-de Rham complex	Ch. 13
∇_q	q-difference or q-derivation operator, with its precise domain displayed at each use	Chs. 3, 13, 15

SYMBOL	MEANING	DEFINED
$\widehat{\mathbb{Z}[q]}$	Habiro completion $\varprojlim_N \mathbb{Z}[q]/((1-q)\cdots(1-q^N))$	Ch. 14
$\mathcal{H}$	short form for $\widehat{\mathbb{Z}[q]}$, also used as a subscript for the Habiro chart and its coefficient objects	Chs. 14–15
$\mathrm{Str}_{\mathfrak{A}}$	proposed all-primes strain operator over the absolute base	Ch. 15
$\mathbf{B}$	generic manufactured base in the comparison of surrogate geometries	Ch. 4
Abs	abstract absolute-base construction of the design vocabulary; glossed but not used as a running symbol	App. G
$\mathbf{B}_{\mathbb{F}_1}$	house symbol $\mathbf{B}_{\mathbb{F}_1}$ for a proposed base beneath $\mathrm{Spec}\mathbb{Z}$; glossed but not used as a running symbol	App. G
$\mathfrak{A}$	proposed absolute arithmetic base	Chs. 3, 11, 15
I_K	Kummer or logarithmic boundary ideal in the local shell package	Ch. 23

Sites, topoi, substrate objects, and analytic legs

SYMBOL	MEANING	DEFINED
$\mathfrak{S}_p$	finite p-adic leg of the global substrate	Ch. 18
$\mathfrak{S}_\infty$	archimedean analytic-gauge leg of the substrate	Chs. 17–18
$\mathfrak{S}$	global substrate obtained from the finite and archimedean leg diagram, subject to the stated descent obligations	Ch. 18
σ	product-formula descent datum on the leg diagram	Ch. 18
$\mathbb{G}_m^{\sharp}$	divided or formal multiplicative group acting on the gauge data	Ch. 18
$\mathfrak{G}_\infty$	archimedean gauge stack	Ch. 17
$\mathcal{T}_\infty$	archimedean test category for analytic gauges	Ch. 17
X_{dR}	de Rham or infinitesimal analytic avatar of X	Ch. 17
$\mathbb{Z}_\blacksquare$, $\mathbb{Q}_{p\blacksquare}$	solid integral and p-adic coefficient objects	Ch. 16
$\mathbb{Z}[[S]]$	completed coefficient ring used in the condensed model	Ch. 16
$\mathcal{M}_{p'}(S)$	prime-to-p measure module on S	Ch. 16
$\mathcal{R}_W(M)$	liquid or condensed Rees object attached to a filtered module	Ch. 27

SYMBOL	MEANING	DEFINED

Gauges, strain, dilation, and trace

SYMBOL	MEANING	DEFINED
$\mathrm{Rees}(M), U(M)$	Rees construction of a filtration and its underlying object	Ch. 10
$\mathbf{Gauge}(M)$	groupoid or category of gauge structures on M	Ch. 10
κ	context-dependent: the arithmetic Kodaira–Spencer class in Chapters 12 and 21, and the strong-limit cardinal controlling size in Chapter 16	Chs. 12, 16, 21
$\mathbb{L}_{B/A}$	cotangent complex of $A \to B$	Ch. 3
$\Omega^1_{B/A}$	module of Kähler differentials of $A \to B$	Ch. 3
$\mathbb{L}_{\mathbb{Z}/\mathbb{Z}}$	zero absolute cotangent complex of $\mathbb{Z}$ over itself, used as the obstruction benchmark	Ch. 58
$\mathbb{L}_{\mathbb{Z}/\mathfrak{S}}$	absolute arithmetic base-direction complex	Ch. 20
$\mathfrak{s}_X = H^0(\mathbb{L}_{X/\mathfrak{S}})$	strain sheaf of X	Ch. 20
$\varepsilon(X/\mathfrak{S})$	strain class represented by the boundary morphism in the cotangent-complex transitivity triangle	Ch. 20
$\kappa(X)$	arithmetic Kodaira–Spencer class of X	Ch. 21
$\kappa_{\log}$	logarithmic Kodaira–Spencer class	Ch. 23
$\mathcal{J}_X$	logarithmic shell of X	Ch. 23
E_q	Tate curve with parameter q	Ch. 24
$\theta(q)$	classical theta value or section on the Tate curve	Chs. 24, 28
Θ	house shorthand for the Θ-line or its dilation carrier; the displayed argument determines the object	Chs. 24, 27–28
Dil	dilation operation or its parameter monoid; glossed but not used as a running symbol	App. G
$\mathcal{G}$	gluing ∞-groupoid of local substrate data	Chs. 25–26
$\pi_0(\mathcal{G}), \pi_1(\mathcal{G}, \mathfrak{g})$	components and based automorphism group of the gluing groupoid	Ch. 25
Θ_λ	dilation by the parameter λ along the Θ-line	Chs. 27–29
$\vartheta(G)$	infinitesimal generator of a dilation on a gauge G; the gauge argument, and only the gauge argument, distinguishes it from the strain degree $\vartheta(X)$ below	Ch. 27
$\mathbf{Gauge}_\Theta$	category of gauges equipped with compatible Θ-dilation	Ch. 30
$\eta(\tau)$	descent comparison attached to a local-to-global trivialisation τ	Ch. 30
$\mathcal{X}$	multiradial descent object	Ch. 31
$c(\lambda, \mu)$	dilation two-cochain measuring the defect of a chosen multiplication law	Ch. 32
$\mathfrak{f}_X$	conductor ideal $\prod_p p^{f_p(X)}$ attached to the arithmetic object X	Ch. 32
A_X	obstruction or rigidity class attached to the dilation cocycle	Ch. 33

SYMBOL	MEANING	DEFINED
$f_!, f^!$	exceptional pushforward and extraordinary pullback	Ch. 34
$\mathbb{D}_f$	relative dualising object for f	Ch. 34
ω_f	dualising line or relative canonical object	Ch. 34
$\mathrm{tr}(f)$, Tr	categorical trace of an endomorphism and the global trace construction	Chs. 35, 37
$\mathrm{Fix}(c)$	fixed locus of a correspondence c	Ch. 37
$t_v(u)$	local trace contribution of u at the place v	Ch. 37
$\mathfrak{d}_{X/Y}$	different over the substrate, represented by $\det \mathbb{L}_{X/Y}$ with substrate metric	Ch. 38
$\mathrm{Ram}(f)$	ramification gauge of an admissible morphism f	Ch. 38

Heights, places, adelic invariants, and curves

SYMBOL	MEANING	DEFINED
$\mathrm{rad}(n)$	product of the distinct rational primes dividing n	Ch. 1
$\mathrm{rad}(abc)$	radical of an abc-triple	Ch. 43
$\mathrm{rad}(F)$	square-free radical of a polynomial F	Ch. 2
$q(a,b,c)$	quality $\log c / \log \mathrm{rad}(abc)$ of a positive abc-triple	Ch. 1
$2g-2$	canonical degree of a smooth projective curve of genus g	Ch. 2
$\Omega^1_{X/k}$	sheaf of Kähler differentials	Ch. 2
$\lVert x \rVert_v$	normalised absolute value at the place v	Ch. 18
$\mathrm{ord}_v(x)$	additive valuation at v, normalised as stated for the corresponding absolute value	Ch. 18
$(K, (\Omega, \mathcal{A}, \nu), (\lvert\cdot\rvert_\omega))$	adelic curve: a field, a measured space of places, and a measurable family of absolute values	Ch. 19
$\overline{E}$	adelic vector bundle E equipped with its family of local norms	Ch. 19
$\widehat{\deg}(\overline{E}), \widehat{\mu}(\overline{E})$	arithmetic degree and normalised slope of an adelic vector bundle	Ch. 19
$\nu_{\overline{E}}$	Harder–Narasimhan probability measure of $\overline{E}$	Ch. 19
$\overline{\mathcal{L}}$	adelic Hodge line bundle	Ch. 19
g_ω	Green function $-\log \lVert s \rVert_\omega$ at the place ω	Ch. 19
c_{dim}	dimensional normalising weight; $c_{\mathrm{dim}} = 12$ for genus one	Ch. 19
w_v, n_w, n_v	product-formula, extension, and archimedean multiplicity weights	Chs. 18, 37
$h_{\mathrm{Tr}}(X)$	trace height obtained from the first variation of $\mathrm{Tr}(\Theta_{e^s})$	Ch. 37
$\text{log-vol}(L, \lVert\cdot\rVert)$	logarithmic covolume contribution of a metrised line or determinant object	Ch. 38
$\lVert\Delta\rVert(\tau)$	Petersson norm of the modular discriminant	Ch. 37
$\vartheta(X)$	strain degree $\deg \det \mathbb{L}_{X/\mathfrak{S}}$	Ch. 38
$q_K(X)$	normalised conductor term in the master inequality	Ch. 39

SYMBOL	MEANING	DEFINED
$r_K(X)$	normalised different term in the master inequality	Ch. 39
d_K, D_K, δ_K	degree, absolute discriminant, and logarithmic root discriminant of K	Chs. 39–40
$h_\Delta(E/K)$	normalised logarithmic minimal-discriminant height	Ch. 41
$n(E/K)$	normalised logarithmic conductor of E/K	Ch. 41
$\mathrm{Szp}(\delta, K), \mathrm{Szp}'(\delta, K)$	discriminant and height forms, respectively, of the Szpiro-type inequality used in the Frey passage	Ch. 42
$H_{\mathrm{mod}}(E)$	modular height $\max\{\lvert c_4(E)\rvert^3, \lvert c_6(E)\rvert^2\}$, in the stated normalisation	Ch. 41
$E^{\mathrm{Frey}}_{a,b}$	Frey–Hellegouarch curve attached to $a + b = c$	Ch. 42
$\mathrm{q}(a, b, c)$	quality $\log c / \log \mathrm{rad}(abc)$, synonymous with $q(a, b, c)$	Ch. 43
$c_4, c_6, \Delta_{\min}, N_E$	standard elliptic invariants, minimal discriminant, and conductor	Chs. 41–42
q_E	Tate parameter of an elliptic curve with split multiplicative reduction	Ch. 22
Φ_E	component group of the Néron model of E	Ch. 22
$\mathfrak{M}_{1,1}$	substrate version of the moduli stack of elliptic curves	Ch. 39

SYMBOL OR OBJECT	MEANING	DEFINED

Uniformity and computational objects

SYMBOL OR OBJECT	MEANING	DEFINED
$\varepsilon_{\mathcal{B}}$	relative strain class for a family over $\mathcal{B}$	Ch. 46
$\kappa_{\mathcal{B}}$	family Kodaira–Spencer class	Ch. 46
$\vartheta_{\mathcal{B}}$	family strain degree	Ch. 46
$\mathrm{def}_{\varepsilon}$	deformation defect used in the uniformity analysis	Ch. 46
$\mu(a,b,c)$	merit statistic for an abc-triple in the search database	Ch. 47
$\mathcal{P}(x,z)$	prescribed search region for triples	Ch. 47
$\mathcal{E}_Q(x)$	extremal envelope at quality threshold Q	Ch. 47
precision-N Habiro record	canonical residue modulo $\prod_{n=1}^{N}(1-q^n)$, with exact integer coefficients	Ch. 48
truncated q-jet	finite Taylor or divided-difference data with an exact truncation certificate	Ch. 48
split gauge record	finite matrix presentation of a split gauge at (p,p^N)	Ch. 49
rational interval $[a,b]$	closed enclosure with $a,b\in\mathbb{Q}$; endpoints and rounding policy form part of the certificate	Ch. 50
complex ball $B(m,r)$	certified disc with rational or exactly encoded midpoint and outward radius	Ch. 50
verification run	versioned execution joining input records, arithmetic checks, model checks, certificates, and obligation status	Ch. 51
computational claim closure	dependency graph from a published computational claim to source, environment, data, and checker	Ch. 52
artefact manifest	canonical record of hashes, media types, schema versions, commands, environment, and outputs	Ch. 52
$\mu_{\infty}(K,\varepsilon)$	unresolved archimedean coercivity loss in the effective master inequality	Chs. 40, 58

How to use this list

A chapter number in the final column locates the first substantive definition, not necessarily the first occurrence. A range indicates that the notation is assembled in stages. Standard symbols such as $\mathbb{R}$, Spec, and Hom retain their conventional meanings, whereas blackboard, fraktur, and script letters carrying an or font belong to the substrate architecture and must not be silently replaced by classical schemes or ordinary groupoids.

There are three recurrent collisions of notation. First, Fontaine's $\theta\colon A_{\mathrm{inf}}\to O_C$, the classical theta function $\theta(q)$, and the strain degree $\vartheta(X)$ are different objects. Second, the tolerance $\varepsilon>0$ is not the strain class $\varepsilon(X/\mathfrak{S})$. Third, $d_K=[K:\mathbb{Q}]$ is the field degree, whereas $\mathfrak{d}_{X/Y}$ is a different. Arguments and type, rather than typography alone, resolve each collision.

Finally, notation introduced inside a conditional construction inherits that construction's status. For example, the formula defining $\varepsilon(X/\mathfrak{S})$ is an ordinary cotangent-complex formula once $\mathfrak{S}$ exists, but the claimed global substrate and some comparisons of its local legs remain

subject to the obligations stated in Chapter 18 and Appendix H. This convention prevents a convenient symbol from laundering an open existence or comparison theorem into an established object.

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Part 0

The Problem and the Obstruction

Chapter 1

The abc Conjecture: Statements, Equivalents, and the Shape of the Difficulty

Dependencies. None.

Summary. The opening chapter fixes the arithmetic problem to which the rest of this volume is addressed. We state the Masser–Oesterlé conjecture in its constant, finiteness, and quality forms; record the leading verified examples; and compare the conjectural power bound with the effective exponential bound of Stewart–Yu. The elliptic, height-theoretic, and Diophantine formulations are compared with every epsilon conversion and effectivity loss visible. We also state at the outset what the volume proves unconditionally and what remains conditional on its ledger of ninety-eight obligations. The conclusion isolates the obstruction: an additive equation is being measured by multiplicative reduced support, while the decisive differential and Riemann–Hurwitz step of Mason–Stothers has no ordinary arithmetic analogue.

1.1 Masser–Oesterlé

For a non-zero integer n, write

$$\operatorname{rad}(n) = \prod_{p \mid n} p,$$

the product being over positive rational primes. An *abc-triple* is a triple (a, b, c) of positive integers satisfying

$$a + b = c, \qquad \gcd(a, b) = 1.$$

The coprimality condition then makes a, b, c pairwise coprime. Put

$$R(a, b, c) = \operatorname{rad}(abc), \qquad q(a, b, c) = \frac{\log c}{\log R(a, b, c)}.$$

The quotient is independent of the base of the logarithm. We call it the *quality* of the triple. These conventions will remain fixed throughout the book.

Example 1.1.1 (Why primitivity is essential). If the gcd condition is removed, no inequality of the conjectural shape can hold. For $m \geq 1$,

$$3^m + 3^m = 2 \cdot 3^m$$

has

$$\operatorname{rad}(3^m \cdot 3^m \cdot 2 \cdot 3^m) = 6,$$

while the third entry tends to infinity. Thus for every fixed ε and every fixed K_ε, the inequality $2 \cdot 3^m \leq K_\varepsilon 6^{1+\varepsilon}$ eventually fails. Dividing a general integer solution by $\gcd(a, b)$ is therefore not cosmetic: it removes an arbitrary common scaling whose multiplicity the radical would otherwise ignore.

Conjecture 1.1.2 (Masser–Oesterlé). *[OPEN] For every $\varepsilon > 0$ there is a constant $K_\varepsilon > 0$ such that for every abc-triple,*

$$c \le K_\varepsilon \operatorname{rad}(abc)^{1+\varepsilon}. \tag{1.1.1}$$

The conjecture is open. The attribution records Masser's formulation at the 1985 London meeting and Oesterlé's closely related formulation through elliptic curves; early published discussion includes [Mas85; ST86].

Proposition 1.1.3 (Symmetric sign form). *[REPROVED : Mas85] Conjecture 1.1.2 is equivalent, with the same constants, to the assertion that for all pairwise coprime non-zero integers x, y, z satisfying $x + y + z = 0$,*

$$\max\{|x|, |y|, |z|\} \le K_\varepsilon \operatorname{rad}(xyz)^{1+\varepsilon}.$$

Proof. In a non-zero relation $x + y + z = 0$, one term has sign opposite to the other two. After permuting and multiplying the relation by -1 if necessary, it has the form $a + b = c$ with $a, b, c > 0$, and the opposite-sign term has the largest absolute value. The radical is unchanged by signs and permutation. The converse follows by writing a positive triple as $a + b + (-c) = 0$. □

The exponent $1+\varepsilon$ has two logically separate features. The 1 is the predicted main exponent; the positive loss makes room for a constant that is uniform over all triples. The record table below already refutes the constant $K = 1$ at exponent 1, while Proposition 1.1.6 proves that no absolute value of K can work at that exponent. Nor does the conjecture prescribe an effective procedure for computing K_ε.

The status of this volume

Warning 1.1.4 (*Programme, not proof*). This book does not prove Conjecture 1.1.2. It constructs an arithmetic analogue of the geometric mechanism behind Mason–Stothers, proves the parts of that construction that are presently accessible, and derives the *abc* inequality only after explicitly assuming the missing interfaces.

The consolidated ledger contains exactly ninety-eight numbered proof obligations. In the present edition all ninety-eight are recorded as open, with an owner, upstream and downstream dependencies, a risk class, and a proposed falsifier. The transitive ledger closure used by Theorem A consists of fifty-nine of those obligations. The later elliptic deduction also names the unallocated interfaces (CM), (MI_Δ), and (SR) rather than treating them as consequences already in hand. Thus neither "ninety-eight" nor "fifty-nine" is a count of proved lemmas: the first is the size of the book-wide audit ledger, and the second is the numbered upstream gate for the master inequality.

The unconditional content of the volume includes definitions, comparison theorems whose stated hypotheses are verified, no-go results, local calculations, and formal implications from the ledger assumptions. The conditional content begins where a statement is marked CONDITIONAL or OPEN. In particular, a proof that the assumptions imply (1.1.1) is not a proof that the assumptions hold. This distinction is maintained in every later chapter and is part of the mathematical claim, not an editorial disclaimer.

Proposition 1.1.5 (Constant, finiteness, and quality forms). *[REPROVED : Mas85] We include the quantifier conversion and the lower bound for the limiting quality, which are often left implicit. The following are equivalent:*

(i) *Conjecture 1.1.2;*

(ii) *for every $\varepsilon > 0$, only finitely many abc-triples satisfy $q(a, b, c) > 1 + \varepsilon$;*

(iii) $\limsup q(a,b,c) \le 1$ *as* $c \to \infty$ *through abc-triples.*
If these conditions hold, the last limsup is exactly 1.

Proof. Assume (i). If $q > 1+\varepsilon$, then $c > R^{1+\varepsilon}$; applying (i) with $\varepsilon/2$ gives $R^{\varepsilon/2} < K_{\varepsilon/2}$. Hence R is bounded. Only finitely many pairwise coprime triples supported on a fixed finite set of primes can solve $a+b=c$ with bounded radical: this is the standard finiteness of the corresponding S-unit equation, or, equivalently here, the finiteness form built into the Masser–Oesterlé statement [BG06]. Thus (ii) follows. Conversely, remove the finite exceptional set in (ii) and let K_ε be the maximum of $c/R^{1+\varepsilon}$ on it, enlarged to at least 1; this gives (i). The equivalence of (ii) and (iii) is the definition of limsup.

For the reverse inequality, use $1+(2^n-1)=2^n$. Its radical is at most $2(2^n-1)<2^{n+1}$, whence $q \ge n/(n+1)$. Therefore the limsup is at least 1. □

The first implication contains a useful quantitative threshold. If a value of $K_{\varepsilon/2}$ is known and $q(a,b,c) > 1+\varepsilon$, then

$$R(a,b,c) < K_{\varepsilon/2}^{2/\varepsilon}. \tag{1.1.2}$$

Thus an *effective abc* constant turns quality finiteness into a finite computation over prime sets below a displayed bound. Without a computable $K_{\varepsilon/2}$, the same argument proves finiteness but gives no search range. This is the first place in the chapter where qualitative truth and effective use diverge, and the distinction persists through the elliptic and Vojta equivalences.

Proposition 1.1.6 (The epsilon loss is necessary). *[REPROVED: ST86] There are abc-triples for which* $c/\operatorname{rad}(abc)$ *is arbitrarily large. Consequently there is no absolute constant* K *such that*

$$c \le K \operatorname{rad}(abc)$$

for every triple.

Proof. Let $p > 2$ be prime and put

$$a = 1, \qquad b = 2^{p(p-1)} - 1, \qquad c = 2^{p(p-1)}.$$

This is a primitive triple. Fermat's congruence gives $2^{p-1} = 1 + kp$ for some integer k. Raising to the pth power,

$$2^{p(p-1)} = (1+kp)^p \equiv 1 \pmod{p^2},$$

because every nonconstant term of the binomial expansion is divisible by p^2. Hence $p^2 \mid b$. Writing $b = p^2 m$ and using $\operatorname{rad}(m) \le m$ gives

$$\operatorname{rad}(abc) = 2\operatorname{rad}(b) \le 2p\operatorname{rad}(m) \le 2pm = \frac{2b}{p} < \frac{2c}{p}.$$

Therefore

$$\frac{c}{\operatorname{rad}(abc)} > \frac{p}{2},$$

which is unbounded as p varies. □

The family also explains the quantifier in the quality formulation. Unboundedness of $c/\operatorname{rad}(abc)$ is much weaker than unboundedness of $c/\operatorname{rad}(abc)^{1+\varepsilon}$ for one fixed $\varepsilon > 0$. The proof gives only the lower bound $p/2$ for the former ratio and gives no infinite family violating any fixed $1+\varepsilon$ exponent. Thus infinitely many failures at exponent 1 are compatible with the conjectural finiteness above every line $q = 1+\varepsilon$. The positive epsilon allows precisely this sub-power room.

The best unconditional scale

The conjecture should be compared with what is actually known for every triple, not only with computations of selected triples. Write $R = \operatorname{rad}(abc)$.

Theorem 1.1.7 (Stewart–Yu). *[KNOWN: SY01] There is an effectively computable absolute constant $C > 0$ such that every abc-triple satisfies*

$$\log c < C R^{1/3} (\log R)^3, \qquad \textit{equivalently} \qquad c < \exp\!\big(C R^{1/3} (\log R)^3\big), \tag{1.1.3}$$

after enlarging C to absorb the finitely many smallest triples.

This is a strong effective theorem: given the factorisation of abc, its right-hand side is a computable number. Its shape is nevertheless separated from Conjecture 1.1.2 by more than an improvement of a constant. The conjecture asks for

$$\log c \le (1+\varepsilon) \log R + \log K_\varepsilon, \tag{1.1.4}$$

which is linear in $\log R$. The Stewart–Yu exponent divided by $\log R$ is

$$C R^{1/3} (\log R)^2 \longrightarrow \infty.$$

Consequently (1.1.3) is larger than R^A for every fixed power A once R is large. It does not imply that qualities are bounded, let alone that their limsup is 1:

$$q(a,b,c) < C R^{1/3} (\log R)^2.$$

The gap is therefore an exponential-to-polynomial gap. Replacing the effective constant C by a smaller value, or shaving a logarithm from (1.1.3), would not cross it.

The proof of Theorem 1.1.7 belongs to the theory of linear forms in logarithms and S-unit equations. Such methods control a fixed collection of prime powers effectively, but their dependence on the number and size of the primes produces the exponential envelope above. The conjecture predicts a uniform bound governed only by the reduced weighted support $\sum_{p \mid abc} \log p$. Obtaining that uniformity is the difficulty, not a technical refinement postponed until after an otherwise complete proof.

Anatomy of the quantifiers

The order of the quantifiers in Conjecture 1.1.2 is

$$\forall\, \varepsilon > 0 \quad \exists\, K_\varepsilon \quad \forall\, (a,b,c).$$

The constant may deteriorate arbitrarily as $\varepsilon \downarrow 0$. Nothing in the conjecture supplies one K valid for every positive ε, and Proposition 1.1.6 rules out the limiting interpretation $c \le K \operatorname{rad}(abc)$. Conversely, choosing a different constant for each triple would make the statement empty; uniformity in the final quantifier is its content.

The strict inequality in the finiteness form is inessential, but changing it requires an epsilon margin. Finiteness of the triples with $q > 1 + \varepsilon/2$ implies finiteness with $q \ge 1 + \varepsilon$, while the reverse implication is immediate after renaming the parameter. Likewise, taking the limsup as $c \to \infty$ or as $R \to \infty$ gives the same statement: bounded R fixes a finite prime set, and the corresponding S-unit equation has only finitely many solutions. These small quantifier moves are legitimate because they spend part of the positive epsilon; they would not be legitimate at exponent exactly one.

Effectivity is an additional quantifier. An effective version asks for an algorithm which, on input a rational $\varepsilon > 0$, terminates with a valid K_ε. The bare conjecture asserts only existence. Consequently an ineffective proof of Conjecture 1.1.2 would settle the qualitative finiteness statement but would not provide a searchable bound for exceptional triples, Fermat exponents, or integral points. The Stewart–Yu theorem has the opposite profile: its bound is effective, but its asymptotic shape is far too large. The desired result must improve the shape without losing control of the quantifiers.

Numerical extremals fix normalisations and exclude the false exponent-1 constant form. The following are the first five verified rows of `code/python/abctk/triples.py`; every displayed sum, radical, and quality is recomputed from the integers.

	identity	$\operatorname{rad}(abc)$	q
1	$2 + 3^{10} \cdot 109 = 23^5$ $2 + 6\,436\,341 = 6\,436\,343$	15 042	1.62991168
2	$11^2 + 3^2 \cdot 5^6 \cdot 7^3 = 2^{21} \cdot 23$ $121 + 48\,234\,375 = 48\,234\,496$	53 130	1.62599052
3	$19 \cdot 1307 + 7 \cdot 29^2 \cdot 31^8 = 2^8 \cdot 3^{22} \cdot 5^4$ $24\,833 + 5\,020\,969\,537\,415\,167 = 5\,020\,969\,537\,440\,000$	4 688 222 070	1.62349026
4	$283 + 5^{11} \cdot 13^2 = 2^8 \cdot 3^8 \cdot 17^3$ $283 + 8\,251\,953\,125 = 8\,251\,953\,408$	1 876 290	1.58075603
5	$1 + 2 \cdot 3^7 = 5^4 \cdot 7$ $1 + 4\,374 = 4\,375$	210	1.56788726

For the first row,

$$\operatorname{rad}(abc) = 2 \cdot 3 \cdot 109 \cdot 23 = 15\,042,$$
$$q = \frac{\log 6\,436\,343}{\log 15\,042} = 1.62991168\ldots.$$

It was found by Reyssat in 1987 and remains the quality record in the data set used by this volume. The table is empirical mathematics, not evidence of a uniform constant: a finite search can refute a proposed bound but cannot prove Conjecture 1.1.2. Sources and larger tables are discussed in [dS26; BB94; Nit96].

Every record row has three independent certificates: the displayed addition, pairwise gcd equal to one, and complete prime factorisations of all three entries. The last is essential. Omitting one prime factor makes the radical too small and the reported quality too large; adding a previously missed factor can only lower the quality. Decimal logarithms are therefore the final step of the verification, not the source of the arithmetic claim.

Example 1.1.8 (What quality measures). The small triple

$$3 + 5^3 = 2^7$$

has $R = 2 \cdot 3 \cdot 5 = 30$ and

$$q(3, 125, 128) = \frac{\log 128}{\log 30} = 1.4265653296\ldots.$$

Its quality exceeds 1 because the size of b and c is carried by the repeated factors 5^3 and 2^7, whereas the radical charges 5 and 2 only once. This is the local pattern shared by high-quality triples; the difficulty is to show that it cannot persist with quality bounded away from 1 along an infinite sequence.

For a fixed ε, one triple gives a lower bound on every admissible constant:

$$K_\varepsilon \geq \frac{c}{R^{1+\varepsilon}} = R^{q(a,b,c)-1-\varepsilon}.$$

The Reyssat triple therefore forces

$$K_{0.1} \geq 163.5321088\ldots, \qquad K_{0.01} \geq 388.6516716\ldots.$$

These numbers are exact consequences of the displayed integers, but they say nothing about triples beyond the search. A record determines a necessary finite constant; the conjecture asserts that the supremum of the same quantity over *all* triples is finite.

What an extremal table certifies

Quality orders triples by the slope $\log c/\log R$, but a proposed constant at a fixed epsilon is tested by the intercept

$$\log c - (1+\varepsilon)\log R.$$

For a finite record set $\mathcal{R}$, define its demanded logarithmic constant by

$$L_{\mathcal{R}}(\varepsilon) = \max_{(a,b,c)\in\mathcal{R}} (\log c - (1+\varepsilon)\log R(a,b,c)). \tag{1.1.5}$$

Then every valid *abc* constant must satisfy $\log K_\varepsilon \geq L_{\mathcal{R}}(\varepsilon)$. The right-hand side is the maximum of finitely many affine decreasing functions of ε; it is therefore continuous, convex, and piecewise linear. Its active triple can change as ε changes.

This distinction matters when reading record lists. The triple of greatest quality need not demand the largest K_ε at every epsilon. A slightly lower-quality triple with much smaller radical can dominate the intercept after the penalty $\varepsilon\log R$ is imposed. Thus the Reyssat triple records the largest known slope, while the full table is needed to calibrate constants. Chapter 43 performs exactly this envelope computation for ten triples and several tolerances.

Equation (1.1.5) is a rigorous lower bound, but only a lower bound. Enlarging the search set can move the graph upwards, never downwards. No finite table supplies an upper bound for it over all triples. This one-sidedness is the precise reason computational evidence can audit a claimed explicit constant but cannot establish the existential conjecture.

1.2 The radical and its failure of functoriality

The radical remembers a set of primes and forgets every multiplicity. In valuation notation,

$$\log \operatorname{rad}(n) = \sum_p \min\{1, v_p(n)\}\log p.$$

Thus it is a *truncated counting function*. The truncation is exactly what makes *abc* strong: replacing $\min\{1, v_p(n)\}$ by $v_p(n)$ would give $\log|n|$ and remove the tension.

Proposition 1.2.1 (Elementary radical calculus). *[REPROVED: Bro00] We record the identities in a form suited to later divisor comparisons. For non-zero integers m, n and $r \geq 1$,*

$$\begin{aligned} \operatorname{rad}(mn) &= \frac{\operatorname{rad}(m)\operatorname{rad}(n)}{\operatorname{rad}(\gcd(m,n))}, \\ \operatorname{rad}(n^r) &= \operatorname{rad}(n), \\ \operatorname{rad}(\operatorname{rad}(n)) &= \operatorname{rad}(n). \end{aligned}$$

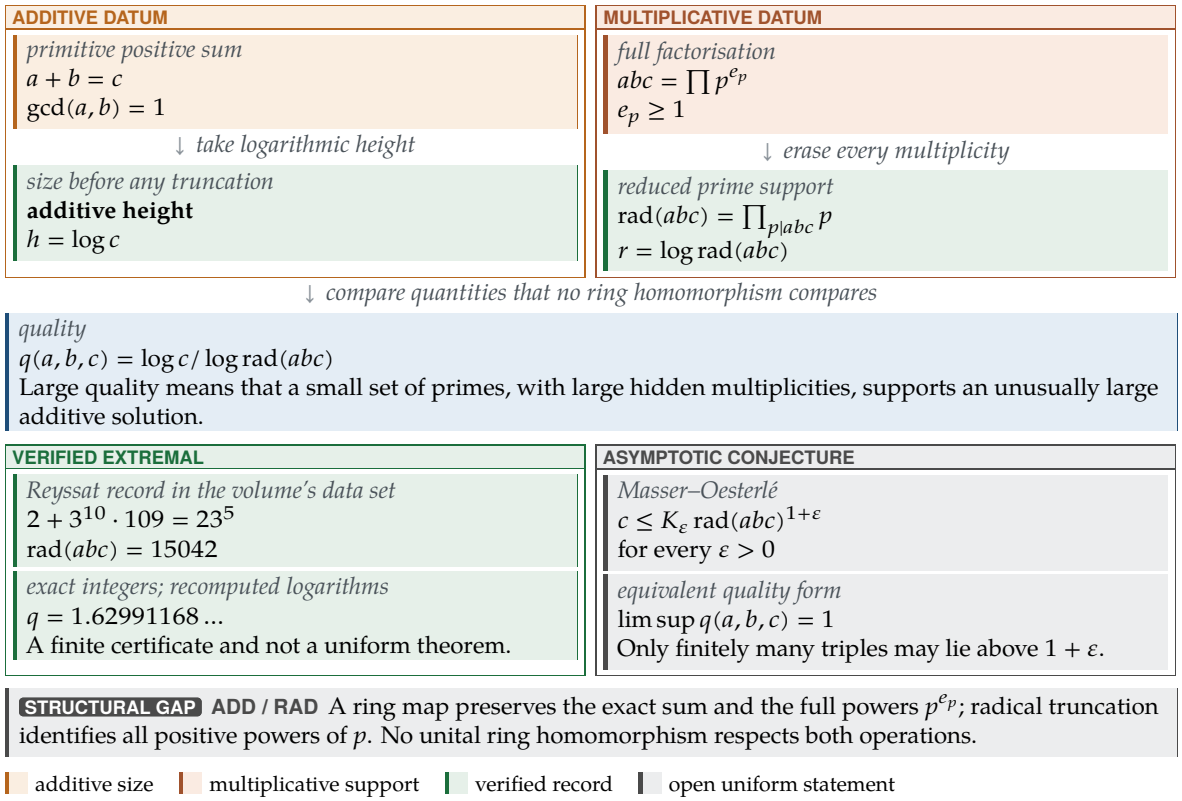


Plate 1.1. The quality quotient compares additive size with reduced multiplicative support.
What it does not show: no unital ring homomorphism respects both operations.

In particular, rad *is multiplicative only on coprime inputs.*

Proof. At each prime p, the indicator of $v_p(mn) > 0$ is the maximum of the two indicators for m and n. For indicators in $\{0, 1\}$ this maximum is their sum minus their minimum, and the minimum is the indicator of $p \mid \gcd(m, n)$. Multiplication over p gives the first identity. The other two follow because replacing a positive valuation by another positive valuation does not change its indicator. □

There are three distinct failures of functoriality. First, radical is not a ring map: $\mathrm{rad}(4) = 2$, whereas $\mathrm{rad}(2)^2 = 4$, and $\mathrm{rad}(8 + 1) = 3$ is not determined by $\mathrm{rad}(8) = 2$ and $\mathrm{rad}(1) = 1$. Secondly, the ideal operation $I \mapsto \sqrt{I}$ does not commute with arbitrary base change. Under $k[s] \to k[t]$, $s \mapsto t^2$, the reduced point (s) pulls back to the non-reduced ideal (t^2), while

$$\sqrt{(s)k[t]} = (t) \neq (t^2) = \sqrt{(s)}k[t].$$

Thirdly, reduction is not additive in exact sequences or divisors: it discards the ramification index that pullback naturally creates.

Geometrically, for $n \neq 0$ the reduced closed fibre

$$(\mathrm{Spec}\,\mathbb{Z}/n\mathbb{Z})_{\mathrm{red}} = \coprod_{p|n} \mathrm{Spec}\,\mathbb{F}_p$$

has Arakelov degree $\log \operatorname{rad}(n)$. This description is useful, but it does not turn radical into a functor preserving tensor products or pullbacks. One must first form the fibre, then reduce it, then count its components. Changing the order changes the object.

What does this cost? Heights linearise products and powers: $\log|xy| = \log|x| + \log|y|$ and $\log|x^r| = r\log|x|$. Radical does the opposite on powers. Consequently no linear, exact, or purely ring-theoretic operation can be inserted into $a + b = c$ and expected to output $\log \operatorname{rad}(abc)$. Every successful reformulation introduces valuations, reduced divisors, conductors, or logarithmic boundary data, and it must prove an additional comparison theorem showing that multiplicities have been truncated without losing control of size.

Scholium 1.2.2 (The price of truncation). *[KNOWN: Voj11] The radical is the arithmetic instance of a truncated counting function. Its non-functoriality is not an inconvenience to be normalised away: it is the source of the conjecture's strength and the reason conductor terms replace raw discriminants in its geometric formulations.*

1.3 Szpiro's conjecture and the discriminant–conductor gap

Let $E/\mathbb{Q}$ be an elliptic curve, let Δ_E be its minimal discriminant, and let

$$N_E = \prod_p p^{f_p(E)}$$

be its conductor. For a global minimal Weierstrass model, write $c_4(E)$ and $c_6(E)$ for the standard invariants. They satisfy

$$c_4(E)^3 - c_6(E)^2 = 1728\Delta_E. \tag{1.3.1}$$

The discriminant measures degeneration with multiplicity; the conductor is a truncated and locally corrected measure of bad reduction. Their logarithmic difference is the first appearance of the gap studied in the rest of this book.

Conjecture 1.3.1 (Szpiro). *[OPEN] For every $\eta > 0$ there is $C_\eta > 0$ such that every elliptic curve $E/\mathbb{Q}$ satisfies*

$$|\Delta_E| \le C_\eta N_E^{6+\eta}. \tag{1.3.2}$$

Conjecture 1.3.2 (Modified Szpiro). *[OPEN] For every $\eta > 0$ there is $C_\eta^{\mathrm{mod}} > 0$ such that*

$$H_{\mathrm{mod}}(E) := \max\{|c_4(E)|^3, |c_6(E)|^2\} \le C_\eta^{\mathrm{mod}} N_E^{6+\eta}. \tag{1.3.3}$$

The exponent is $6+\eta$ in both statements. Modified Szpiro is stronger: by (1.3.1), $1728|\Delta_E| \le 2H_{\mathrm{mod}}(E)$. The converse comparison is false at the level of individual models because c_4^3 and c_6^2 may be large and nearly cancel.

The exact equivalence with *abc* uses the modified statement, not merely the raw discriminant statement. We separate the local elliptic input from the elementary exponent bookkeeping.

Lemma 1.3.3 (Oesterlé height extraction). *[KNOWN: Gol02] Assume the abc inequality with exponent $1+\delta$, where $0 < \delta < 1/5$. Then there is a constant A_δ, independent of E, such that*

$$H_{\mathrm{mod}}(E) \le A_\delta (H_{\mathrm{mod}}(E)^{5/6} N_E)^{1+\delta}. \tag{1.3.4}$$

The lemma is the standard local-minimality step in the equivalence. One applies *abc* to (1.3.1) after removing the common divisor. The radicals of c_4 and c_6 contribute at most $|c_4c_6| \le H_{\mathrm{mod}}^{5/6}$; Tate's local table assigns the remaining common-divisor contribution to N_E, including the bounded exceptional factors at 2 and 3. This gives (1.3.4); see [Gol02; Bar23].

The exponent 5/6 is not a loss hidden in the local table. It is the sum of the two invariant weights:

$$|c_4| \le H_{\mathrm{mod}}^{1/3}, \qquad |c_6| \le H_{\mathrm{mod}}^{1/2}, \qquad |c_4c_6| \le H_{\mathrm{mod}}^{5/6}.$$

After the three terms of (1.3.1) are divided by their common gcd, their radical is supported on the primes dividing $c_4c_6\Delta$. Away from 2 and 3, minimality ensures that any residual common prime is a prime of bad reduction and is charged once to N_E. The finitely many wild cases contribute an absolute local factor. Thus the radical has the schematic bound

$$\mathrm{rad}(\text{primitive invariant triple}) \ll H_{\mathrm{mod}}^{5/6} N_E,$$

which is exactly the input of (1.3.4). The restriction $\delta < 1/5$ is also structural: it makes $1 - (5/6)(1+\delta) = (1-5\delta)/6$ positive, so that the power of H_{mod} can be moved to the left. At $\delta = 1/5$ this particular rearrangement ceases to give an upper bound.

Proposition 1.3.4 (The precise Szpiro implication chain). ***[REPROVED: Gol02]*** *We make the two epsilon conversions explicit and distinguish the original from the modified statement. Over* $\mathbb{Q}$*,*

$$abc \iff \text{modified Szpiro} \implies \text{Szpiro}.$$

Applying original Szpiro directly to the Frey curve gives only

$$|abc|^{1/3} \ll_\varepsilon \mathrm{rad}(abc)^{1+\varepsilon}, \tag{1.3.5}$$

not the full height bound $c \ll_\varepsilon \mathrm{rad}(abc)^{1+\varepsilon}$.

Proof. Assume *abc*. From Lemma 1.3.3,

$$H^{(1-5\delta)/6} \le A_\delta N_E^{1+\delta}, \qquad H = H_{\mathrm{mod}}(E).$$

Hence

$$H \le A_\delta^{6/(1-5\delta)} N_E^{6(1+\delta)/(1-5\delta)}.$$

Given $\eta > 0$, choose

$$\delta = \frac{\eta}{36+5\eta}; \qquad \frac{6(1+\delta)}{1-5\delta} = 6+\eta.$$

This is modified Szpiro. The invariant identity then gives original Szpiro.

Conversely, start from an *abc*-triple and attach the Frey curve

$$E_{a,b} : y^2 = x(x-a)(x+b).$$

For the displayed model,

$$c_4 = 16(a^2+ab+b^2), \qquad \Delta = 16a^2b^2c^2,$$

and $\frac{3}{4}c^2 \le a^2+ab+b^2 \le c^2$. Minimalisation changes these estimates only by bounded powers of 2, while $N_{E_{a,b}} \ll \mathrm{rad}(abc)$. Modified Szpiro therefore yields

$$c^6 \ll H_{\mathrm{mod}}(E_{a,b}) \ll_\eta \mathrm{rad}(abc)^{6+\eta}.$$

Taking sixth roots and setting $\eta = 6\varepsilon$ proves *abc*.

If one assumes only original Szpiro, the same curve gives $(abc)^2 \ll_\eta \operatorname{rad}(abc)^{6+\eta}$. Taking sixth roots gives (1.3.5). No passage from this symmetric product bound to the full modified height bound has been used or asserted. □

The Frey bridge under a microscope

The algebra in the reverse implication is short enough to audit directly. Expanding

$$E_{a,b}\colon y^2 = x(x-a)(x+b) = x^3 + (b-a)x^2 - abx$$

gives

$$b_2 = 4(b-a), \qquad b_4 = -2ab, \qquad b_6 = 0, \qquad b_8 = -a^2b^2.$$

Hence

$$\begin{aligned} c_4 &= b_2^2 - 24b_4 = 16(a^2 + ab + b^2), \\ c_6 &= -b_2^3 + 36b_2b_4 - 216b_6 = 32(a-b)(2a+b)(a+2b), \\ \Delta &= -b_2^2b_8 - 8b_4^3 = 16a^2b^2(a+b)^2. \end{aligned}$$

Since $c = a + b$ and $ab \le c^2/4$,

$$\frac{3}{4}c^2 \le a^2 + ab + b^2 = c^2 - ab \le c^2. \tag{1.3.6}$$

Thus the displayed c_4 has size exactly c^2 up to absolute factors, and its cube has size c^6. This is the origin of the sixth root in (1.5.1).

The conductor computation is equally structural. At an odd prime $p \mid abc$, pairwise coprimality says that p divides exactly one of a, b, or c. The formula for c_4 is then a p-adic unit, while

$$v_p(\Delta) = 2v_p(abc).$$

The fibre is multiplicative of type $\mathrm{I}_{2v_p(abc)}$ and has conductor exponent one. Therefore the odd part of N_E is exactly the odd part of $\operatorname{rad}(abc)$. At 2, minimalisation and wild reduction introduce only the finite list of factors computed in Chapter 42; in particular

$$N_E \le 16 \operatorname{rad}(abc).$$

The same minimalisation can divide c_4 by a bounded power of 2, but it cannot change the exponent of c in (1.3.6).

Consequently modified Szpiro places the two quantities needed by *abc* on opposite sides of one inequality:

$$c^6 \ll H_{\mathrm{mod}}(E_{a,b}) \ll_\eta N_E^{6+\eta} \ll \operatorname{rad}(abc)^{6+\eta}.$$

Original Szpiro replaces the first term by $|\Delta| = 16(abc)^2$, which is why it yields only the symmetric product bound. The Frey curve is therefore a faithful translator: its conductor truncates prime multiplicities, and its modular height retains additive size. It does not manufacture the missing estimate between them.

The distinction is structural. Δ_E records the difference $c_4^3 - c_6^2$; H_{mod} records the size of the two terms before cancellation. The conductor must control height, not merely the residual error after cancellation.

1.4 Vojta's height inequality

Let $X = \mathbb{P}^1_K$ over a number field K and $D = [0] + [1] + [\infty]$. Then $K_X + D \sim O_{\mathbb{P}^1}(1)$. For $P \in X(\overline{K}) \setminus D$, let $h_{K_X+D}(P)$ be the corresponding logarithmic height, let $N_D^{(1)}(P)$ be the truncated finite-place counting function, and let

$$d_K(P) = \frac{1}{[K(P) : \mathbb{Q}]} \log |\mathrm{Disc}(K(P)/K)|$$

after the customary fixed normalisation.

Conjecture 1.4.1 (Vojta for the three-point line). *[OPEN] Fix a degree bound r and an ample divisor A on X. For every $\varepsilon > 0$, outside a proper closed subset and for $[K(P) : K] \le r$,*

$$h_{K_X+D}(P) \le N_D^{(1)}(P) + d_K(P) + \varepsilon h_A(P) + O_{K,r,\varepsilon}(1). \tag{1.4.1}$$

We call (1.4.1) the $(1,1)$-*form* in this volume: the truncated boundary term and the normalised discriminant term each occur with coefficient one. This terminology concerns the two arithmetic coefficients; it is not the Hodge type of a differential form. The epsilon is attached to an auxiliary height, rather than inserted by changing either coefficient.

Proposition 1.4.2 (The three-point specialisation). *[REPROVED: Voj11] We spell out the local counting identity. For $K = \mathbb{Q}$, $r = 1$, and $P = a/c$ arising from an abc-triple,*

$$\begin{aligned} h_{K_X+D}(P) &= \log c + O(1), \\ N_D^{(1)}(P) &= \sum_{p \mid abc} \log p = \log \mathrm{rad}(abc), \qquad d_{\mathbb{Q}}(P) = 0. \end{aligned}$$

Consequently (1.4.1) is equivalent, after reparametrising ε, to the abc conjecture.

Proof. At a finite prime p, the reduction of a/c meets 0, 1, or ∞ precisely when p divides a, $b = c - a$, or c, respectively. Pairwise coprimality makes these alternatives disjoint, and truncation assigns the weight $\log p$ once. This proves the counting identity. Since $K_X + D$ has degree 1, its height is the usual height of a/c, namely $\log c$ for $0 < a < c$. Finally $K(P) = \mathbb{Q}$, so the relative discriminant term vanishes. Thus (1.4.1) reads $(1 - \varepsilon) \log c \le \log R + O_\varepsilon(1)$; replacing ε by $\varepsilon/(1+\varepsilon)$ gives $\log c \le (1+\varepsilon) \log R + O_\varepsilon(1)$. The reverse conversion is identical. □

What changes over a number field

The rational specialisation suppresses two pieces of data that reappear over a number field. Let $K/\mathbb{Q}$ have degree d, and let $P = (a : c) \in \mathbb{P}^1(K) \setminus \{0, 1, \infty\}$ with $b = c - a$. Replacing K by the field of definition of P, assume $K = \mathbb{Q}(P)$. At each finite prime $\mathfrak{p}$, choose locally primitive homogeneous coordinates. The truncated three-point count is

$$N^{(1)}_{0,1,\infty}(P) = \frac{1}{d} \sum_{\mathfrak{p}:\, P \bmod \mathfrak{p} \in \{0,1,\infty\}} \log \mathrm{N}\mathfrak{p}.$$

If a, b, and c are integral and their principal ideals are pairwise coprime, this is

$$\frac{1}{d} \log \mathrm{N}\,\mathrm{rad}_K(abc),$$

where rad_K denotes the product of the distinct prime ideals dividing the three principal ideals. Local primitive coordinates are essential in general: a nontrivial class group may prevent one global choice from simultaneously displaying every gcd as 1.

Regard the same point as an algebraic point of $\mathbb{P}^1_{\mathbb{Q}}$. Its normalised field-discriminant term is

$$\delta_K = \frac{1}{d}\log|D_K|.$$

The bounded-degree three-point inequality therefore has the schematic specialisation

$$(1-\varepsilon)h(P) \le N^{(1)}_{0,1,\infty}(P) + \delta_K + O_{d,\varepsilon}(1). \tag{1.4.2}$$

For a fixed K, the discriminant may be absorbed into the constant. In a statement uniform over fields of degree d, it must remain visible. This is the same base term that later appears as r_K in Theorem A and as the canonical degree $2g-2$ over a function field.

Two cautions follow. First, rational *abc* does not formally imply the number-field statement by extending scalars: a prime of $\mathbb{Q}$ splits and ramifies, the radical becomes a prime-ideal divisor, and the field discriminant enters (1.4.2). Secondly, bounded-degree Vojta is stronger than the collection of fixed-field statements, because its constant and exceptional set must be uniform as K varies subject only to a degree bound. Chapter 44 keeps these distinctions explicit. The rational equivalence proved above is the degree-one fibre of this larger conjectural picture, not a substitute for it.

The general bounded-degree statement is substantially stronger than rational *abc*: it retains field discriminants, exceptional sets, and uniformity in algebraic degree. What matters here is its architecture. A log-canonical height is bounded by reduced boundary intersection plus ramification of the field of definition. This is the arithmetic form of the function-field inequality proved in Chapter 2.

1.5 Equivalences and implications

The logical map may now be stated without conflating conjectures. Write V for the Vojta $(1,1)$ statement on $(\mathbb{P}^1, 0+1+\infty)_{\mathbb{Q}}$. Then

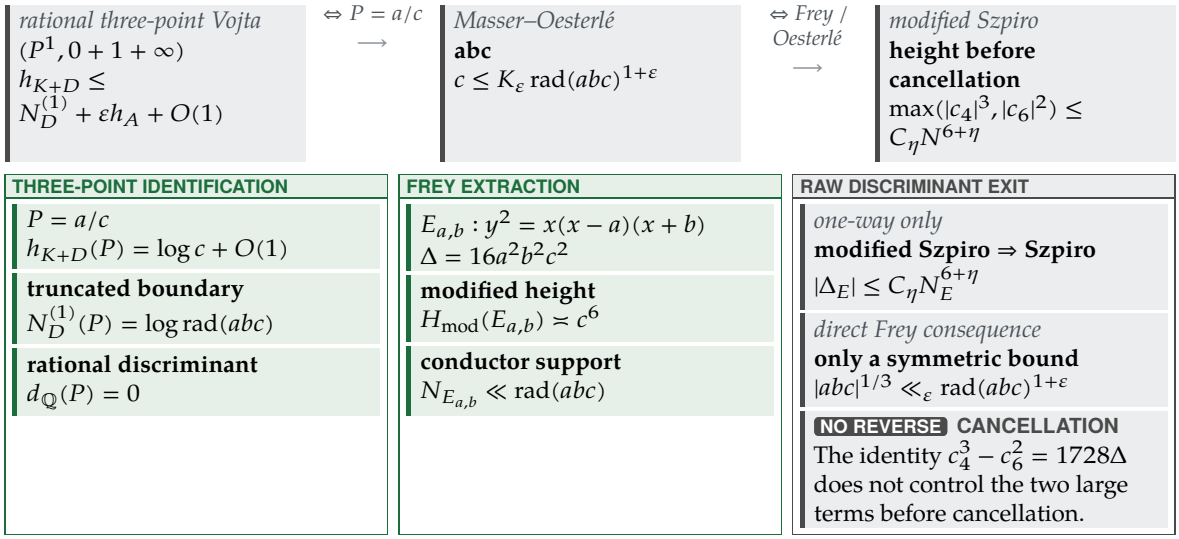
$$\begin{array}{ccccc} \mathrm{V} & \Longleftrightarrow & abc & \Longleftrightarrow & \text{modified Szpiro} \\ & & \Downarrow & & \Downarrow \\ & & \text{quality finiteness} & & \text{original Szpiro.} \end{array}$$

The first equivalence is restricted to rational points; bounded-degree Vojta contains more information. The lower-right arrow has no reverse arrow in this diagram. Mason–Stothers, proved in the next chapter, is a function-field analogue, not a premise from which the number-field conjecture follows.

Epsilon and effectivity bookkeeping

The double arrows above concern existence of constants. They do not by themselves say that those constants can be computed. The elementary parameter changes are, however, completely explicit. Writing η for a Szpiro tolerance, δ for the tolerance fed into *abc*, and ε for the final *abc* tolerance, the two elliptic directions use

$$\begin{aligned} abc \Longrightarrow \text{modified Szpiro}: &\quad \delta = \frac{\eta}{36+5\eta}, \\ \text{modified Szpiro} \Longrightarrow abc: &\quad \eta = 6\varepsilon. \end{aligned} \tag{1.5.1}$$



Logical scope of the arrows

Source	Target	Conversion	Scope
Vojta (1,1)	abc	$P = a/c$ and three truncated fibres	equivalence for rational points
abc	modified Szpiro	$\delta = \eta/(36+5\eta)$	exponent $6+\eta$
modified Szpiro	abc	$\eta = 6\varepsilon$ and sixth roots	Frey curves
modified Szpiro	original Szpiro	$1728\lvert\Delta\rvert \le 2H_{\mathrm{mod}}$	one-way
abc	quality finiteness	$q > 1+\varepsilon$ only finitely often	equivalent quality form

open conjectural interface proved reduction or conversion

Plate 1.2. The exact equivalences and the one-way passage to the raw discriminant inequality.
What it does not show: the identity $c_4^3 - c_6^2 = 1728\Delta$ does not control the two large terms before cancellation.

The first formula is forced by $6(1+\delta)/(1-5\delta) = 6+\eta$; the second is forced by taking a sixth root of the Frey-curve height inequality. In particular, no hidden limiting argument occurs in either conversion.

Proposition 1.5.1 (Transport of constants). *[REPROVED : GoI02] The classical implication chain has the following quantitative content.*

(i) *If an effective abc constant K_δ is supplied, then the local calculation in Lemma 1.3.3 and the first substitution in (1.5.1) produce an effective modified-Szpiro constant at tolerance η.*

(ii) *If an effective modified-Szpiro constant is supplied at tolerance 6ε, then the Frey construction, Tate's algorithm, and the sixth root produce an effective abc constant at tolerance ε.*

(iii) *Modified Szpiro gives original Szpiro effectively, with $C_\eta = C_\eta^{\mathrm{mod}}/864$ admissible in the normalisations of (1.3.2) and (1.3.3).*

(iv) *An effective rational three-point Vojta inequality, including an effective exceptional set and $O(1)$ term, gives effective abc, and effective abc gives the corresponding effective rational-point inequality. The epsilon conversion is $\tau = \varepsilon/(1+\varepsilon)$, where τ is the coefficient of the auxiliary height in (1.4.1).*

Proof. For (i), the proof of Proposition 1.3.4 gives

$$H_{\mathrm{mod}}(E) \le A_\delta^{6/(1-5\delta)} N_E^{6(1+\delta)/(1-5\delta)}.$$

The local constant A_δ is obtained by finitely many gcd operations and the local minimality table from K_δ; hence it is computable when K_δ is. The stated choice of δ changes the exponent to $6+\eta$.

For (ii), the invariants of $E_{a,b}$ and its conductor are computed by finite integer arithmetic and Tate's algorithm. Modified Szpiro gives $c^6 \ll_\varepsilon R^{6+6\varepsilon}$ with a displayed constant, and taking the positive sixth root is effective. For (iii), (1.3.1) gives $1728|\Delta_E| \le 2H_{\mathrm{mod}}(E)$, hence the factor $1/864$.

For (iv), the rational specialisation of (1.4.1) is

$$(1-\tau)\log c \le \log R + B_\tau.$$

Taking $\tau = \varepsilon/(1+\varepsilon)$ gives $\log c \le (1+\varepsilon)\log R + (1+\varepsilon)B_\tau$. Every exceptional rational point is one explicit triple and may be absorbed if the exceptional set is effective. Reversing the algebra gives the other direction. □

Thus the familiar equivalences do not intrinsically destroy effectivity: they transport whatever effectivity is present in their input. The conjectures are usually stated only with existential constants, so no algorithm follows from their bare statements. In the conditional chain developed in this volume, the specific loss occurs earlier, in the archimedean coercivity modulus entering the master constant. Until that modulus is computable, the later substitutions and sixth roots remain effective operations applied to an ineffective input. Chapters 40 and 43 isolate this point as a separate obligation rather than attributing the loss to the Frey curve.

Logical status at a glance

The principal statements encountered in the opening part of the book occupy different logical levels:

Statement	Relation to rational *abc*	Present status
Stewart–Yu	Unconditional exponential upper bound	Proved and effective
Mason–Stothers	Exact function-field analogue	Proved; it does not imply the number-field case
Original Szpiro	Follows from modified Szpiro; gives only the symmetric Frey bound	Open over number fields
Modified Szpiro	Equivalent over $\mathbb{Q}$, with explicit epsilon transport	Open
Three-point Vojta over $\mathbb{Q}$	Equivalent on rational points	Open
Theorem A of this volume	Feeds original Szpiro after (CM) and (MI_Δ)	Conditional on fifty-nine numbered upstream obligations
Height-form application	Feeds modified Szpiro and then *abc*	Conditional on further stated interfaces

The table prevents two common invalid inferences. An analogy is not an implication: the proof of Mason–Stothers supplies a design test, not a proof over $\mathbb{Q}$. A weaker elliptic consequence is not an equivalence: original Szpiro cannot replace modified Szpiro merely because both display the exponent six. Every double arrow in this chapter has been justified with a parameter conversion; every single arrow is intentionally single.

Proposition 1.5.2 (Conditional asymptotic Fermat consequence). *[REPROVED : GT02] We give the uniform boundedness argument after the usual radical estimate. If abc holds, then there is n_0 such that for every $n \ge n_0$ the equation*

$$x^n + y^n = z^n$$

has no solution in positive pairwise coprime integers.

Proof. Assume a solution and apply *abc* to (x^n, y^n, z^n). For a fixed $0 < \varepsilon < 1/3$,

$$z^n \le K_\varepsilon \operatorname{rad}(xyz)^{1+\varepsilon} \le K_\varepsilon (xyz)^{1+\varepsilon} < K_\varepsilon z^{3(1+\varepsilon)}.$$

If $n > 3(1+\varepsilon)$, then $z \le K_\varepsilon^{1/(n-3(1+\varepsilon))}$; in particular all such solutions have $z \le B$, for a constant B independent of n. For $1 \le x, y < z \le B$,

$$\left(\frac{x}{z}\right)^n + \left(\frac{y}{z}\right)^n \le 2\left(1 - \frac{1}{B}\right)^n < 1$$

when n is sufficiently large, contradicting the equation after division by z^n. □

This implication is historically important but no longer needed to establish Fermat's last theorem, which is known by modular methods. Its role is to show that exponent 1 in *abc* is strong enough to dominate high perfect powers.

A second consequence detects squarefree prime divisors in exponential sequences. Fix an integer $u > 1$. A prime $p \nmid u$ is *Wieferich to base u* if $u^{p-1} \equiv 1 \pmod{p^2}$.

Proposition 1.5.3 (Non-Wieferich primes from *abc*). *[REPROVED: Sil88] We reproduce the squarefree-part argument proving infinitude. If abc holds, then infinitely many primes are non-Wieferich to base u.*

Proof. For $n \ge 1$, write

$$u^n - 1 = U_n V_n, \quad U_n = \prod_{v_p(u^n-1)=1} p, \quad V_n = \prod_{v_p(u^n-1)\ge 2} p^{v_p(u^n-1)}.$$

Since $\operatorname{rad}(V_n) \le V_n^{1/2}$, application of *abc* to $1 + (u^n - 1) = u^n$ gives, for $0 < \varepsilon < 1$,

$$u^n \ll_{u,\varepsilon} (U_n V_n^{1/2})^{1+\varepsilon} \le (U_n^{1/2} u^{n/2})^{1+\varepsilon}.$$

Therefore

$$U_n \gg_{u,\varepsilon} u^{n(1-\varepsilon)/(1+\varepsilon)},$$

so the product of the primes occurring to exact exponent one is unbounded.

Let an odd prime p divide U_n, and put $r = \operatorname{ord}_p(u)$. Then $r \mid n$. If p were Wieferich, the congruence $u^{p-1} \equiv 1 \pmod{p^2}$ and $p \nmid (p-1)/r$ would force $u^r \equiv 1 \pmod{p^2}$; raising to the power n/r would give $u^n \equiv 1 \pmod{p^2}$, contrary to $v_p(u^n - 1) = 1$. Thus every odd prime in U_n is non-Wieferich. A finite set of such primes would bound U_n, which is impossible. □

Silverman's sharper counting argument gives $\#\{p \le X : p \text{ non-Wieferich to base } u\} \gg_u \log X$ under *abc* [Sil88]. Other consequences need more height theory. In particular, *abc* yields Roth-type approximation statements and strong forms of Mordell's theorem; we use the established reductions rather than reproduce them here [Fra99; Elk91]. These are outgoing implications: they do not provide an easier route back to *abc*.

1.6 The additive–multiplicative gap

The equation $a + b = c$ is additive and exact. The quantity $\mathrm{rad}(abc)$ is multiplicative, local, and reduced. The conjecture asks the first datum to control the second after all multiplicities have been erased. This is the conceptual core, rather than a defect of notation.

Why the natural estimates stop

Several first attempts fail for reasons that are worth distinguishing. The inequality

$$\mathrm{rad}(abc) \le abc \le \frac{1}{4}c^3$$

uses $ab \le c^2/4$, but it points in the wrong direction: it gives a lower bound of cube-root scale for c in terms of the radical. Replacing the radical by the full product has restored multiplicity and thereby removed the problem.

Fixing the set S of primes dividing abc is much stronger. Then $a/c + b/c = 1$ is an S-unit equation, and the theory of linear forms in logarithms gives finiteness and, in fixed data, effective bounds [Bak98; Gyo22]. But abc does not fix S: the set of primes varies with the triple, and the conjectural cost of that variation is only

$$\sum_{p \in S} \log p.$$

Known S-unit estimates depend much more strongly on the cardinality and the largest elements of S. The proof of Proposition 1.1.5 invokes S-unit finiteness only *after* the radical has been bounded. Using the same theorem to obtain that bound would be circular.

Prime-by-prime congruences do not close the gap either. If $p^m \mid a$, then $b \equiv c \pmod{p^m}$, and pairwise coprimality ensures that p divides neither b nor c. This is strong local information, but its cost is tied to the exponent m that the radical has deliberately discarded. A global argument must compare all such congruences simultaneously with the archimedean size of a, b, and c. Linear forms in logarithms perform such a comparison with the exponential loss of Theorem 1.1.7; they do not yield the coefficient one of (1.1.4).

Nor can computation bridge the quantifiers. A finite search may increase the known lower bound for K_ε, and a single triple may refute a proposed explicit value. It cannot show that

$$\sup_{a+b=c} \frac{c}{\mathrm{rad}(abc)^{1+\varepsilon}}$$

is finite. Statistical scarcity of high-quality triples is compatible both with the conjecture and with a counterexample appearing beyond every searched range.

Finally, the elliptic reformulation does not make cancellation disappear. Original Szpiro controls Δ_E, hence the difference $c_4^3 - c_6^2$. The *abc* height is recovered only when the two terms are controlled before they cancel, which is why modified Szpiro occurs in the double arrow of Proposition 1.3.4. Passing to elliptic curves organises the missing comparison geometrically; it does not prove it for free.

Proposition 1.6.1 (No ring-homomorphic radical). *[REPROVED: Bro00] We isolate the elementary incompatibility used as the starting obstruction of the book. There is no unital ring homomorphism $T\colon \mathbb{Z} \to \mathbb{Z}$ satisfying $T(n) = \mathrm{rad}(n)$ for every positive integer n.*

Proof. Every unital ring endomorphism of $\mathbb{Z}$ fixes 1, hence fixes every integer; therefore $T(4) = 4$. Radical truncation requires $T(4) = \mathrm{rad}(4) = 2$, a contradiction. Equivalently, a ring

homomorphism preserves the passage from 2 to 2^2, whereas radical identifies all positive powers of 2. □

The point is not that addition and multiplication cannot coexist. A ring homomorphism preserves $a + b = c$ and the full multiplicities in abc; radical asks it also to destroy those multiplicities. No single ring homomorphism does both.

The exact function-field step with no arithmetic analogue

Over $k[t]$, geometry supplies a bridge that is not a ring homomorphism: the derivation d/dt. It is additive, satisfies Leibniz, and lowers the multiplicity of every zero by one. The divisor of a differential then converts that local loss into a global degree inequality. This is precisely the Mason–Stothers mechanism of Chapter 2. Over $\mathbb{Z}$, the four pieces needed for that bridge—a smaller base, a derivation, non-zero relative differentials, and a ramification-degree formula—are absent in ordinary scheme theory; Chapter 3 proves the obstruction.

The irreplaceable algebraic line of the polynomial proof is

$$\frac{F}{\operatorname{rad}(F)} \mid F', \qquad \deg F' \leq \deg F - 1, \tag{1.6.1}$$

with $F' \neq 0$ for a nonconstant F in characteristic zero. The first half turns every multiplicity m into $m - 1$; the second says that the object carrying all those lost multiplicities is globally smaller than F. From $A + B = C$, additivity and Leibniz then produce one non-zero Wronskian divisible by the three multiplicity quotients. That one object is the certificate linking the additive relation to reduced support.

For integers, every ordinary $\mathbb{Z}$-derivation of $\mathbb{Z}$ is zero. More generally, a number field has no non-zero derivation over $\mathbb{Q}$, because it is algebraic and separable. Regarding a, b, and c as constant polynomials merely gives $a' = b' = c' = 0$ and a zero Wronskian. Introducing an auxiliary parameter and interpolating the three integers produces a derivative that depends on the interpolation, not an invariant of the triple. Fermat-quotient operators, p-derivations, and arithmetic derivatives can retain useful prime-power information, but they have modified addition or Leibniz laws and begin one prime at a time. An all-place descent and a global size formula must be proved before any of them can replace (1.6.1).

The geometric proof identifies the same missing line without coordinates. For $f = A/C\colon X \to \mathbb{P}^1$, the reduced inverse image of $\{0, 1, \infty\}$ counts the radical, while Riemann–Hurwitz bounds the ramification above those three fibres by the fixed canonical degree $2g(X) - 2$. The rational number a/c is a point of $\mathbb{P}^1(\mathbb{Q})$, not a nonconstant map from an arithmetic curve lying over a smaller ordinary base. There is therefore no ordinary relative differential $d(a/c)$ whose local orders are the exponents discarded by $\operatorname{rad}(abc)$ and whose divisor has a fixed global degree. This is the precise step that does not transfer from Mason–Stothers.

Vojta's three-point inequality predicts what an arithmetic replacement must look like:

$$\begin{aligned}\text{height} \leq{}& \text{truncated boundary count} + \text{field discriminant} \\ &+ \varepsilon\,\text{height} + O(1).\end{aligned}$$

Over a function field, the corresponding assertion is exact Riemann–Hurwitz, with the canonical term $2g - 2$ and no epsilon loss. Over a number field, obtaining the truncated count with coefficient one is the unproved step. The later substrate, strain, and trace constructions are designed to manufacture this comparison; their proof obligations record exactly where the manufacture is not yet complete.

STAGE 1 · CERTIFY THE INPUT

1.1 Read positive integers a, b, c and check $a + b = c$ exactly.
reject if false

1.2 Compute $\gcd(a, b)$; primitivity then gives pairwise coprimality.
reject unless $\gcd = 1$

1.3 Retain the equality and gcd computations as the primitive certificate.
integer arithmetic only

↓ *primitive abc-triple*

STAGE 2 · FACTOR AND MEASURE

2.1 Factor a, b and c and reconstruct each integer from its prime powers.
factorisation must be exact

2.2 Discard exponents only after verification and form $\mathrm{rad}(abc) = \prod_{p \mid abc} p$.
reduced support

2.3 Compute $q = \log(c)/\log(\mathrm{rad}(abc))$ with declared precision.
quality is a diagnostic

2.4 Compare the certified value with the finite record table.
no extrapolation

↓ *same data, three interfaces*

integer interface
height / support
$\log c$ versus $\log \mathrm{rad}(abc)$

elliptic interface
Frey invariants
$c_4 = 16(a^2 + ab + b^2)$,
$\Delta = 16a^2b^2c^2$

Vojta interface
$P = a/c$
$N_D^{(1)}(P) = \log \mathrm{rad}(abc)$

↓ *verified report*

Example output: current record in the packaged data

Equality	$\mathrm{rad}(abc)$	quality	Status
$2 + 3^{10} \cdot 109 = 23^5$	15042	1.62991168...	finite verified record

STOP UNIFORMITY The pipeline can refute a proposed bound on its search range. It cannot prove that one constant K_ε works for every primitive sum, and no ring-homomorphic radical crosses this boundary.

exact validation · factor-and-measure stage · classical interface · conjectural boundary

Plate 1.3. The verification pipeline for an abc-triple, and the boundary beyond which conjecture begins.

What it does not show: a proof of the uniform bound: the pipeline can refute a proposed bound on its search range, but cannot prove that one constant K_ε works for every primitive sum.

The rest of the monograph should therefore be read as a search for a *geometric comparison mechanism*, not for a clever manipulation of the integer equation. Any proposed mechanism must retain addition, see reduced prime support, measure multiplicity before truncating it, and recover ordinary Mason–Stothers over a function field. Failing the last test would produce an analogy, not an arithmetic transposition.

Notes and References

Sources. The formulation in Conjecture 1.1.2 follows Masser's 1985 problem list and the Oesterlé–Masser terminology recorded by Stewart and Tijdeman [Mas85; ST86]. The equivalence between the constant and finiteness forms is standard; broad accounts are Browkin's survey and the chapter on *abc* in Bombieri–Gubler [Bro00; BG06]. The record values in §1.1 are taken from the exact integer table shipped with this project, whose external data source is de Smit's maintained archive [dS26]. The Browkin–Brzeziński and Nitaj papers document the computational search tradition and its normalisations [BB94; Nit96; Nit93]. The effective unconditional estimate (1.1.3) is Theorem 1 of Stewart–Yu [SY01]; its factor $R^{1/3}(\log R)^3$ is quoted rather than replaced by the weaker shorthand $R^{1/3+\varepsilon}$. The elementary prime family in Proposition 1.1.6 is included to distinguish the false exponent-one constant statement from the conjectural $1+\varepsilon$ statement.

The elliptic invariant identity, Frey curve, and modified-Szpiro equivalence in §1.3 are classical; Goldfeld gives a convenient modular account [Gol02]. Barrios uses the modified Szpiro ratio in the exact $\max(|c_4|^3, |c_6|^2)$ normalisation adopted here [Bar23]. The warning that raw Szpiro gives only the symmetric product estimate by direct substitution is included because the two versions are often conflated. Vojta's dictionary and the truncated height inequality are treated in [Voj87a; Voj11]; the general height conventions may be found in [BG01]. The non-Wieferich theorem is Silverman's [Sil88]. The Roth-type and Mordell consequences cited at the end of §5 are due, in the forms used here, to van Frankenhuijsen and Elkies [Fra99; Elk91]. Three normalisations are consequential. Quality uses c, rather than $\max(|a|, |b|, |c|)$, because the chapter works with positive sums. Modified Szpiro uses invariants of a global minimal model, so a change of integral presentation cannot manufacture height. The Vojta term is truncated at one at every finite place; replacing it by the untruncated intersection number would recover size rather than radical. Each proof in the chapter preserves these choices explicitly.

Attribution of the new material. No theorem in this chapter is claimed as new. The proofs tagged REPROVED are self-contained reorganisations of known arguments: the quality quantifier conversion, the exact epsilon substitutions $\eta = 6\varepsilon$ and $\delta = \eta/(36+5\eta)$, the bounded-base argument for asymptotic Fermat, and the squarefree-part proof of the non-Wieferich consequence. The separation of original from modified Szpiro, the three-level account of radical non-functoriality, and the final additive–multiplicative diagnosis are expository syntheses for this volume. The table was generated and checked by the project's machine-assisted arithmetic code and then independently compared with the cited tables. The three plate briefs were likewise prepared in the machine-assisted editorial process described in Chapter 56. Machine assistance supplied organisation and exact arithmetic, not a new claim about the truth of *abc*. In particular, the implication diagram was checked arrow by arrow: no converse is inferred from a shared exponent, and the raw Szpiro inequality is not silently substituted for modified Szpiro. Constants hidden by $\ll$ occur only in known implication statements and never alter the displayed limiting exponent.

Alternative treatments. One may begin with elliptic curves and present *abc* as a corollary of a Szpiro-type height bound. That route makes the exponent 6 memorable but risks hiding the difference between $|\Delta|$ and modified height. One may instead begin with Vojta's general conjecture; then the three-point line is a one-line specialisation, but the elementary arithmetic problem arrives buried under height notation. A third route starts from consequences—Fermat, Catalan-type equations, or effective Mordell—and advertises the conjecture by its reach. We have chosen the integer equation first because it makes the radical's loss of multiplicity visible before any geometry is introduced. The elliptic and Vojta formulations then answer, rather than precede, the question of what geometric object should replace that loss. A fourth organisation, through S-unit equations, makes finiteness for fixed prime support primary. It is valuable for effectivity, but it can conceal the uniformity problem: *abc* must remain controlled while the set S itself varies. The radical formulation keeps that moving support in view and is therefore better suited to the architecture of this volume.

Historical remarks. Masser proposed the conjecture publicly in 1985, reporting a suggestion of Oesterlé; Stewart and Tijdeman's 1986 paper is among the first published studies. Reyssat found the quality-1.6299 triple in 1987. The same decade saw Frey's elliptic-curve insight, the modular route to Fermat, and Silverman's 1988 application to Wieferich primes. During the 1990s, Browkin, Brzeziński, Nitaj, and others developed systematic searches for high-quality triples, while Elkies and Vojta placed *abc* within increasingly geometric height theories. Distributed searches later enlarged the verified range without surpassing Reyssat's quality. Robert, Stewart, and Tenenbaum proposed a refined sub-power correction to the exponent-1 scale [RST14]; it is a sharpening of the conjectural asymptotic, not an established replacement for Conjecture 1.1.2. The lower-case name "abc" became standard because no variable is privileged. Its history has repeatedly moved between integer sums, elliptic curves, and height inequalities; that movement explains why the unqualified word "Szpiro" still requires careful interpretation. Mochizuki announced a proof by inter-universal Teichmüller theory in 2012, and the four principal papers appeared in 2021 [Moc21a; Moc21c]. The claimed proof has not received broad acceptance; the Scholze–Stix report records a central objection [SS18]. This volume therefore follows the standard current status and labels *abc* open. Its own conditional architecture is independent of any claim that the IUT argument is complete.

Further reading. Browkin's survey [Bro00] is a compact entry to formulations and examples. Bombieri–Gubler [BG01] places *abc* inside the modern theory of heights. Goldfeld [Gol02] explains the elliptic and modular interfaces, including Szpiro. Vojta [Voj11] develops the Nevanlinna–Diophantine dictionary behind the $(1,1)$-form. De Smit's tables [dS26] are the appropriate source for current computational records, while Robert–Stewart– Tenenbaum [RST14] should be read for the conjectured fine-scale distribution beyond the coarse $1+\varepsilon$ statement. For the best unconditional scale obtainable from linear forms in logarithms, the Stewart–Yu paper [SY01] is indispensable: its exponential bound also quantifies how far present methods remain from a power $R^{1+\varepsilon}$.

Chapter 2

Mason–Stothers and the Anatomy of an Easy Theorem

Dependencies. Chapter 1.

Summary. The polynomial *abc* theorem is proved three times: by a Wronskian, by Riemann–Hurwitz applied to a/c, and by a local logarithmic-differential count. The proofs are elementary only because a complete relative geometric package is already present. We isolate its four indispensable inputs: a base Spec$k$, a derivation, a non-zero sheaf of relative differentials, and a degree–ramification formula. The higher-genus statement is the exact function-field Vojta inequality, with correction $2g-2$. The final checklist hands these inputs to Chapter 3, where their four arithmetic failures are proved and unfolded into eight design requirements for the later substrate.

2.1 The theorem and three proofs

Let k be a field of characteristic zero. For a non-zero polynomial $F \in k[t]$, let

$$\operatorname{rad}(F) = \prod_{P \mid F} P$$

up to multiplication by $k^\times$, the product extending over the distinct monic irreducible factors of F. Thus $\deg \operatorname{rad}(F)$ is the number of distinct roots of F over an algebraic closure, counted with residue degrees over k.

Theorem 2.1.1 (Mason–Stothers). *[REPROVED: GAL20] We give three complete proofs, separating polynomial divisibility, global ramification, and local logarithmic differentiation. Let non-zero $A, B, C \in k[t]$ have no common non-constant factor, suppose that not all are constant, and assume*

$$A + B = C.$$

Then

$$\max\{\deg A, \deg B, \deg C\} \leq \deg \operatorname{rad}(ABC) - 1. \tag{2.1.1}$$

Because $A + B = C$, a common factor of any two polynomials divides the third. The stated primitivity therefore implies pairwise coprimality. Put $R = \operatorname{rad}(ABC)$ and $r = \deg R$.

First proof: the polynomial Wronskian

Wronskian proof. Set

$$W = A'B - AB'.$$

We first prove that W is non-zero, including the assertion about constants which is often hidden in this argument. Let $f \in k(t)$ satisfy $f' = 0$, and write $f = P/Q$ with coprime $P, Q \in k[t]$ and $Q \neq 0$. The quotient rule gives

$$P'Q - PQ' = 0.$$

Hence P divides $P'Q$. Since $\gcd(P, Q) = 1$, Euclid's lemma gives $P \mid P'$. If P were non-constant, characteristic zero would imply that $P' \neq 0$ and $\deg P' = \deg P - 1$, which is incompatible with $P \mid P'$. Thus P is constant. Interchanging P and Q shows that Q is constant. We have proved

$$\ker\left(\frac{d}{dt} : k(t) \longrightarrow k(t)\right) = k. \tag{2.1.2}$$

If $W = 0$, then

$$\left(\frac{A}{B}\right)' = \frac{A'B - AB'}{B^2} = 0,$$

so (2.1.2) gives $A = \lambda B$ for some $\lambda \in k^\times$. Pairwise coprimality then forces A and B to be constant: any non-constant irreducible factor of one would divide the other. The relation $C = A + B$ would make C constant as well, contrary to the hypothesis. Therefore $W \neq 0$.

We next account for every multiplicity. Factor a non-zero polynomial as

$$F = u \prod_{i=1}^{s} P_i^{e_i}, \qquad u \in k^\times,$$

where the P_i are distinct monic irreducibles and $e_i \geq 1$. Fix i and write $F = P_i^{e_i} G_i$ with $P_i \nmid G_i$. Differentiation gives

$$F' = P_i^{e_i - 1}(e_i P_i' G_i + P_i G_i').$$

Thus $P_i^{e_i - 1} \mid F'$ for every i. The powers belonging to distinct P_i are coprime, so their product divides F':

$$\frac{F}{\operatorname{rad}(F)} \mid F'. \tag{2.1.3}$$

Notice that only divisibility is needed here; the later non-vanishing of the Wronskian is what prevents (2.1.3) from becoming vacuous.

Put

$$A_0 = \frac{A}{\operatorname{rad}(A)}, \qquad B_0 = \frac{B}{\operatorname{rad}(B)}, \qquad C_0 = \frac{C}{\operatorname{rad}(C)}.$$

The polynomial A_0 divides both A and A' by (2.1.3); hence it divides both $A'B$ and AB', and therefore $A_0 \mid W$. The same argument gives $B_0 \mid W$. The additive relation supplies the remaining divisibility with no omitted sign calculation:

$$\begin{aligned} A'C - AC' &= A'(A + B) - A(A' + B') = A'B - AB' = W, \\ B'C - BC' &= B'(A + B) - B(A' + B') = AB' - A'B = -W. \end{aligned}$$

Since C_0 divides both C and C', either identity shows that $C_0 \mid W$.

The polynomials A, B, C are pairwise coprime. Therefore no irreducible polynomial can divide two of A_0, B_0, C_0, and these three quotients are pairwise coprime. Their product consequently divides W. Moreover, pairwise coprimality gives

$$\operatorname{rad}(ABC) = \operatorname{rad}(A)\operatorname{rad}(B)\operatorname{rad}(C)$$

up to a unit of k, so the divisibility is exactly

$$\frac{ABC}{R} \mid W. \tag{2.1.4}$$

Write $a = \deg A$, $b = \deg B$, and $c = \deg C$. Since $W \neq 0$, (2.1.4) may be compared by degree. From its first expression,

$$\deg W \leq \max\{(a-1)+b, a+(b-1)\} = a+b-1,$$

where the same formula remains valid if exactly one of A, B is constant. On the other hand,

$$\deg\left(\frac{ABC}{R}\right) = a+b+c-r.$$

Thus

$$a+b+c-r \leq a+b-1,$$

and cancellation gives $c \leq r-1$.

The other two bounds use the same non-zero certificate but a different degree estimate. The identity $W = A'C - AC'$ gives

$$a+b+c-r \leq a+c-1,$$

so $b \leq r-1$. The identity $-W = B'C - BC'$ gives

$$a+b+c-r \leq b+c-1,$$

so $a \leq r-1$. Hence

$$\max\{a,b,c\} \leq r-1,$$

which is (2.1.1). □

Second proof: the three-fibre degree count

Riemann–Hurwitz proof. After extending k algebraically, neither polynomial degrees nor r changes. Consider

$$f = \frac{A}{C} \colon \mathbb{P}^1 \longrightarrow \mathbb{P}^1, \qquad 1-f = \frac{B}{C}.$$

Since A and C are coprime, the degree of this finite separable map is

$$d = \max\{\deg A, \deg C\} = \max\{\deg A, \deg B, \deg C\}.$$

Let $S = f^{-1}\{0,1,\infty\}$ and $s = \#S$. Its finite points are precisely the distinct roots of ABC; the point at infinity may contribute once. Therefore $s \leq r+1$.

For each of the three target points, the ramification indices in its fibre sum to d. The ramification contribution above the three-point set is thus

$$\sum_{x \in S} (e_x - 1) = 3d - s.$$

Riemann–Hurwitz on $\mathbb{P}^1$ gives total ramification $2d-2$, so $2d-2 \geq 3d-s$. Hence $d \leq s-2 \leq r-1$, which is (2.1.1). □

Proposition 2.1.2 (The exact three-fibre defect). *[REPROVED: Har77] Let $f\colon X \to \mathbb{P}^1$ be a finite separable map of degree d from a smooth projective curve of genus g, let $S = f^{-1}\{0,1,\infty\}$, and let R_f^{out} be the part of the ramification divisor supported outside S. Then*

$$d = \#S + 2g - 2 - \deg R_f^{\mathrm{out}}. \tag{2.1.5}$$

For a primitive polynomial triple, write $\epsilon_\infty = 1$ when the point $\infty \in \mathbb{P}^1$ belongs to S and $\epsilon_\infty = 0$ otherwise. Then

$$\deg \mathrm{rad}(ABC) - 1 - d = \deg R_f^{\mathrm{out}} + 1 - \epsilon_\infty. \tag{2.1.6}$$

Proof. Above each of $0, 1, \infty$, the sum of the ramification indices is d. Hence the part of the ramification divisor supported on S has degree $3d - \#S$. Splitting the full divisor into its inside and outside parts in Riemann–Hurwitz gives

$$\begin{aligned} 2g - 2 &= -2d + \deg R_f \\ &= -2d + (3d - \#S) + \deg R_f^{\mathrm{out}} \\ &= d - \#S + \deg R_f^{\mathrm{out}}. \end{aligned}$$

Rearrangement proves (2.1.5). For a polynomial triple, the finite points of S are exactly the distinct roots of ABC, so $\#S = \deg \mathrm{rad}(ABC) + \epsilon_\infty$. Substitution with $g = 0$ yields

$$d = \deg \mathrm{rad}(ABC) + \epsilon_\infty - 2 - \deg R_f^{\mathrm{out}},$$

which is equivalent to (2.1.6). □

The map need not be a Belyi map: it may have branch points away from $0, 1, \infty$. The proof uses only the three designated fibres. Proposition 2.1.2 records both sources of slack. Equality in the polynomial inequality requires no ramification outside the three fibres *and* requires the source point at infinity to lie over $0, 1$, or ∞. The first condition is the Belyi condition; the second is the affine-chart condition which makes the polynomial radical miss exactly one point rather than none.

Third proof: the Stothers–Zannier local differential

Logarithmic-differential proof. Retain $f = A/C$ and work on $X = \mathbb{P}^1_{\bar{k}}$. Write n_0, n_1, n_∞ for the numbers of distinct points in the fibres over $0, 1, \infty$, so that $s = n_0 + n_1 + n_\infty$. Consider the meromorphic differential

$$\omega = \frac{df}{f(1-f)}.$$

At a point where f has a zero of order e, a local parameter z gives $f = z^e u$ with u a unit. Since $e \neq 0$ in k, $\mathrm{ord}(df) = e - 1$, and ω has a simple pole. The same calculation with $1 - f$ gives a simple pole at each point over 1. If f has a pole of order e, then $\mathrm{ord}(df) = -e - 1$ and $\mathrm{ord}(f(1-f)) = -2e$, so $\mathrm{ord}(\omega) = e - 1$. Away from the three fibres, ω has no pole.

The poles of ω therefore have total degree $n_0 + n_1$, while its zeros above ∞ have total degree at least

$$\sum_{f(x)=\infty} (e_x - 1) = d - n_\infty.$$

Every meromorphic differential on $\mathbb{P}^1$ has divisor of degree -2. Thus

$$-2 = \deg \mathrm{div}(\omega) \geq (d - n_\infty) - (n_0 + n_1) = d - s.$$

It follows that $d \le s-2 \le r-1$. □

The three arguments are different presentations. The first never leaves $k[t]$ and compares divisibility with polynomial degree. The second counts ramification of a finite map globally. The third studies one logarithmic differential point by point and uses only the degree of its canonical divisor. Their agreement will matter: it identifies the common geometric mechanism beneath the one-line algebra.

Example 2.1.3 (Sharpness and inseparability). [KNOWN: GAL20] For $n \ge 1$,

$$t^n + (1-t^n) = 1$$

has maximum degree n and $\deg \operatorname{rad}(t^n(1-t^n)) = n+1$, so equality holds. In characteristic p, the same identity with $n = p$ becomes $t^p + (1-t)^p = 1$; its radical has degree 2, while the maximum degree is p. Here every differential used above vanishes. Characteristic zero, or an explicit separability hypothesis excluding simultaneous pth powers, is therefore essential.

Example 2.1.4 (One triple, all three certificates). Take

$$A = t^3, \qquad B = (1-t)^3, \qquad C = A + B = 3t^2 - 3t + 1.$$

The quadratic C has discriminant -3, so it is squarefree in characteristic zero and shares no root with $t(1-t)$. Therefore

$$R = \operatorname{rad}(ABC) = t(1-t)(3t^2-3t+1), \qquad \deg R = 4.$$

The largest polynomial degree is 3, and the bound $3 \le 4-1$ is an equality.

For the first certificate,

$$\begin{aligned} W &= A'B - AB' \\ &= 3t^2(1-t)^3 - t^3(-3(1-t)^2) \\ &= 3t^2(1-t)^2((1-t)+t) \\ &= 3t^2(1-t)^2. \end{aligned}$$

On the other hand,

$$\frac{ABC}{R} = t^2(1-t)^2.$$

Thus the multiplicity quotient divides W with quotient 3. No factor has been hidden: t^2 records the excess multiplicity of $A = t^3$ and $(1-t)^2$ records that of $B = (1-t)^3$; the squarefree polynomial C contributes no excess factor. Here

$$\deg(ABC/R) = 4 = \deg W,$$

so the divisibility-to-degree step is exact. The coarser estimate $\deg W \le \deg A + \deg B - 1 = 5$ has one degree of cancellation, but the alternative expression $-W = B'C - BC'$ has upper bound $\deg B + \deg C - 1 = 4$ and gives the sharp bound on $\deg A$.

For the second certificate, put $f = A/C$. Its degree is 3. The fibre over 0 consists of $t = 0$ with ramification index 3, and the fibre over 1 consists of $t = 1$ with ramification index 3.

The fibre over ∞ consists of the two roots of C and the source point at infinity, each with ramification index 1. Consequently

$$\#S = 1 + 1 + 3 = 5, \qquad \sum_{x \in S} (e_x - 1) = 2 + 2 = 4.$$

Riemann–Hurwitz gives total ramification $2d - 2 = 4$. Hence there is no ramification outside S, and

$$d = \#S - 2 = 3.$$

The map is a Belyi map, and $\epsilon_\infty = 1$, so both defect terms in (2.1.6) vanish.

For the local-differential certificate, the quotient rule and the computed Wronskian give

$$df = \frac{3t^2(1-t)^2}{C^2}\,dt, \qquad f(1-f) = \frac{t^3(1-t)^3}{C^2}.$$

Therefore

$$\frac{df}{f(1-f)} = \frac{3\,dt}{t(1-t)}. \tag{2.1.7}$$

This form has a simple pole at 0 and at 1. At infinity, with $z = 1/t$,

$$\frac{dt}{t(1-t)} = -\frac{dz}{z-1},$$

which is regular and non-zero at $z = 0$. Thus its divisor has two simple poles, no zeros, and degree -2, exactly as required. The factorisation divisibility, the ramification partition, and the divisor of the logarithmic differential are three presentations of the same numerical equality.

Proposition 2.1.5 (The characteristic-p kernel). *[REPROVED : Har77] Let k have characteristic $p > 0$. Then*

$$\ker\left(\frac{d}{dt} : k(t) \longrightarrow k(t)\right) = k(t^p).$$

Consequently the Wronskian of A and B vanishes precisely when $A/B \in k(t^p)$.

Proof. Put $K = k(t^p)$. The element t satisfies $T^p - t^p \in K[T]$, and $1, t, \dots, t^{p-1}$ form a basis of $k(t)$ over K. Every $f \in k(t)$ therefore has a unique expression

$$f = \sum_{i=0}^{p-1} a_i t^i, \qquad a_i \in K.$$

Every element of K has derivative zero, so

$$f' = \sum_{i=1}^{p-1} i\, a_i t^{i-1}.$$

The elements $1, t, \dots, t^{p-2}$ are linearly independent over K, and $i \neq 0$ in k for $1 \leq i \leq p-1$. Hence $f' = 0$ if and only if $a_i = 0$ for all $i > 0$, which is equivalent to $f \in K$. The final assertion follows from

$$\left(\frac{A}{B}\right)' = \frac{A'B - AB'}{B^2}.$$

□

This proposition locates the failure more precisely than the phrase "Frobenius kills derivatives". In characteristic zero the constants field is the full kernel, as in (2.1.2); in characteristic p the kernel is enlarged by an entire purely inseparable rational subfield. The example

$$t^p + (1-t)^p = 1$$

has

$$\max\{\deg t^p, \deg(1-t)^p, 0\} = p, \qquad \deg \operatorname{rad}(t^p(1-t)^p) = 2.$$

The asserted characteristic-zero inequality would read $p \leq 1$, which is false. Here $f = t^p$ factors through the absolute Frobenius $t \mapsto t^p$, its differential is zero, and the induced map of curves is purely inseparable. Neither the Wronskian divisor nor the separable Riemann–Hurwitz count used above exists in the stated form.

If $A'B - AB' \neq 0$, however, the entire polynomial divisibility argument still works in characteristic p: the relation $F/\operatorname{rad}(F) \mid F'$ remains true, the common Wronskian is non-zero, and degrees can be compared. Thus the operative hypothesis is separability of A/B, not characteristic zero as a bare label. Characteristic zero is the convenient uniform assumption which makes every nonconstant rational function separable.

Hasse derivatives explain why replacing d/dt by a higher derivative is not a uniform repair. They are defined by

$$D^{(m)}(t^n) = \binom{n}{m} t^{n-m}.$$

For example $D^{(p)}(t^p) = 1$, so the first invisible Frobenius power becomes visible at order p. But t^{p^2} is killed by every $D^{(m)}$ with $0 < m < p^2$, and the product rule is

$$D^{(m)}(fg) = \sum_{i+j=m} D^{(i)}(f) D^{(j)}(g),$$

not the first-order Leibniz identity which produced one Wronskian. Choosing an order after inspecting the input can prove separable variants, but it does not supply one functorial differential with kernel exactly the constants. This is the positive-characteristic version of the design constraint carried into the arithmetic chapters.

Example 2.1.6 (Why primitivity and non-constancy are hypotheses). Fix $m \geq 1$ and take

$$A = t^m, \qquad B = t^m, \qquad C = 2t^m.$$

Then $A + B = C$, but the triple has the common factor t^m. Its radical is, up to a non-zero scalar,

$$\operatorname{rad}(ABC) = t,$$

so the conclusion of Mason–Stothers would be $m \leq 0$. This is false. At the Wronskian level the reason is equally explicit: $A'B - AB' = 0$, because $A/B = 1$ is constant, although the three polynomials are not constant. In the proof of Theorem 2.1.1, pairwise coprimality was what converted "A/B is constant" into "A and B are constant". Removing the common factor first turns the displayed relation into $1 + 1 = 2$, where there is no polynomial height left to bound.

The exclusion of an all-constant triple is also forced by the sharp constant. For $1 + 1 = 2$ one has maximum degree 0 and radical degree 0, so (2.1.1) would read $0 \leq -1$. The theorem concerns a nonconstant map to the three-point line; a constant projective point has degree zero but does not satisfy the ramification count $3d - \#S$ used in the proof. Finally, requiring each of A, B, C to be non-zero prevents one of the three target fibres from disappearing and

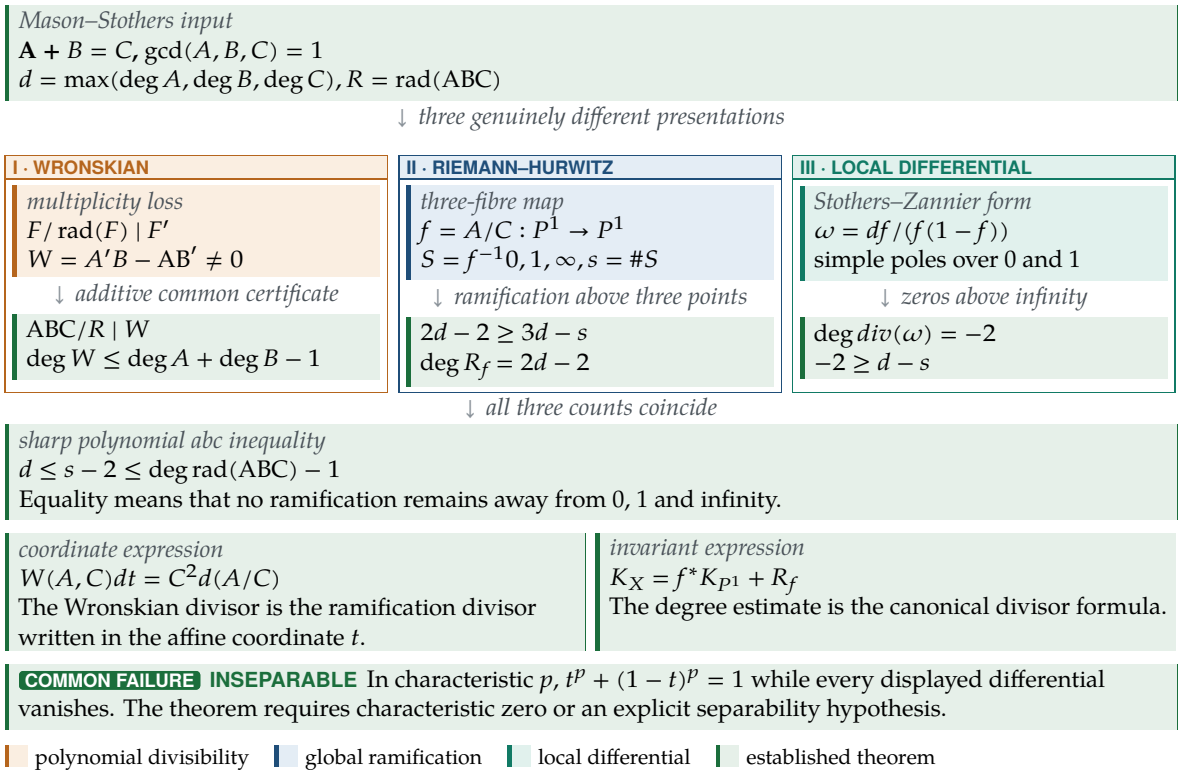


Plate 2.1. Three proofs of Mason–Stothers and their common separability hinge.
What it does not show: the theorem in characteristic p: there $t^p + (1-t)^p = 1$ while every displayed differential vanishes, so characteristic zero or an explicit separability hypothesis is required.

keeps A/C and $df/[f(1-f)]$ defined. Thus every elementary hypothesis has a visible job in at least one of the three proofs.

2.2 The Wronskian mechanism

The first proof can be decomposed into four operations, each of which will have an arithmetic counterpart demanded later.

Multiplicity becomes a derivative divisor. If F vanishes to order m at α, then F' vanishes to order at least $m-1$. In characteristic zero it is exactly $m-1$ unless further cancellation comes from the unit factor. Globally this says

$$\frac{F}{\operatorname{rad}(F)} \mid F'.$$

Differentiation sees every repeated factor but charges it one unit less. The radical is what remains after those lost units are removed.

Addition creates a common certificate. From $A + B = C$ one obtains

$$A'B - AB' = A'C - AC' = B'C - BC'.$$

Thus the three multiplicity quotients divide one non-zero polynomial. The Wronskian is not merely an estimate; it is a common certificate generated by the additive relation.

Coprimality permits multiplication. Knowing separately that $A/\operatorname{rad}(A)$, $B/\operatorname{rad}(B)$, and $C/\operatorname{rad}(C)$ divide W is insufficient unless these factors are coprime. Primitivity turns three local divisibilities into the single global divisibility $ABC/\operatorname{rad}(ABC) \mid W$. This is the point at which the additive equation and multiplicative support first meet.

Degree closes the inequality. The derivative costs one degree:

$$\deg(A'B - AB') \le \deg A + \deg B - 1.$$

Divisibility supplies the opposite inequality. Cancelling the two degrees that occur on both sides leaves exactly the third polynomial bounded by the reduced support, with the sharp constant -1.

Lemma 2.2.1 (Wronskian as a relative differential). *[REPROVED: Har77] We restate the polynomial certificate intrinsically. On the affine line $X = \mathbb{A}^1_k$,*

$$W(A, C)\, dt = C^2 d(A/C)$$

as a section of $\Omega^1_{k(t)/k}$ after clearing denominators. Its vanishing orders are the ramification orders of A/C away from poles.

Proof. The quotient rule gives $d(A/C) = (A'C - AC')C^{-2}dt$. Multiplication by C^2 gives the displayed identity. If $A/C = uz^e$ in a local parameter z, with u a unit, then

$$d(A/C) = z^{e-1}(eu + zu')\, dz.$$

The coefficient eu is a unit because $\operatorname{char} k = 0$ and $u(0) \neq 0$. Hence $d(A/C)$ has order exactly $e - 1$, which is the local ramification contribution. □

The notation d/dt depends on the coordinate, but the differential df does not. A change of parameter rescales the local expression by the transition function of $\Omega^1_{X/k}$. This is why the proof globalises to an arbitrary curve and why a bare operator, without a sheaf in which its values live, is not enough.

2.3 Riemann–Hurwitz as the true content

Theorem 2.3.1 (Riemann–Hurwitz for curves). *[KNOWN: Har77] Let $f\colon X \to Y$ be a finite separable morphism of degree d between smooth projective geometrically connected curves over an algebraically closed field of characteristic zero. If $R_f = \sum_x (e_x - 1)[x]$ is the ramification divisor, then*

$$K_X \sim f^* K_Y + R_f, \qquad 2g(X) - 2 = d(2g(Y) - 2) + \deg R_f.$$

The Mason inequality is the case $Y = \mathbb{P}^1$ in which three fibres are charged at once. If $S = f^{-1}\{0, 1, \infty\}$, then

$$\deg(R_f|_S) = 3d - \#S.$$

Since the remaining ramification divisor is effective,

$$2g(X) - 2 = -2d + \deg R_f \ge -2d + (3d - \#S),$$

and therefore

$$d \le \#S + 2g(X) - 2. \tag{2.3.1}$$

For $X = \mathbb{P}^1$, this is $d \leq \#S - 2$. The polynomial radical counts all finite members of S and misses at most the point at infinity, producing $d \leq \deg \operatorname{rad}(ABC) - 1$.

This derivation explains the sharpness condition. Equality in (2.3.1) holds precisely when R_f is supported over $0, 1, \infty$. In other words, the extremal maps are Belyi maps relative to the chosen three points. For $f = t^n$, all ramification lies over 0 and ∞, and the identity in Example 2.1.3 is extremal. Additional branch points create positive slack.

Principle 2.3.2 (The geometric content of polynomial *abc*). *[REPROVED : Mue93] We make explicit the invariant statement concealed by the coordinate proof. Mason–Stothers is the inequality obtained by comparing the log-canonical degree of a three-point pullback with the canonical divisor formula for a separable map of curves.*

Proof. The reduced divisor $S = f^{-1}\{0, 1, \infty\}_{\mathrm{red}}$ has degree $\#S$, while $\deg K_X = 2g(X) - 2$. Thus the right side of (2.3.1) is $\deg(K_X + S)$. The left side is the height $\deg f$. Riemann–Hurwitz gives the comparison, and the polynomial statement is its genus-zero affine chart. □

The Wronskian proof and the Riemann–Hurwitz proof are consequently not rival explanations. The degree estimate for W is a coordinate expression of the canonical divisor degree; the divisibility by multiplicity quotients is a coordinate expression of ramification. The theorem is easy because these expressions agree exactly.

2.4 What the proof uses that arithmetic lacks

We now perform a dependency audit. None of the following four inputs is optional; removing any one breaks a visible line of all three proofs.

Criterion 2.4.1 (The four consumed ingredients). *[REPROVED : GAL20] The delta from the usual proof is an explicit separation of its geometric dependencies. The Mason–Stothers mechanism consumes:*

(F1) *a genuine base* Spec*k beneath the curve X;*
(F2) *a non-zero k-derivation, locally d/dt;*
(F3) *the non-zero relative cotangent sheaf* $\Omega^1_{X/k}$ *in which differentials and Wronskians live;*
(F4) *a degree and ramification formula, equivalently Riemann–Hurwitz or the rule* $\deg \operatorname{div}(\omega) = 2g(X) - 2$.

Proof. (F1) determines the constants and the word "relative". Without it there is no specified kernel for a derivation. (F2) converts multiplicity m into $m - 1$ and turns $A + B = C$ into a common Wronskian. (F3) makes the coordinate expressions glue and supplies a canonical divisor. (F4) converts local orders into a global numerical inequality. In the first proof these appear as constants, differentiation, $W\,dt$, and polynomial degree; in the second as the structural morphism, separability, the cotangent map, and Riemann–Hurwitz; in the third as the same base, df, the meromorphic 1-form $df/[f(1-f)]$, and the degree of its divisor. □

Each item has a precise arithmetic failure.

Failure of (F1): no lower ordinary base. Every scheme is canonically a $\mathbb{Z}$-scheme, so Spec$\mathbb{Z}$ is already the terminal ordinary scheme. One can map it into many chosen ambient schemes, but ordinary scheme theory supplies no canonical terminal layer beneath it playing the role of Spec*k* beneath a curve. This distinction is proved in Chapter 3; it is not the claim that Spec$\mathbb{Z}$ has no morphisms to other schemes.

Failure of (F2): no arithmetic d/dt. For every $\mathbb{Z}$-module M,

$$\mathrm{Der}_{\mathbb{Z}}(\mathbb{Z}, M) = 0.$$

A derivation must kill 1, hence every integer. Difference operators, Fermat-quotient operators, and Frobenius lifts can be useful, but none is the ordinary derivation consumed above without additional structure and a new Leibniz rule.

Failure of (F3): the identity cotangent sheaf vanishes. The relative objects are

$$\Omega^1_{\mathbb{Z}/\mathbb{Z}} = 0, \qquad \mathbb{L}_{\mathbb{Z}/\mathbb{Z}} \simeq 0.$$

This says that the arithmetic base has no tangent direction relative to itself. It does not say that $\Omega^1_{X/\mathbb{Z}}$ vanishes for every arithmetic scheme X, nor that arithmetic deformation theory is empty.

Failure of (F4): no matching relative curve formula. Arithmetic discriminant formulae, conductors, and Riemann–Hurwitz theorems for maps of arithmetic curves certainly exist. What is missing is the Chapter 2 configuration in which $\mathrm{Spec}\mathbb{Z}$ itself is a relative curve over a lower base, with a non-zero relative differential whose divisor has a degree and whose reduced support is $\log \mathrm{rad}(abc)$. Without (F1)–(F3), the desired instance of (F4) has no source object.

2.4.1 A failed arithmetic Wronskian, computed

It is useful to see the obstruction on an actual coprime sum. Consider

$$3 + 125 = 128. \tag{2.4.1}$$

The three integers are pairwise coprime and

$$\mathrm{rad}(3 \cdot 125 \cdot 128) = 2 \cdot 3 \cdot 5 = 30, \qquad \frac{3 \cdot 125 \cdot 128}{30} = 1600.$$

Thus 1600 is the arithmetic multiplicity quotient which a literal Wronskian translation would have to place inside a non-zero certificate.

Let $D\colon \mathbb{Z} \to M$ be an ordinary derivation into any $\mathbb{Z}$-module. The Leibniz rule gives

$$D(1) = D(1 \cdot 1) = D(1) + D(1),$$

so $D(1) = 0$. Additivity then gives $D(n) = nD(1) = 0$ for $n \geq 0$, and $D(-n) = -D(n) = 0$. Hence D is identically zero. Applied to (2.4.1), every putative Wronskian

$$W_D(x, y) = D(x)y - xD(y)$$

is zero. The divisibility $1600 \mid 0$ is formally true but useless: there is no degree or logarithmic size of a non-zero certificate to compare with 1600. This is why saying that the integer Wronskian "vanishes" is not a minor loss in a familiar proof. It removes the inequality-producing object.

One might instead use the *arithmetic derivative* ∂_{ar} on positive integers, defined by

$$\partial_{\mathrm{ar}}(p) = 1 \quad \text{for primes } p, \qquad \partial_{\mathrm{ar}}(xy) = x\,\partial_{\mathrm{ar}}(y) + y\,\partial_{\mathrm{ar}}(x).$$

Unique factorisation gives the explicit formula

$$\partial_{\mathrm{ar}}(n) = n \sum_{p \mid n} \frac{v_p(n)}{p}. \tag{2.4.2}$$

This operator detects prime multiplicities and obeys the ordinary product rule, but it is not additive. For (2.4.1),

$$\partial_{\mathrm{ar}}(3) = 1, \qquad \partial_{\mathrm{ar}}(125) = 75, \qquad \partial_{\mathrm{ar}}(128) = 448.$$

Consequently

$$\partial_{\mathrm{ar}}(3) + \partial_{\mathrm{ar}}(125) = 76 \neq 448 = \partial_{\mathrm{ar}}(128).$$

The two identities which were equal in the polynomial proof now give

$$\begin{aligned} W_{\partial_{\mathrm{ar}}}(3, 125) &= 1 \cdot 125 - 3 \cdot 75 = -100, \\ W_{\partial_{\mathrm{ar}}}(3, 128) &= 1 \cdot 128 - 3 \cdot 448 = -1216. \end{aligned}$$

They are neither equal nor opposite, and 1600 does not divide the first. The arithmetic derivative therefore repairs multiplicative sensitivity while destroying the additive identity that made one common certificate possible. The smaller relation $1 + 8 = 9$ already shows the same defect: $\partial_{\mathrm{ar}}(1) + \partial_{\mathrm{ar}}(8) = 12$ whereas $\partial_{\mathrm{ar}}(9) = 6$.

Fermat quotients, p-derivations, and Frobenius lifts deliberately replace one or both ordinary rules by twisted formulae. They are not refuted by this example. Rather, the calculation specifies what any replacement must recover after its twists are assembled: it must interact with addition strongly enough to produce a common non-zero object, and it must retain enough multiplicity information to compare that object with a reduced support.

2.4.2 The three exact resources and the proposed substitutes

The four structural items (F1)–(F4) can be regrouped according to the three places where the numerical proof actually closes. This sharper list is the one used by the later architecture.

(a) A derivation with exactly the constants as kernel. The polynomial proof uses not merely a map called a derivative, but the exactness at the middle term of

$$0 \longrightarrow k \longrightarrow k(t) \xrightarrow{d} \Omega^1_{k(t)/k}. \tag{2.4.3}$$

The injective first arrow specifies the constants. Exactness says that $df = 0$ forces $f \in k$; this was the non-vanishing argument for W. Locally, the formula

$$d(z^e u) = z^{e-1}(eu + zu')\, dz$$

then removes exactly one copy of a repeated factor. Both statements are needed. A multiplicity-sensitive operator with a larger kernel fails on simultaneous constants, as Frobenius does in characteristic p; an operator with the right kernel but no additive compatibility does not turn $A + B = C$ into one certificate.

The proposed arithmetic replacement is deliberately multi-stage. Chapter 6 begins at a fixed prime with a p-derivation and its Frobenius lift. These obey explicit twisted addition and multiplication laws rather than the ordinary Leibniz rule. Chapters 7 and 8 place that Frobenius in relative prismatic geometry, and Chapter 9 packages the filtered Frobenius data as F-gauges. This is the *prismatic Frobenius* part of the substitute.

It is still only one prime at a time. The all-prime candidate begins with the Jackson q-derivative in Chapter 13. On the framed Laurent line over its generic q-base, that chapter proves that its kernel is the constants; at a root of unity it also computes the enlarged kernel, so the failure locus is visible rather than suppressed. The Habiro completion of Chapter 14 stores all cyclotomic specialisations in one ring. Finally Chapter 15 asks that the operator and its prismatic Frobenius realisations descend to the absolute base. Thus the proposed replacement for (2.4.3) is not one mysterious "arithmetic derivative". It is a q-difference operator, its cyclotomic family, and comparison with all fixed-prime Frobenius gauges.

There is an important residual distinction. The framed calculation in Chapter 13 has an exact constants theorem. The global descent obligations in Chapter 15 establish, if solved, existence, coordinate independence, comparison, and non-vanishing of the descended strain operator; they do not presently contain a separate theorem that its kernel on every arithmetic object is exactly the global constants. No later chapter may silently import that stronger conclusion from the classical line.

(b) An additive degree with no unrecorded infinite-place term. For non-zero polynomials,

$$\deg(FG) = \deg F + \deg G,$$

and $G \mid F$ with $F \neq 0$ implies $\deg G \leq \deg F$. These two elementary facts are the entire numerical engine of the first proof. Geometrically they come from the exact degree map on divisors of a proper curve:

$$\deg\colon \operatorname{Div}(X) \longrightarrow \mathbb{Z}, \qquad \deg \operatorname{div}(f) = 0.$$

Every place of a function field is non-archimedean, including the place called infinity on $\mathbb{P}^1$. The pole order at that place is part of the same divisor theory as the finite zero orders. There is no additional archimedean metric term whose normalisation must be chosen after the calculation.

For a positive integer n, the corresponding equality is the product formula

$$\log n = \sum_{p} v_p(n) \log p. \tag{2.4.4}$$

The left side is the archimedean size; the right side records all finite multiplicities. The logarithmic radical is only the truncation

$$\log \operatorname{rad}(n) = \sum_{p \mid n} \log p.$$

Thus the arithmetic problem asks for control of the archimedean quantity in (2.4.4) by a finite support count which has forgotten the $v_p(n)$. Omitting the infinite place would not simplify the problem; it would delete the quantity to be bounded.

Chapter 16 supplies the coefficient categories in which finite solid objects and real liquid objects can be placed in one diagram. Chapter 17 proposes the archimedean gauge which is meant to turn a real norm into a categorical weight. Chapter 18 proposes the all-place gluing, and Chapter 19 supplies the established product-formula notion of adelic degree together with the conditional lift from norms to gauges. This is the book's substitute for the exact polynomial degree step: not the claim that the archimedean place disappears, but a mechanism intended to include it without an untracked correction.

(c) Riemann–Hurwitz and genus bookkeeping. The global proofs consume the identity

$$2g(X) - 2 = (\deg f)(2g(Y) - 2) + \deg R_f. \tag{2.4.5}$$

Its force lies in all three terms being degrees in the same theory. The ramification divisor is effective, the canonical degrees are exact integers, and the three chosen fibres contribute $3 \deg f - \#S$. There is no asymptotic constant and no loss depending on f. In the logarithmic-differential proof, the same resource is the exact equation

$$\deg \operatorname{div}(\omega) = 2g(X) - 2.$$

The arithmetic substitute is again spread over several chapters. Chapter 34 states the operations and duality needed to form relative cotangent objects and their determinants. Chapter 36 constructs the trace formalism needed for a choice-independent numerical output, and Chapter 37 identifies the intended degree with height on the objects for which the comparison is available. Chapter 38 then formulates the direct analogue of (2.4.5),

$$\vartheta(X) = (\deg f)\vartheta(Y) + \deg \mathfrak{d}^{\mathfrak{S}}_{X/Y},$$

with the substrate different replacing R_f. Its function-field specialisation is proved to recover (2.4.5); its all-place arithmetic form is conditional. The estimate of that different, not the formal identity alone, is what enters Chapter 39.

Warning 2.4.2. The arrows in the preceding three paragraphs are a programme map, not a proof that the classical resources have already been replaced.

For (a), descent and comparison of the all-prime operator remain conditional on PO-008 and PO-011–PO-015, and an exact global kernel-equals-constants statement is not presently a numbered obligation. For (b), the coefficient categories are known, but the archimedean gauge, its finite-place compatibility, the global substrate, and the lift of adelic degree remain conditional on PO-017–PO-036. For (c), the arithmetic Riemann–Hurwitz identity used later is conditional on PO-062, PO-063, PO-065–PO-068, and PO-070–PO-076. Consequently this chapter proves the function-field theorem and diagnoses the arithmetic interfaces; it does not prove an arithmetic Mason–Stothers theorem or the *abc* conjecture.

These are four independent checkpoints, not four phrasings of "there is no derivative". A proposed repair may create an operator while lacking global duality, or create a cotangent object at one prime while lacking all-place degree. The later architecture must pass each checkpoint separately.

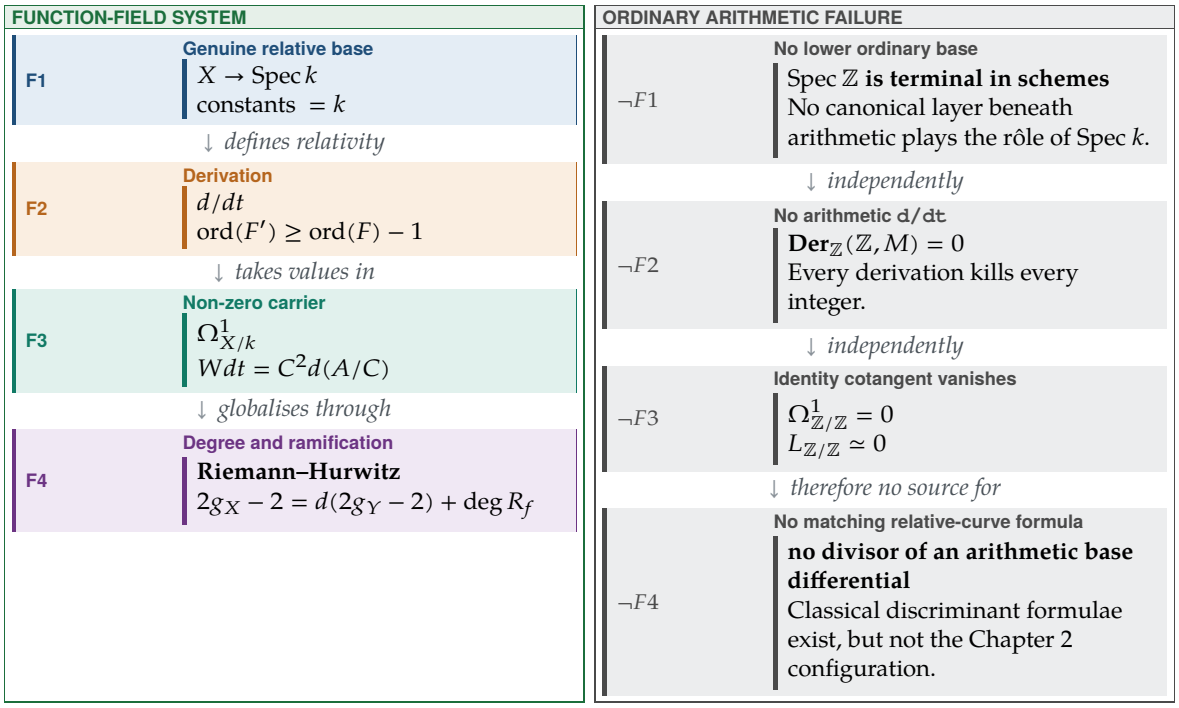
2.5 Higher genus and the Vojta analogue

The local-differential proof already contains the correct generalisation. Let X/k be a smooth projective geometrically connected curve of genus g over an algebraically closed field of characteristic zero, and let $f \in k(X) \setminus k$. Define its height by

$$h_X(f) = \deg(f\colon X \to \mathbb{P}^1)$$

and its truncated three-point count by

$$N^{(1)}_{0,1,\infty}(f) = \#(f^{-1}\{0, 1, \infty\}).$$



Where each line of the proof uses the system

Input	Local task	Global task	Failure if removed
F1 · Spec k	specifies constants	defines relative geometry	no kernel for an operator
F2 · d/dt	turns m into m-1	builds the Wronskian	multiplicity remains invisible
F3 · $\Omega^1_{X/k}$	carries df	glues coordinate formulae	no canonical divisor
F4 · degree / RH	counts local orders	closes the inequality	no height-support comparison

HAND-OFF CHAPTER 3 These four arrows point to the absolute obstruction and its design specification. They do not point directly to an asserted arithmetic solution.

classical mechanism | missing arithmetic structure | book-level dependency audit

Plate 2.2. The four classical inputs and the four arithmetic failures proved in Chapter 3.

What it does not show: an arithmetic solution: the four arrows point to the absolute obstruction and its design specification, not to an asserted arithmetic proof.

Theorem 2.5.1 (Function-field Vojta–Mason inequality). *[REPROVED : Mue93] We derive the genus correction directly from ramification. For every nonconstant f as above,*

$$h_X(f) \leq N^{(1)}_{0,1,\infty}(f) + 2g - 2. \tag{2.5.1}$$

Proof. Put $d = h_X(f)$ and $S = f^{-1}\{0, 1, \infty\}$. Above each of the three target points, the sum of ramification indices is d, so $\deg(R_f|_S) = 3d - \#S$. Riemann–Hurwitz gives

$$2g - 2 = -2d + \deg R_f \geq -2d + (3d - \#S) = d - \#S.$$

Rearranging gives (2.5.1). Equivalently, the proof with $df/[f(1-f)]$ uses $\deg \operatorname{div}(\omega) = 2g - 2$ instead of -2. □

For a function-field equation $a + b = c$ in $k(X)$, divide by c and take $f = a/c$. The reduced set S is the set of places at which the valuation vector of (a, b, c) is not constant after common scaling. Thus (2.5.1) is precisely

$$\text{height} \leq \text{truncated boundary count} + 2g - 2.$$

It is the function-field $(1, 1)$-form of Vojta's inequality from Section 1.4: the conductor coefficient is one, the canonical correction is exact, and there is no ε-loss.

2.5.1 The valuation form of the theorem

We spell out the preceding translation because it is the form which survives in Diophantine geometry. For each closed point $x \in X$, let v_x be the normalised discrete valuation. Given non-zero $a, b, c \in k(X)$ with $a + b = c$, put

$$\nu_x(a, b, c) = (v_x(a), v_x(b), v_x(c))$$

and define the reduced boundary

$$\Sigma(a, b, c) = \{x \in X : v_x(a), v_x(b), v_x(c) \text{ are not all equal}\}. \tag{2.5.2}$$

Multiplying the triple by $u \in k(X)^\times$ adds $v_x(u)$ to all three entries, so (2.5.2) depends only on the projective triple $[a : b : c]$. This is the function-field analogue of removing $\gcd(a, b, c)$ before taking an integer radical.

The valuation inequality

$$v_x(a + b) \geq \min\{v_x(a), v_x(b)\}$$

has equality whenever the two valuations differ. Since $a + b = c$, the minimum of the three integers $v_x(a), v_x(b), v_x(c)$ is therefore attained at least twice. Subtract this common minimum. Exactly one of the following four patterns remains:

normalised valuations	value of $f = a/c$	point of the target
$(0, 0, 0)$	$f(x) \notin \{0, 1, \infty\}$	outside the boundary
$(m, 0, 0),\ m > 0$	$f(x) = 0$	0
$(0, m, 0),\ m > 0$	$f(x) = 1$	1
$(0, 0, m),\ m > 0$	$f(x) = \infty$	∞.

For the second row, $v_x(a/c) = m > 0$. For the third, $v_x(1 - f) = v_x(b/c) = m > 0$, with the harmless sign $1 - f = b/c$. For the fourth, $v_x(a/c) = -m < 0$. Conversely each of these three target values forces the corresponding valuation pattern. Hence

$$\Sigma(a, b, c) = f^{-1}\{0, 1, \infty\}_{\text{red}}. \tag{2.5.3}$$

The height of f can also be read valuation by valuation:

$$h_X(f) = \sum_{x \in X} \max\{0, -v_x(f)\} \deg(x) = \sum_{x \in X} \max\{0, v_x(c) - v_x(a)\} \deg(x). \tag{2.5.4}$$

The first sum is the degree of the polar divisor of f, hence the degree of the map $f\colon X \to \mathbb{P}^1$. The second follows from $v_x(f) = v_x(a) - v_x(c)$. Over an algebraically closed field all residue

degrees are one. Over a general constant field the same formula, with $\deg(x) = [k(x) : k]$, is the correct weighted version.

Define the truncated conductor of the triple by

$$N_X^{(1)}(a,b,c) = \sum_{x \in \Sigma(a,b,c)} \deg(x). \tag{2.5.5}$$

Equations (2.5.3)–(2.5.5) turn Theorem 2.5.1 into the invariant statement

$$h_X(a/c) \le N_X^{(1)}(a,b,c) + 2g(X) - 2. \tag{2.5.6}$$

Every multiplicity in the boundary has been truncated to one, while the height retains the full pole multiplicity. This is precisely the tension between size and radical in the integer conjecture.

For $X = \mathbb{P}^1$ and $A, B, C \in k[t]$, the finite summands of (2.5.5) are the distinct irreducible factors of ABC, weighted by their residue degrees. The point at infinity belongs to the boundary exactly when the three orders

$$-\deg A, \quad -\deg B, \quad -\deg C$$

are not all equal after the possible leading-term cancellation imposed by $A + B = C$. Thus

$$N_{\mathbb{P}^1}^{(1)}(A,B,C) = \deg \operatorname{rad}(ABC) + \epsilon_\infty.$$

Substitution into (2.5.6) gives

$$\deg(A/C) \le \deg \operatorname{rad}(ABC) + \epsilon_\infty - 2,$$

and the elementary estimate $\epsilon_\infty \le 1$ gives Mason–Stothers. This calculation also explains why the theorem is projectively invariant even though the displayed polynomial radical is tied to one affine chart.

The three quantities now line up with the language used later:

function field	number field	role
$h_X(a/c)$	$h(a:b:c)$	untruncated size
$N_X^{(1)}(a,b,c)$	$\log \operatorname{rad}(abc)$	truncated bad support
$2g(X) - 2$	canonical/discriminant term	base complexity.

The first two correspondences are exact definitions. The third is an analogy until the arithmetic Riemann–Hurwitz and height comparisons of Chapters 38 and 39 are supplied with their stated proof obligations.

The correction is not an artefact. It is $\deg K_X$. On an elliptic curve X it vanishes. For example, on

$$y^2 = x(x-1)(x-\lambda), \qquad \lambda \notin \{0,1\},$$

the map $x\colon X \to \mathbb{P}^1$ has degree 2. The fibres over $0, 1, \infty$ contain three distinct points, so the theorem gives $2 \le 3$. Each of those three points has ramification index 2. The fourth branch point lies over λ and also has index 2. Thus

$$\deg R_x = 4, \qquad \deg(R_x|_S) = 3, \qquad \deg R_x^{\mathrm{out}} = 1.$$

Riemann–Hurwitz reads

$$0 = 2g(X) - 2 = -2 \cdot 2 + 4,$$

while the exact defect formula reads

$$2 = \#S + 2g(X) - 2 - \deg R_x^{\mathrm{out}} = 3 + 0 - 1.$$

The missing unit in the inequality is therefore exactly the ramification over λ, not an unspecified error. If all ramification lay over the three-point boundary, equality would hold.

For $X = \mathbb{P}^1$ and a polynomial triple, infinity contributes at most one point to S. Hence

$$d \leq \#S - 2 \leq \deg \operatorname{rad}(ABC) - 1,$$

recovering Theorem 2.1.1 with its exact -1. This is the function-field limit that every arithmetic construction later in the book must recover. Part V's arithmetic Riemann–Hurwitz is modelled on (2.5.1), with the finite-prime conductor and the archimedean contribution replacing the degree of a reduced boundary divisor.

2.6 A checklist of the missing structure

The proof audit can now be used as a hand-off rather than a slogan. Its four primary questions are:

(C1) What object lies beneath arithmetic and plays the role of Spec$k$?
(C2) What operator replaces d/dt, with an explicit additive and Leibniz law?
(C3) In what non-zero relative cotangent object do its values live?
(C4) What global degree, duality, and ramification theorem converts local multiplicity loss into a conductor–height inequality?

A proposal that cannot answer all four has not transposed the Mason proof.

For use in Chapter 3, the four questions unfold into the following eight checks. The wording is deliberately aligned with the design specification proved there.

(H1) *Non-trivial relativity:* the arithmetic object maps to a base that is not merely itself under another name.
(H2) *A controlled cotangent direction:* the relative cotangent or strain complex is non-zero and admits support, base change, and duality.
(H3) *A derivation-like operator:* an additive or explicitly twisted Leibniz rule is available for forming a Wronskian replacement.
(H4) *Simultaneity of places:* finite primes and the archimedean place belong to one descended object rather than an unrelated product.
(H5) *Six operations and duality:* pullback, pushforward, tensor, internal Hom, and their exceptional variants support a relative formula.
(H6) *A choice-independent trace:* the numerical degree does not depend on choosing identifications between coefficient lines.
(H7) *Compatibility with Diophantine size:* reduced support becomes conductor, while degree or trace becomes height or discriminant.
(H8) *The function-field limit:* over $k(X)$ the construction becomes $\Omega^1_{X/k}$, ordinary differentiation, and (2.5.1) with its exact coefficient and $2g-2$ term.

The expansion from four to eight is forced by arithmetic globality. (C1), (C2), and (C3) become (H1), (H3), and (H2). The single classical degree step (C4) splits into all-place descent, six operations, invariant trace, and height–conductor compatibility, namely (H4)–(H7). Finally (H8) prevents a formally impressive construction from changing the theorem it was designed to transpose.

Chapter 3 proves that the ordinary choices Spec$\mathbb{Z}$, d, $\Omega^1_{\mathbb{Z}/\mathbb{Z}}$, and classical relative degree fail at the four primary checkpoints. It then turns (H1)–(H8) into the book's design requirements. This chapter supplies no arithmetic repair and makes no Diophantine claim. It identifies, with the function-field theorem as a test, what a repair would have to contain.

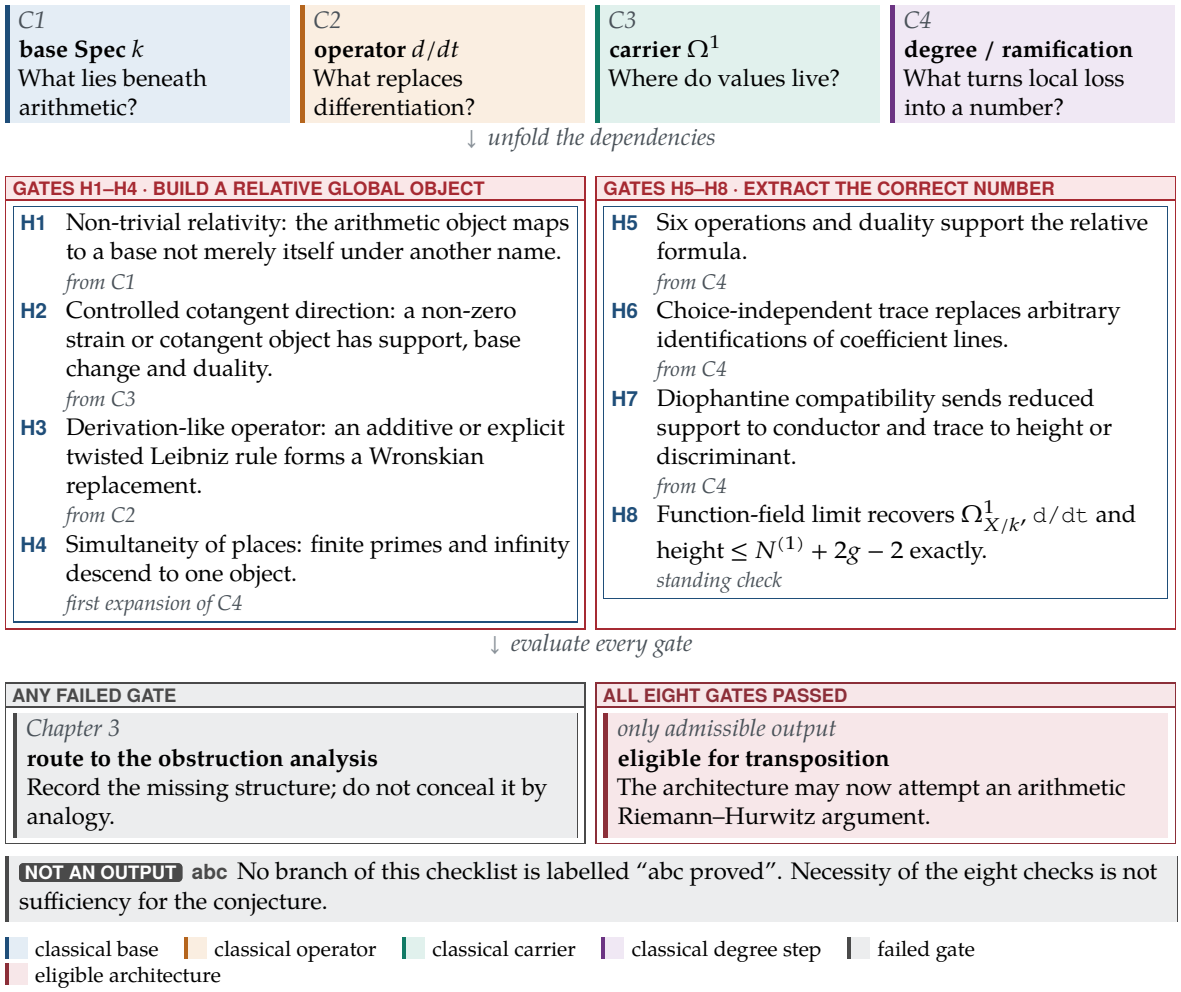


Plate 2.3. The eight checks obtained by unfolding the four inputs of the classical proof.
What it does not show: no branch of this checklist is labelled "abc proved".

Notes and References

Sources. The polynomial theorem is associated independently with Mason and Stothers; Stothers' work appeared in 1981 and Mason's function-field treatment in the early 1980s. The statement and elementary derivative proof are reviewed, with applications, by Gallardo [GAL20]. The higher-genus and generalised Fermat context is represented in the present bibliography by Mueller [Mue93]. The Riemann–Hurwitz formula, divisors of meromorphic differentials, and the identity $\deg K_X = 2g - 2$ are standard algebraic-curve results; our conventions follow Hartshorne and Liu [Har77; LE02a]. The three-point height interpretation belongs to the Vojta dictionary discussed in [Voj87a; Voj11]. Two terminological points matter. The map A/C is called Belyi only when it has no branch value outside $0, 1, \infty$; the general proof merely counts three fibres. Moreover, $\deg \operatorname{rad}(ABC)$ is invariant under extension to an algebraic closure even though irreducible factors split, because residue degrees become the number of geometric roots. These facts justify the passage between the polynomial and geometric proofs without changing either side of the inequality.

Theorem 2.1.1 is proved rather than merely cited because the rest of the monograph depends on the proof's inputs, not only its conclusion. The first proof is the familiar Mason

Wronskian argument. The second is the Riemann–Hurwitz three-fibre count. The third is the logarithmic-differential form often used in S-unit and function-field treatments: the local orders of $df/[f(1-f)]$ reproduce the same ramification inequality without beginning from the global cover formula. Theorem 2.5.1 is the curve form of function-field *abc*, and its $2g-2$ is the canonical degree, not an error term hidden in asymptotic notation.

Attribution of the new material. No mathematical theorem in this chapter is claimed as new. The contribution of the volume is the comparative arrangement of three proofs and the explicit dependency audit in Criterion 2.4.1. The labels (F1)–(F4) and the hand-off (H1)–(H8) are editorial interfaces: they state which classical objects are consumed and how Chapter 3 will test an arithmetic replacement. Their necessity for this particular proof architecture follows line by line from the reproduced proofs; they are not asserted to classify all possible approaches to *abc*. The local example on the Legendre curve and the plate briefs were prepared during the machine-assisted comparative process described in Chapter 56, then checked against the divisor and ramification formulae by the author. Machine assistance did not supply an unproved bridge from the function field to the integers. The dependency audit was checked by removing each item in turn: without the base there is no relative derivation; without the derivation multiplicities persist; without the cotangent sheaf local formulae do not glue; and without degree or trace no numerical bound emerges. This is an audit of the reproduced argument, not evidence for a new arithmetic theorem.

Alternative treatments. The shortest presentation occupies half a page: differentiate, show $ABC/\operatorname{rad}(ABC)$ divides one Wronskian, and compare degrees. It is ideal for a first encounter but poor preparation for arithmetic transposition because it makes the base and cotangent sheaf invisible. At the opposite extreme, one can state a general S-unit theorem for function fields, including weighted places over a non-algebraically closed constant field and several summands. That generality clarifies applications but obscures the exact three-point geometry. A Nevanlinna-theoretic treatment replaces Riemann–Hurwitz by the Second Main Theorem and the logarithmic derivative lemma. It is conceptually close to Vojta's programme, but the present chapter needs the algebraic identity with no analytic error term. We therefore retain all three elementary forms and postpone Nevanlinna theory to the height-theoretic parts of the volume. A fourth treatment uses valuations throughout and states the result directly for S-units in $k(X)$. It is efficient in higher genus, but it suppresses the visible polynomial Wronskian that motivates the later search for an arithmetic operator. In positive characteristic one may also formulate separable or Hasse-derivative variants; those answer a different obstruction from the absolute-base problem pursued here.

Historical remarks. Stothers reached the polynomial inequality through work on polynomial identities and modular functions; Mason recognised and developed its Diophantine function-field significance. The name "Mason–Stothers theorem" now reflects those independent routes. The result became famous as the function-field *abc* theorem because it has the same formal shape as the Masser–Oesterlé conjecture and because it immediately constrains polynomial Fermat equations. The word "easy" in the chapter title is therefore relative. The calculation is short only after the theory of derivations, differentials, divisors, and ramification has been built. In positive characteristic, Frobenius shows exactly where that ease stops: simultaneous pth powers have zero derivative, and a separability correction is mandatory. This failure was not a peripheral exception; it revealed that the derivative is the theorem's active input. Priority is therefore best described as a convergence of viewpoints: Stothers supplied a ramification-rich polynomial setting, Mason made the Diophantine function-field theorem explicit, and later expositions compressed the argument into the Wronskian form now commonly taught. The

paired name records that convergence rather than a claim that the published proofs were identical.

Further reading. Gallardo [GAL20] gives a concise modern account of the polynomial theorem and examples. Mueller [Mue93] treats function-field *abc* in relation to generalised Fermat equations. Hartshorne [Har77] is the standard source for differentials, divisors, and Riemann–Hurwitz on curves, while Liu [LE02a] supplies an arithmetic-curve perspective that is especially useful before Chapter 3. Vojta [Voj11] explains why the reduced three-point count is the function-field prototype of a much wider Diophantine height inequality. Readers interested in computation should compare exact factorisation of ABC, local ramification partitions of A/C, and the Wronskian divisor on the same examples: agreement of those three certificates is the most useful practical check on conventions.

Chapter 3

The Absolute Obstruction: Why $\mathrm{Spec}\mathbb{Z}$ Has No Base

Dependencies. Chapter 2.

Summary. The four devices isolated in Chapter 2 all require a direction along a base. This chapter proves that ordinary arithmetic supplies no such direction: $\mathbb{Z}$ is initial among commutative rings, $\mathrm{Spec}\mathbb{Z}$ is terminal among schemes, and the cotangent complex of the identity base is zero. A no-go theorem shows that adding 1-categorical structure cannot alter this conclusion while the lift of $\mathbb{Z}$ remains initial. We then classify three possible escapes and extract eight requirements for the substrate constructed in Parts I–II. The requirements are a design contract, not a proof of any Diophantine conjecture.

3.1 Initiality and its consequences

Throughout, CRing denotes commutative unital rings and unit-preserving homomorphisms. This convention matters: without a unit, the zero map would destroy the universal property on which the argument rests. Write $\underline{\mathbb{Z}}$ when it is useful to regard the integers as an object rather than as a set. We index the notation and the obstruction at their first appearance.

Proposition 3.1.1 (The initial ring). *[REPROVED : LE02a] The usual universal-property statement is recalled with the tensor and geometric consequences needed below. For every $R \in \mathrm{CRing}$,*

$$\mathrm{Hom}_{\mathrm{CRing}}(\mathbb{Z}, R) = \{\iota_R\}, \qquad \iota_R(n) = n \cdot 1_R.$$

Consequently $\mathbb{Z}$ is initial in CRing *and is the unit for the symmetric monoidal product $\otimes_{\mathbb{Z}}$:*

$$\mathbb{Z} \otimes_{\mathbb{Z}} R \xrightarrow{\sim} R.$$

Proof. A unital homomorphism $f\colon \mathbb{Z} \to R$ must satisfy $f(1) = 1_R$. Additivity then forces $f(n) = n1_R$ for $n \geq 0$ and $f(-n) = -f(n)$. Thus $f = \iota_R$, and the displayed Hom-set is a singleton. The map $\mathbb{Z} \otimes_{\mathbb{Z}} R \to R$, $n \otimes r \mapsto nr$, is inverse to $r \mapsto 1 \otimes r$. These maps respect multiplication and units, proving the second assertion. This is the elementary affine foundation of the relative language of schemes [LE02a]. □

Counterexample 3.1.2 (The unit-preserving hypothesis is load-bearing). If morphisms are not required to preserve 1, a multiplicative additive map $f\colon \mathbb{Z} \to R$ is determined by the idempotent $e = f(1)$ and has the form $f(n) = ne$. For $R = \mathbb{Z} \times \mathbb{Z}$, the four idempotents

$$(0,0), \quad (1,0), \quad (0,1), \quad (1,1)$$

therefore give four distinct non-unital ring maps from $\mathbb{Z}$ to R. The integers are not initial in that category; the zero rng is.

This does not manufacture a useful arithmetic base. It discards the distinguished unit which makes every ordinary ring a $\mathbb{Z}$-algebra and hence changes the object being modelled.

The example explains why Proposition 3.1.1 states the morphism convention before invoking initiality.

Initiality has two distinct readings. Algebraically, every ring already contains its prime subring. Categorically, there is no choice in the map from $\mathbb{Z}$. The latter is stronger than the assertion that ordinary derivations of $\mathbb{Z}$ happen to vanish: it says that no parameter has entered the category in the first place. Adding more rings does not create a new morphism out of an object that remains initial.

Corollary 3.1.3 (The terminal arithmetic scheme). *[REPROVED: LE02a] The scheme* $\mathrm{Spec}\mathbb{Z}$ *is terminal in the category* Sch *of schemes. Equivalently, every scheme* X *has a unique structural morphism*

$$X \longrightarrow \mathrm{Spec}\mathbb{Z}.$$

Proof. For an affine scheme $X = \mathrm{Spec}R$, contravariance turns the unique homomorphism $\mathbb{Z} \to R$ of Proposition 3.1.1 into a unique morphism $\mathrm{Spec}R \to \mathrm{Spec}\mathbb{Z}$. On intersections of affine opens the maps agree because they are unique, so they glue. Conversely, any morphism $X \to \mathrm{Spec}\mathbb{Z}$ restricts to the stated affine morphism on every affine open of X and hence equals the glued map. □

The word *base* must now be used carefully. There are many non-canonical maps from $\mathrm{Spec}\mathbb{Z}$ to schemes: a section $\mathrm{Spec}\mathbb{Z} \to \mathbb{A}^1_{\mathbb{Z}}$ is one. Such an ambient scheme is not a base *beneath arithmetic* in the universal sense; it is extra geometry chosen over arithmetic. If a scheme B is required to have the same universal property as the absolute recipient of every scheme, then B is terminal and hence uniquely isomorphic to $\mathrm{Spec}\mathbb{Z}$. Thus a strictly smaller *absolute base*, retaining the universal role of a base, cannot be another ordinary scheme. It must either change the ambient geometry or relax the universal property.

This qualification prevents a common but consequential overstatement. The obstruction is not that $\mathrm{Spec}\mathbb{Z}$ admits no morphism to a non-trivial scheme. It is that ordinary scheme theory contains no canonical terminal layer below it. Chapter 2 needs precisely such a layer: a direction intrinsic to arithmetic, rather than a variable adjoined by hand.

3.2 The vanishing of $\Omega_{\mathbb{Z}}$ and $\mathbb{L}_{\mathbb{Z}}$

Let $A \to B$ be a ring homomorphism. The module of Kähler differentials $\Omega^1_{B/A}$ corepresents A-derivations from B, while the cotangent complex $\mathbb{L}_{B/A}$ records derived square-zero deformation theory. We use cohomological notation in which the conormal module of a regular quotient occurs in degree -1, denoted by a shift $[1]$.

Proposition 3.2.1 (Identity bases have no differential direction). *[REPROVED: GR17a] For every commutative ring* A*,*

$$\Omega^1_{A/A} = 0, \qquad \mathbb{L}_{A/A} \simeq 0.$$

In particular,

$$\Omega^1_{\mathbb{Z}/\mathbb{Z}} = 0, \qquad \mathbb{L}_{\mathbb{Z}/\mathbb{Z}} \simeq 0, \qquad \mathbb{L}_{\mathbb{Z}_p/\mathbb{Z}_p} \simeq 0.$$

Proof. An A-derivation $d\colon A \to M$ is required to vanish on the image of the base ring A. Since that image is all of the source, $d = 0$. The universal derivation is therefore the zero map into the zero module, proving $\Omega^1_{A/A} = 0$.

For the derived assertion, take the constant simplicial A-algebra resolution of A over itself. Applying relative Kähler differentials degreewise gives the zero simplicial module, whose normalisation is zero. Equivalently, the identity morphism has a unique lift through every

square-zero extension over itself, in every derived degree, so its representing cotangent object is zero. Setting $A = \mathbb{Z}$ and $A = \mathbb{Z}_p$ gives the displayed special cases. The same proof works for the continuous or derived p-complete cotangent theory because the map is still the identity. This is the normalisation axiom for cotangent complexes [GR17a]. □

Proposition 3.2.2 (The ordinary derivation calculation). *[REPROVED : GR17a] For every $\mathbb{Z}$-module M,*

$$\mathrm{Der}(\mathbb{Z}, M) = 0.$$

This remains true even if "derivation" is initially read without specifying a base ring.

Proof. The Leibniz rule gives $d(1) = d(1 \cdot 1) = 2d(1)$, hence $d(1) = 0$. Additivity then gives $d(n) = nd(1) = 0$ for positive n and $d(-n) = -d(n) = 0$. Thus every derivation vanishes. Equivalently, the prime subring is contained in the constants of every derivation on a unital ring. □

The calculation is stronger than the absence of a preferred derivation: there is no non-zero ordinary derivation to choose. A later arithmetic operator must therefore change either the source object or the Leibniz law, not merely select a distinguished point from a non-empty family.

Proposition 3.2.3 (Every square-zero extension of $\mathbb{Z}$ splits uniquely). *[NEW] Let*

$$0 \longrightarrow M \longrightarrow E \xrightarrow{\pi} \mathbb{Z} \longrightarrow 0$$

be an extension of unital commutative rings with $M^2 = 0$. There is a unique unital ring section $s\colon \mathbb{Z} \to E$. Under the resulting decomposition $E \simeq \mathbb{Z} \oplus M$, multiplication is

$$(n, m)(n', m') = (nn',\ nm' + n'm). \tag{3.2.1}$$

Proof. Initiality supplies the unique unital map $s(n) = n1_E$. Since $\pi(1_E) = 1$, one has $\pi s = \mathrm{id}_{\mathbb{Z}}$. Every $e \in E$ decomposes uniquely as $s(\pi(e)) + (e - s(\pi(e)))$, with the second term in M. Expanding a product and using $M^2 = 0$ gives (3.2.1). No second section can exist because there is only one unital map out of $\mathbb{Z}$. □

Thus the vanishing of the cotangent complex is visible without resolutions: there are no non-trivial square-zero extension classes and no choices of splitting. Derived language packages this rigidity in a form stable under later base change and descent.

The p-adic statement is deliberately included. Completion changes the topology and the available limits, but $\mathbb{Z}_p$ relative to itself still has no base direction. Merely replacing $\mathbb{Z}$ by one of its completions does not solve the problem.

The first useful contrast appears only after the base is made strictly smaller than the object being studied.

Proposition 3.2.4 (The residue field remembers the quotient). *[REPROVED : GR17a] For every prime p,*

$$\Omega^1_{\mathbb{F}_p/\mathbb{Z}} = 0, \qquad \mathbb{L}_{\mathbb{F}_p/\mathbb{Z}} \simeq \mathbb{F}_p[1].$$

Proof. Write $I = (p) \subset \mathbb{Z}$ and $B = \mathbb{Z}/I = \mathbb{F}_p$. The conormal presentation for a quotient gives the two-term model

$$\mathbb{L}_{B/\mathbb{Z}} \simeq [\, I/I^2 \longrightarrow \Omega^1_{\mathbb{Z}/\mathbb{Z}} \otimes_{\mathbb{Z}} B \,],$$

with I/I^2 in degree -1. The second term is zero by Proposition 3.2.1. Multiplication by the class of p gives an isomorphism $B \xrightarrow{\sim} I/I^2$: its kernel is zero because $ap \in (p^2)$ precisely when $a \in (p)$. Hence $\mathbb{L}_{B/\mathbb{Z}} \simeq B[1]$. Its degree-zero homology is zero, which recovers $\Omega^1_{B/\mathbb{Z}} = 0$. □

The shift is not cosmetic. Ordinary differentials miss the operation of killing p, whereas the cotangent complex records it as a derived normal direction. This is the first model for the strategy used later: retain $\mathbb{Z}$ as the arithmetic object, but place it relative to something different.

Counterexample 3.2.5 (*The residue-field extension does not split*). The reduction

$$\mathbb{Z}/p^2\mathbb{Z} \longrightarrow \mathbb{F}_p$$

has square-zero kernel $p\mathbb{Z}/p^2\mathbb{Z} \simeq \mathbb{F}_p$, but it has no unital ring section. Indeed, a section would send 1 to 1; applying it to the identity $p \cdot 1 = 0$ in $\mathbb{F}_p$ would force $p = 0$ in $\mathbb{Z}/p^2\mathbb{Z}$, a contradiction. This unsplit extension is the deformation detected by $\mathbb{L}_{\mathbb{F}_p/\mathbb{Z}} \simeq \mathbb{F}_p[1]$. Ordinary Kähler differentials miss it because they retain only degree zero.

Proposition 3.2.6 (Base-changing the identity is not taking a quotient). *[NEW] The three cotangent complexes*

$$\mathbb{L}_{\mathbb{Z}/\mathbb{Z}} \otimes^{\mathbf{L}}_{\mathbb{Z}} \mathbb{F}_p, \qquad \mathbb{L}_{\mathbb{F}_p/\mathbb{F}_p}, \qquad \mathbb{L}_{\mathbb{F}_p/\mathbb{Z}}$$

are respectively

$$0, \qquad 0, \qquad \mathbb{F}_p[1].$$

The non-zero third object is not obtained by base-changing the identity map; it belongs to the different morphism $\mathbb{Z} \to \mathbb{F}_p$.

Proof. The first two vanish by identity normalisation and exact pullback of the zero object. The third is Proposition 3.2.4. Cotangent complexes are attached to arrows, so changing only the coefficient ring of an identity arrow and replacing the arrow by a quotient are distinct operations. □

Proposition 3.2.7 (All principal arithmetic quotients). *[REPROVED: GR17a] Let $n \geq 2$ and put $B = \mathbb{Z}/n\mathbb{Z}$. Then*

$$\Omega^1_{B/\mathbb{Z}} = 0, \qquad \mathbb{L}_{B/\mathbb{Z}} \simeq B[1]. \tag{3.2.2}$$

For $n = p^r$, the only non-zero homology module has $\mathbb{Z}_p$-length r.

Proof. The element n is a non-zero-divisor in $\mathbb{Z}$, so the quotient is regular of codimension one. Its conormal module $(n)/(n^2)$ is freely generated over B by the class of n, giving (3.2.2). When $n = p^r$, the module B has the filtration

$$0 \subset (p^{r-1}) \subset \cdots \subset (p) \subset B$$

with r successive quotients $\mathbb{F}_p$. □

Thus the cotangent complex distinguishes the identity map from every non-trivial quotient and records the thickness of a prime-power fibre by length. It still does not make $\mathbb{Z}$ relative to a lower ordinary base: the direction belongs to the quotient map $\mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$, not to $\mathbb{Z} \to \mathbb{Z}$.

Proposition 3.2.8 (An adjoined parameter and its special fibre). *[REPROVED: GR17a] Let $A = \mathbb{Z}[t]$. Then*

$$\mathbb{L}_{A/\mathbb{Z}} \simeq A\,dt \tag{3.2.3}$$

in degree zero. For any integer a, the specialisation $s_a\colon A \to \mathbb{Z}$, $t \mapsto a$, has

$$\mathbb{L}_{\mathbb{Z}/A} \simeq \mathbb{Z}[1], \tag{3.2.4}$$

generated by the conormal class of $t - a$.

Proof. The polynomial algebra is smooth of relative dimension one, so its cotangent complex is the free module generated by dt. The kernel of s_a is the principal regular ideal $(t - a)$; the regular-quotient calculation identifies its conormal module with $\mathbb{Z}$ in degree -1. □

Corollary 3.2.9 (The parameter is the entire derivation datum). *[REPROVED: GR17a] For every $\mathbb{Z}[t]$-module M, evaluation at t gives an isomorphism*

$$\mathrm{Der}_{\mathbb{Z}}(\mathbb{Z}[t], M) \xrightarrow{\sim} M, \qquad d \longmapsto d(t).$$

Proof. Given $m \in M$, the formula $d_m(f) = f'(t)m$ defines a $\mathbb{Z}$-derivation and sends t to m. Conversely, the Leibniz rule gives $d(t^r) = rt^{r-1}d(t)$, so a derivation is determined by that one value. □

Taking $M = \mathbb{Z}[t]$ and $d(t) = 1$ recovers ∂_t. Taking any other m rescales or twists that direction by a coefficient. The family therefore carries an entire module of derivations, whereas Proposition 3.2.2 says the arithmetic fibre carries none. Specialising t does not select one of these ambient derivations or prove that its value is independent of a lift. That distinction is structurally decisive.

This is a genuine direction, but it was adjoined by hand. Different choices of coordinate and section change its presentation. In particular, putting $t = q$ and $a = 1$ gives

$$\mathbb{L}_{\mathbb{Z}/\mathbb{Z}[q]} \simeq \mathbb{Z}[1]$$

with conormal generator $q - 1$, the elementary normal direction underlying a q-deformation. The conormal class alone does not construct a global q-connection; it records the parameter whose divided difference later uses.

Proposition 3.2.10 (Transitivity for the $q = 1$ fibre). *[NEW] For*

$$\mathbb{Z} \longrightarrow \mathbb{Z}[q] \xrightarrow{q\mapsto 1} \mathbb{Z},$$

the transitivity triangle is

$$\mathbb{Z}\,dq \longrightarrow 0 \longrightarrow \mathbb{Z}[1] \xrightarrow{\sim} (\mathbb{Z}\,dq)[1]. \tag{3.2.5}$$

Thus the family has a degree-zero parameter direction, the arithmetic fibre has a degree-one conormal direction inside that family, and the absolute cotangent complex of the fibre over $\mathbb{Z}$ remains zero.

Proof. Base change of $\mathbb{L}_{\mathbb{Z}[q]/\mathbb{Z}} \simeq \mathbb{Z}[q]\,dq$ along $q \mapsto 1$ gives $\mathbb{Z}\,dq$. The middle term is $\mathbb{L}_{\mathbb{Z}/\mathbb{Z}} = 0$, and Proposition 3.2.8 gives the third term $\mathbb{Z}[1]$. The connecting map is therefore the canonical equivalence required to make the triangle a cofiber sequence. □

Equation (3.2.5) prevents two opposite mistakes. The normal class at $q = 1$ is not zero merely because the absolute fibre has zero cotangent, but neither is it an ordinary derivation of the constants. It is the boundary created by embedding the constants in a chosen family.

The completed analogue makes the same distinction. For $A = \mathbb{Z}_p[[t]]$, the continuous cotangent module over $\mathbb{Z}_p$ is the free complete module $A\,dt$, and the fibre $t = 0$ has $\mathbb{L}_{\mathbb{Z}_p/A} \simeq \mathbb{Z}_p[1]$. Yet $\mathbb{L}_{\mathbb{Z}_p/\mathbb{Z}_p} = 0$. Completion preserves the direction supplied by t; it does not create one when no parameter is present.

Counterexample 3.2.11 (*A chosen family does not canonically differentiate its fibre*). The derivation ∂_t on $\mathbb{Z}[t]$ restricts to zero on the constant subring $\mathbb{Z}$. If one first chooses lifts of an integer to the family, the answer depends on that choice: the two lifts

$$2, \qquad 2 + t$$

have the same specialisation at $t = 0$, but their derivatives specialise to 0 and 1. Hence an ambient derivative does not induce a derivative on the fibre unless a lift or connection is supplied and its independence proved.

There is also no invariant orientation of the normal line from the bare family. The automorphism $t \mapsto -t$ fixes the section $t = 0$ and sends the conormal generator t to $-t$. A choice of generator therefore breaks a symmetry. Later descent and trace constructions must account for this automorphism rather than silently identify the two orientations.

Counterexample 3.2.12 (*A p-derivation is not an ordinary derivation*). Fix a prime p. The identity Frobenius lift on $\mathbb{Z}$ defines

$$\delta_p(n) = \frac{n - n^p}{p},$$

which is integral by Fermat's congruence. For $p = 3$,

$$\delta_3(1) = 0, \qquad \delta_3(2) = -2.$$

Hence $\delta_3(1 + 1) \neq \delta_3(1) + \delta_3(1)$. The operator satisfies the nonlinear p-derivation identities because $n \mapsto n^p + p\delta_p(n) = n$ is a ring endomorphism, but it is not a point of $\mathrm{Der}(\mathbb{Z}, \mathbb{Z})$. This is why the δ-ring escape route does not contradict Proposition 3.2.2; it changes the Leibniz and additivity laws [Bui05].

The corresponding Kodaira–Spencer warning is equally precise. If a Kodaira–Spencer construction is fed the tangent complex of the identity base $\mathrm{Spec}\mathbb{Z} \to \mathrm{Spec}\mathbb{Z}$, then its base-direction source is zero. This does *not* say that $\mathbb{L}_{X/\mathbb{Z}}$ vanishes for an arithmetic scheme X, nor that arithmetic schemes have no deformation theory. It says that there is no deformation *of the arithmetic base along itself*.

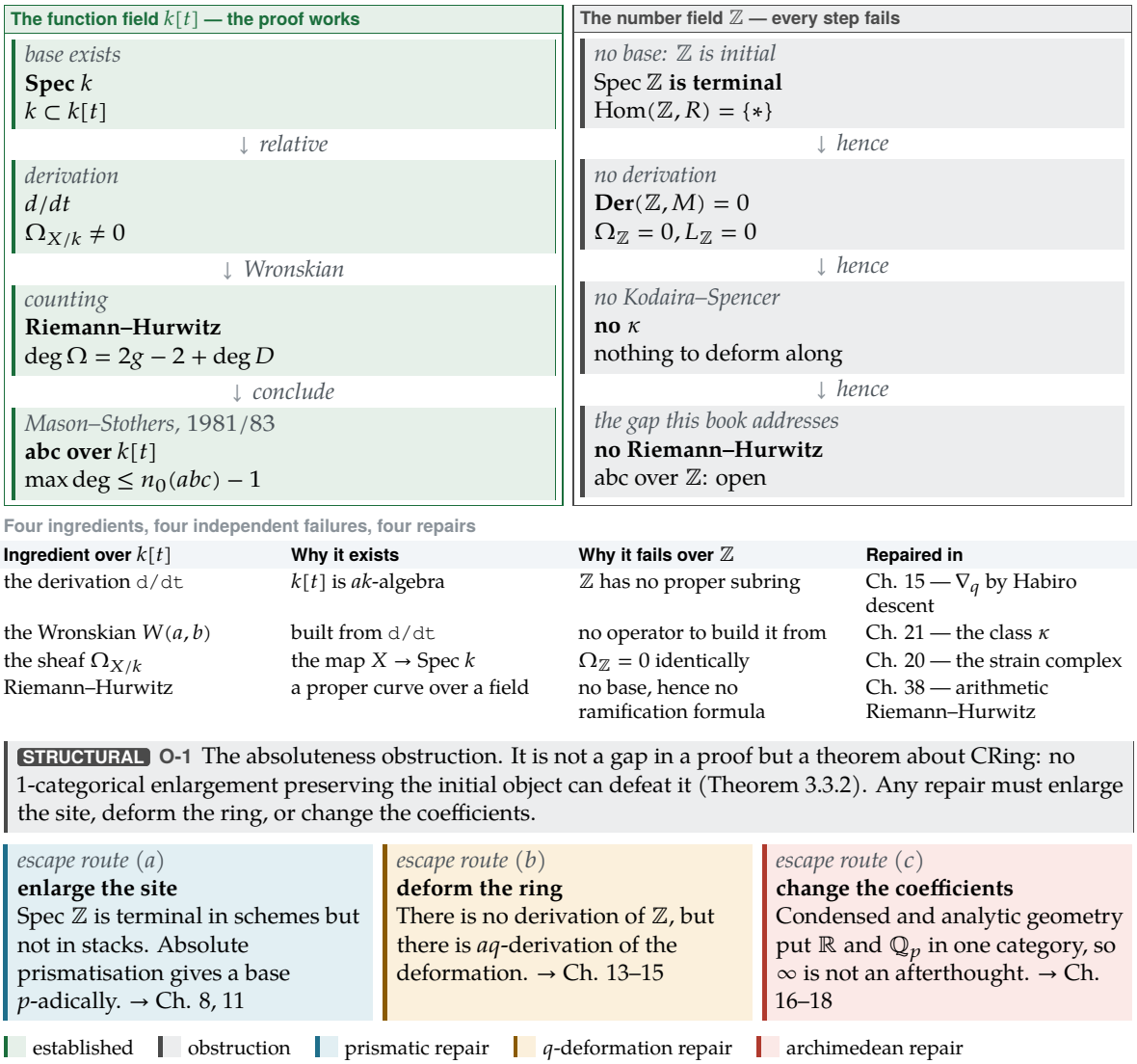
Example 3.2.13 (Non-zero geometry over a zero base direction). Let

$$B = \mathbb{Z}[x]/(x^2 - 2).$$

Then

$$\mathbb{L}_{B/\mathbb{Z}} \simeq \left[B \cdot (x^2 - 2) \xrightarrow{2x} B\,dx \right]. \tag{3.2.6}$$

After inverting 2, the element $2x$ is a unit and the complex vanishes, reflecting that $\mathbb{Q}(\sqrt{2})/\mathbb{Q}$ is separable. Modulo 2, the derivative vanishes and the special fibre acquires a non-zero cotangent complex. Thus the arithmetic scheme has ramification and deformation theory



Four ingredients, four independent failures, four repairs

Ingredient over $k[t]$	Why it exists	Why it fails over $\mathbb{Z}$	Repaired in
the derivation `d/dt`	$k[t]$ is ak-algebra	$\mathbb{Z}$ has no proper subring	Ch. 15 — ∇_q by Habiro descent
the Wronskian $W(a, b)$	built from `d/dt`	no operator to build it from	Ch. 21 — the class κ
the sheaf $\Omega_{X/k}$	the map $X \to \mathrm{Spec}\, k$	$\Omega_{\mathbb{Z}} = 0$ identically	Ch. 20 — the strain complex
Riemann–Hurwitz	a proper curve over a field	no base, hence no ramification formula	Ch. 38 — arithmetic Riemann–Hurwitz

STRUCTURAL **O-1** The absoluteness obstruction. It is not a gap in a proof but a theorem about CRing: no 1-categorical enlargement preserving the initial object can defeat it (Theorem 3.3.2). Any repair must enlarge the site, deform the ring, or change the coefficients.

escape route (a)
enlarge the site
Spec $\mathbb{Z}$ is terminal in schemes but not in stacks. Absolute prismatisation gives a base p-adically. → Ch. 8, 11

escape route (b)
deform the ring
There is no derivation of $\mathbb{Z}$, but there is aq-derivation of the deformation. → Ch. 13–15

escape route (c)
change the coefficients
Condensed and analytic geometry put $\mathbb{R}$ and $\mathbb{Q}_p$ in one category, so ∞ is not an afterthought. → Ch. 16–18

established · obstruction · prismatic repair · q-deformation repair · archimedean repair

Plate 3.1. Initiality, dual terminality, and the first non-zero relative cotangent direction.

What it does not show: a gap in a proof that a better argument could close: the absoluteness obstruction is a theorem about CRing, and no 1-categorical enlargement preserving the initial object can defeat it (Theorem 3.3.1).

relative to $\mathbb{Z}$ even though $\mathbb{L}_{\mathbb{Z}/\mathbb{Z}} = 0$. The missing object is the tangent of the base itself, not all relative geometry above it.

Likewise, Riemann–Hurwitz remains valid for finite maps of curves over a field, and discriminant formulae exist in arithmetic geometry. What is absent is the Chapter 2 configuration in which Spec$\mathbb{Z}$ itself is a relative curve over a lower base and ramification is measured by a relative cotangent sequence. The slogan "there is no Riemann–Hurwitz" always means this specific missing relative mechanism.

Finally, the expression $\Omega^1_{\mathbb{Z}/\mathbb{F}_1}$ is presently not an object of ordinary commutative algebra. There is no commutative ring $\mathbb{F}_1$ and no ring map $\mathbb{F}_1 \to \mathbb{Z}$ to which the usual universal property can be applied. Monoids, blueprints, Λ-rings, and tropical semirings supply different candidate categories, with different notions of modules and square-zero extensions. Until one specifies such a category and proves that it carries a cotangent theory, writing $\Omega^1_{\mathbb{Z}/\mathbb{F}_1}$ conceals rather than solves the obstruction.

3.3 A no-go theorem for 1-categorical repairs

We now isolate exactly what can and cannot be changed. A bare 1-category does not possess a cotangent complex, so the hypotheses must include a cotangent theory rather than smuggle one into the conclusion.

Definition 3.3.1 (A 1-categorical algebraic repair). A 1-*categorical algebraic repair of* CRing consists of:

(R1) a locally small 1-category C and a faithful functor $U\colon C \to \mathrm{CRing}$;

(R2) an initial object $\underline{\mathbb{Z}}_C$ such that $U(\underline{\mathbb{Z}}_C) = \mathbb{Z}$, with its canonical ring structure;

(R3) for every arrow $A \to B$ in C, a relative cotangent object $\mathbb{L}^C_{B/A}$ in a stable tangent category $\mathcal{T}_B$;

(R4) identity normalisation $\mathbb{L}^C_{A/A} \simeq 0$, and a square-zero interpretation in which maps from $\mathbb{L}^C_{B/A}$ corepresent the derived space of A-linear sections of split square-zero extensions of B.

The *absolute* cotangent object in the repair is, by definition,

$$\mathbb{L}^C_B := \mathbb{L}^C_{B/\underline{\mathbb{Z}}_C}.$$

The faithfulness of U ensures that the repair has retained the underlying ring and its maps. Identity normalisation is not optional: a theory that assigns a non-zero cotangent object to an identity arrow is not a cotangent-complex theory in the usual deformation-theoretic sense.

Obstruction 3.3.2 (No-go theorem for initial lifts). [NEW] *Let $(C, U, \underline{\mathbb{Z}}_C, \mathbb{L}^C)$ be a 1-categorical algebraic repair in the sense of Definition 3.3.1. Then*

$$\mathbb{L}^C_{\underline{\mathbb{Z}}_C} \simeq 0.$$

Moreover, for every tangent coefficient M, the set of C-derivations of $\underline{\mathbb{Z}}_C$ into the split square-zero extension by M is a singleton.

Proof. Because $\underline{\mathbb{Z}}_C$ is initial, its structure map from the initial object of C is the identity. Therefore

$$\mathbb{L}^C_{\underline{\mathbb{Z}}_C} = \mathbb{L}^C_{\underline{\mathbb{Z}}_C/\underline{\mathbb{Z}}_C} \simeq 0$$

by identity normalisation.

For the second assertion, let $\pi\colon \underline{\mathbb{Z}}_C \oplus M \to \underline{\mathbb{Z}}_C$ be a split square-zero extension. Initiality gives exactly one morphism $s\colon \underline{\mathbb{Z}}_C \to \underline{\mathbb{Z}}_C \oplus M$. The composite πs is an endomorphism of the initial object, hence is the unique endomorphism, namely the identity. Thus s is a section and is the only section. Under the square-zero interpretation, it is the zero derivation. □

Counterexample 3.3.3 (*Dropping identity normalisation only renames the problem*). Fix a non-zero module V and declare, by fiat, that the "cotangent object" of every identity map is V. This produces a non-zero symbol at $\mathbb{Z} \to \mathbb{Z}$, but it cannot corepresent deformation theory. The identity map has a unique section through every split square-zero extension over itself, whereas maps out of V generally form a non-trivial set or space. The proposed object therefore predicts deformations which do not exist. Identity normalisation in Definition 3.3.1 is the condition excluding this tautological evasion.

The proof shows why the theorem is stable under a wide class of apparent repairs. One may add an endomorphism, a filtration, a topology, a grading, chosen Frobenius lifts, or

descent data. If the resulting object over $\mathbb{Z}$ is still initial and the cotangent object is still measured relative to that initial object, the answer remains zero.

Corollary 3.3.4 (Stability under added algebraic structure). *[NEW] Suppose a category of structured rings admits a faithful forgetful functor to* CRing*, the canonical structured $\mathbb{Z}$ is initial, and its cotangent theory satisfies identity normalisation. Then no choice-free derivation of the initial structured $\mathbb{Z}$ is produced internally by that category.*

Proof. Apply Obstruction 3.3.2. A choice-free internal derivation would determine a non-zero point in the represented derivation functor, whereas that functor is terminal at the initial object. □

Proposition 3.3.5 (Two ways of adding structure). *[NEW] Consider the following two categories.*
(a) *In the category of rings equipped with a chosen endomorphism, the initial object is $(\mathbb{Z}, \mathrm{id})$.*
(b) *In the category of rings equipped with a chosen element, the initial object is $(\mathbb{Z}[t], t)$.*
Thus an endomorphism adds structure while retaining the initial arithmetic ring, whereas a freely chosen element changes the initial object to a genuine one-parameter family.

Proof. Every unital ring endomorphism fixes the prime subring, so the unique map $\mathbb{Z} \to R$ commutes with $\mathrm{id}_{\mathbb{Z}}$ and the chosen endomorphism of R. This proves (a). For (b), a morphism $(\mathbb{Z}[t], t) \to (R, r)$ is uniquely determined by sending t to r; the universal property of the polynomial ring gives existence and uniqueness. □

Part (a) lies inside Obstruction 3.3.2: the absolute cotangent of the initial structured object remains the cotangent of an identity. Part (b) escapes because the initial object is no longer $\mathbb{Z}$; its relative cotangent over $\mathbb{Z}$ is the line computed in Proposition 3.2.8. The comparison makes precise the difference between decorating arithmetic and adjoining a direction.

Proposition 3.3.6 (Higher-categorical initiality is equally rigid). *[REPROVED : Lur09a] Let C be an ∞-category with initial object I. For every arrow $E \to I$, the space of sections $I \to E$ is contractible. Consequently passing from sets of morphisms to mapping spaces does not create an infinitesimal direction at I; any cotangent theory with identity normalisation still has $\mathbb{L}_{I/I} \simeq 0$.*

Proof. Initiality says that both $\mathrm{Map}_C(I, E)$ and $\mathrm{Map}_C(I, I)$ are contractible. The section space is the homotopy fibre over id_I of the map between these two mapping spaces. A map between contractible spaces is an equivalence, so this fibre is contractible. The cotangent assertion is identity normalisation. □

Thus higher categories are relevant only when they change the arithmetic coefficient object, the ambient moduli problem, or the tangent theory. Higher homotopies attached to an object which remains initial do not by themselves evade the obstruction.

Criterion 3.3.7 (A no-go audit for a proposed absolute base). *[NEW] To test a proposed repair before using it downstream, perform the following four checks.*
(1) *Identify its initial coefficient object and compute its underlying ordinary ring.*
(2) *Write the actual morphism whose cotangent object is being called "absolute"; determine whether it is an identity, quotient, specialisation, or map to a moduli stack.*
(3) *Exhibit the represented square-zero extension problem and verify identity normalisation.*
(4) *If a generator or operator is chosen, compute its transformation under automorphisms of the presentation.*

If (1)–(3) return the initial lift of $\mathbb{Z}$, its identity map, and an ordinary normalised cotangent theory, then the proposed absolute cotangent object is zero. A non-zero answer is admissible only after the calculation identifies which input changed.

Proof. The zero conclusion is Obstruction 3.3.2, or Proposition 3.3.6 in an ∞-category. The quotient and family computations of Propositions 3.2.7 and 3.2.8 show how a legitimate non-zero answer records the changed morphism. The sign change in Counterexample 3.2.11 explains the final automorphism check. □

Two limitations should be recorded. First, the theorem does not say that every structured category is useless. A δ-ring can supply a Frobenius-lift operator, and a Λ-ring can organise such operators for many primes. It says only that their *absolute internal cotangent complex at an initial object* is still zero. Their useful directions arise from a non-trivial family, a special fibre, or a comparison with another coefficient object. Second, merely passing to presheaves by Yoneda does not help: Yoneda preserves terminal objects. A successful stack must contain new moduli or automorphisms for which the image of $\mathrm{Spec}\mathbb{Z}$ is not terminal.

3.4 Three escape routes: enlarge the site, deform the ring, change the coefficients

Principle 3.4.1 (The escape trichotomy). *[NEW] Any construction that produces a non-zero intrinsic arithmetic base direction must violate at least one hypothesis of Obstruction 3.3.2. Up to the role played by the violated hypothesis, there are three routes:*

(a) *change the geometric ambient category so that* $\mathrm{Spec}\mathbb{Z}$ *is no longer terminal;*
(b) *replace the isolated ring $\mathbb{Z}$ by a non-constant deformation or family;*
(c) *change the coefficient and tangent category so that ordinary faithful 1-categorical ring-valued deformation theory no longer controls the cotangent object.*

These routes are necessary alternatives, not individually sufficient constructions.

Proof. If none of (a)–(c) occurs, the arithmetic object is still the initial lift of $\mathbb{Z}$ in a faithful 1-categorical enlargement, and its tangent theory has the usual identity normalisation. The hypotheses of Obstruction 3.3.2 then hold, so the cotangent object is zero. Taking the contrapositive gives the trichotomy. The final sentence follows because violating a necessary hypothesis does not by itself construct a global cotangent complex, trace, or height theory. □

Route (a): enlarge the site. One seeks a site or stack $\mathfrak{A}$ and a morphism

$$\mathrm{Spec}\mathbb{Z} \longrightarrow \mathfrak{A}$$

which is not an equivalence and for which the image of $\mathrm{Spec}\mathbb{Z}$ is not terminal in the relevant geometric category. The additional objects must encode actual arithmetic moduli, not merely presheaves on the old category. Absolute prismatisation and the Cartier–Witt stack, developed in Chapters 8 and 11, are the principal instances. Prismatic geometry supplies moduli of Cartier–Witt divisors and Frobenius-compatible thickenings [BS22]. What this route buys is a geometric source for relative cotangent complexes and descent. Its cost is stackiness: automorphisms, effectivity of descent, and compatibility across different primes must all be proved.

Route (b): deform the ring. Fixing a prime p, a δ-structure replaces a missing ordinary derivation by a Frobenius lift $\phi(x) = x^p + p\delta(x)$. A q-deformation instead places arithmetic in a family with special value $q = 1$ and a difference or q-derivation transverse to that fibre. The relevant object is no longer the identity $\mathbb{Z} \to \mathbb{Z}$, but a diagram $A \to \mathbb{Z}$ with a genuine kernel, parameter, or specialisation. Chapters 6 and 13–15 develop these mechanisms. This route buys a calculable operator and arithmetic jets. It costs uniformity: a p-derivation is initially one-prime-at-a-time, while a q-family requires convergence, specialisation, and descent before it can serve all places simultaneously.

Route (c): change the coefficients. In spectral algebra the sphere spectrum is initial, whereas $H\mathbb{Z}$ is not; the relative object $\mathbb{L}_{H\mathbb{Z}/\mathbb{S}}$ is therefore not constrained by the initiality of $\mathbb{Z}$ in ordinary rings. In condensed and analytic geometry, global sections are not a faithful description of the object, and topology at the archimedean place becomes categorical data. Chapters 16–17 use this route. Higher categories by themselves do not help: an initial object still has a contractible mapping space. The gain occurs only when the arithmetic coefficient object ceases to be initial or when the forgetful functor loses the faithfulness assumed in Obstruction 3.3.2. The benefit is access to homotopies, duality, and the infinite place; the cost is a substantially more delicate notion of module, completion, and trace [Lur17; Sch22].

Criterion 3.4.2 (What constitutes an escape certificate). *[NEW] A proposed escape from Obstruction 3.3.2 is mathematically checkable only if it supplies all four items below.*

(i) *The new ambient object and the precise morphism replacing the identity $\mathbb{Z} \to \mathbb{Z}$.*
(ii) *A computation of the resulting relative cotangent or difference direction on at least one explicit chart.*
(iii) *The transformation law under changes of parameter, framing, or presentation, together with the descent statement needed to remove those choices.*
(iv) *A statement of which later operations—trace, norm, positivity, or all-place gluing—remain conditional.*

A name for an absolute geometry, by itself, satisfies none of these items.

Proof. Item (i) identifies the hypothesis of the no-go theorem which has changed. Proposition 3.2.8 and Counterexample 3.4.3 show that item (ii) cannot be inferred merely from non-equivalence of objects. The sign change in Counterexample 3.2.11 proves the need for (iii). Finally, a cotangent direction alone supplies none of the six-operation or numerical conclusions in (iv), as the independent gates of Criterion 3.5.1 record. □

Counterexample 3.4.3 (A non-trivial target can still have zero cotangent). Let $A = \mathbb{Z} \times \mathbb{Z}$ and let $A \to \mathbb{Z}$ be the first projection. The induced morphism

$$i\colon \mathrm{Spec}\mathbb{Z} \longrightarrow \mathrm{Spec}A$$

is not an equivalence: the target is the disjoint union of two copies of Spec$\mathbb{Z}$. Nevertheless $\mathbb{Z}$ is an idempotent direct summand of A, so the projection is finite étale and

$$\mathbb{L}_{\mathbb{Z}/A} \simeq 0.$$

Thus enlarging the target and satisfying the non-equivalence requirement does not by itself create strain. Route (a) must produce a morphism with a non-zero relative tangent direction; a disconnected decorative component is insufficient.

central no-go theorem
$L_{\mathbb{Z}/\mathbb{Z}} \simeq 0$
If structured $\mathbb{Z}$ remains initial and the cotangent theory is identity-normalised, its intrinsic derivation space is still contractible.

↓ *a successful repair must violate at least one hypothesis*

ROUTE A · ENLARGE THE SITE

new arithmetic moduli
$\operatorname{Spec}\mathbb{Z} \to \mathfrak{A}$
not an equivalence

↓ *realised p-adically by*

prismatisation and WCart
Cartier–Witt divisors and Frobenius-compatible thickenings.

gain
relative cotangent geometry
Moduli and descent become available.

cost
stacky effectivity
Automorphisms, descent and compatibility across primes must be proved.

ROUTE B · DEFORM THE RING

replace an identity by a family
$A \to \mathbb{Z}$
$\phi(x) = x^p + p\delta(x), q \to 1$

↓ *arithmetic and q-jets*

$\delta-$ and $q-$ calculi
Explicit twisted Leibniz laws replace the absent ordinary derivation.

gain
calculable operator
A transverse direction detects arithmetic variation.

cost
uniformity and descent
$A\delta$-operator is one-prime-at-a-time; aq-family needs convergence and specialisation.

ROUTE C · CHANGE COEFFICIENTS

leave faithful ring-valued deformation theory
spectral and condensed coefficients
$S \to H\mathbb{Z}, \mathbb{R}$ and $\mathbb{Q}_p$

↓ *higher and analytic structure*

modules beyond global sections
Topology at infinity becomes categorical data.

gain
homotopy, duality, infinity
The archimedean place enters the same formalism.

cost
delicate completion and trace
Nuclearity, finiteness and dualisability become non-trivial.

OPERATIONS THAT DO NOT LEAVE THE OBSTRUCTION

Yoneda alone
preserves terminal objects

↓ ↩

completion alone
retains the identity base

↓ ↩

added rigid structure
still zero if structured $\mathbb{Z}$ is initial

↓ ↩

$L_{\mathbb{Z}/\mathbb{Z}} = 0$
the loop closes

NECESSARY, NOT SUFFICIENT **TRICHOTOMY** Breaking *a* no-go hypothesis creates room for a direction. It does not by itself construct an all-place cotangent complex, a trace, or a height theorem.

geometric exit · deformation exit · coefficient exit · obstruction or cost · chapter-level trichotomy

Plate 3.2. The three ways to leave the initial-object obstruction, with their gains and costs.

What it does not show: a construction: breaking a no-go hypothesis creates room for a direction, but does not by itself construct an all-place cotangent complex, a trace, or a height theorem.

3.5 The design specification

The trichotomy determines what the rest of the book must construct. We now state the contract without claiming that satisfying it proves the abc conjecture. The symbol $\mathfrak{S}$ is reserved for the global substrate of Chapter 18.

Criterion 3.5.1 (Mason–Stothers transposition requirements). *[NEW] For the proof architecture of Chapter 2 to admit an arithmetic transposition, the proposed substrate $\mathfrak{S}$ must meet all eight requirements below.*

(D1) Non-trivial relativity. *There is a morphism $i\colon \mathrm{Spec}\mathbb{Z} \to \mathfrak{S}$ which is not an equivalence. The object $\mathfrak{S}$ is not merely* Spec$\mathbb{Z}$ *presented in a larger notation.*

(D2) A non-zero, controlled cotangent direction. *The strain complex*

$$\mathbb{L}_{\mathbb{Z}/\mathfrak{S}}$$

is non-zero and belongs to a coefficient category in which support, base change, duality, and a usable finiteness condition can be formulated.

(D3) A derivation-like operator. *There is an operator, ultimately denoted ∇_q, satisfying an explicit Leibniz or twisted-Leibniz rule, functorial enough to form an arithmetic Wronskian or its categorical replacement.*

(D4) Simultaneity of places. *The construction treats every finite prime and the archimedean place in one object. A product of unrelated p-adic answers is insufficient unless a proved descent datum glues them.*

(D5) Six operations and duality. *The relevant categories over $\mathfrak{S}$ carry pullback, pushforward, exceptional pullback, exceptional pushforward, tensor product, and internal Hom, together with the base-change and projection identities needed for a relative Riemann–Hurwitz argument.*

(D6) A choice-independent trace. *Dualisable objects admit a categorical trace* Tr*, invariant under the groupoid of gluing choices. Numerical comparison must not depend on choosing an identification between two copies of the coefficient line.*

(D7) Compatibility with Diophantine size. *The support and norm of the resulting deformation classes recover conductor terms at bad primes, while their trace recovers logarithmic height or discriminant terms. Positivity and normalisation must be stated, not inferred from analogy.*

(D8) The function-field limit. *After replacing the arithmetic base by a smooth curve over a field, the substrate direction becomes $\Omega^1_{X/k}$, the operator becomes the usual derivation, the trace formula becomes Riemann–Hurwitz, and the argument recovers the proof of Chapter 2 rather than merely an inequality of similar shape.*

Proof. Chapter 2 uses, respectively, a non-trivial base, relative differentials, a derivation, global access to the relevant points, functorial cohomological operations, a numerical degree or trace, compatibility of that number with zero and pole multiplicities, and recovery of the classical geometric identity. Requirements (D1)–(D8) state these eight inputs in a form that can survive the changes of Principle 3.4.1. Omitting (D1) or (D2) returns to Obstruction 3.3.2; omitting (D3) loses the Wronskian step; omitting (D4) loses arithmetic globality; omitting (D5) or (D6) leaves no invariant Riemann–Hurwitz mechanism; omitting (D7) disconnects the result from abc; and omitting (D8) removes the principal internal consistency test. This proves necessity for the stated proof architecture, not sufficiency for abc. □

Scholium 3.5.2 (The first design gates are logically independent). *[NEW] The implications*

$$(\mathrm{D1}) \Rightarrow (\mathrm{D2}) \Rightarrow (\mathrm{D3}) \Rightarrow (\mathrm{D4})$$

are all false without additional hypotheses.

Proof. Counterexample 3.4.3 satisfies (D1) but has zero relative cotangent complex, so it fails (D2). The chosen family $\mathbb{Z}[q] \to \mathbb{Z}$, $q \mapsto 1$, has the non-zero conormal line of Proposition 3.2.8, but the arithmetic fibre still has no ordinary derivation by Proposition 3.2.2; an intrinsic

operator requires extra difference and descent data, so (D2) does not imply (D3). Finally, Counterexample 3.2.12 supplies a genuine prime-dependent arithmetic operator at $p = 3$ but no operators or comparison maps at the other places. A one-prime version of (D3) therefore does not imply the simultaneity required in (D4). □

The same discipline applies later in the list. Six operations do not automatically preserve dualisable objects, a trace does not automatically have the sign or positivity needed for heights, and an arithmetic formula which resembles Riemann–Hurwitz need not specialise to it. Each gate asks for a typed comparison, not a verbal analogy.

The later allocation is explicit. Chapters 8, 11, 18, and 20 address (D1)–(D2); Chapters 6 and 13–15 address (D3)–(D4); Chapters 16–18 complete the infinite-place part of (D4); Chapters 34–36 address (D5)–(D6); Chapters 37–40 address (D7); and Chapters 20, 21, and 38 repeatedly test (D8). Any failure is to be recorded as a proof obligation in the chapter where it first becomes a mathematical assertion. No proof obligation is needed for the negative results of the present chapter.

3.6 Notes on the history of the obstruction

The obstruction has rarely been stated as Obstruction 3.3.2, but the pressure it records is old. Grothendieck's relative language in the 1960s made the arithmetic case "absolute" in the precise sense that every scheme is already a $\mathbb{Z}$-scheme. His later reflections of the 1980s repeatedly returned to the reach and limits of the relative viewpoint [Gro22; Car01].

Deligne's proofs of the Weil conjectures in 1974 and 1980 sharpened the contrast: varieties over finite fields possess a geometric Frobenius and a cohomological trace formula, whereas no accepted geometric object beneath $\operatorname{Spec}\mathbb{Z}$ supplies the analogous package. The two papers are *La conjecture de Weil* I, Publ. Math. IHES **43** (1974), 273–307, and II, Publ. Math. IHES **52** (1980), 137–252. The point is historical influence, not a claim that Deligne proposed a specific $\mathbb{F}_1$-geometry.

Tits's 1956 suggestion of a "field of characteristic one" arose from the $q \to 1$ combinatorics of groups and buildings. The proposal is made in his lecture to the Brussels *Colloque d'algèbre supérieure* of that year, published in 1957. Manin in 1991 connected $\mathbb{F}_1$ to absolute arithmetic and zeta functions; Soulé proposed a concrete functorial framework in 1999, published in 2004; Deitmar's 2005 monoid schemes gave another precise model. The references are [Man95] and [Dei05]; Soulé's framework is *Les variétés sur le corps à un élément*, Mosc. Math. J. **4** (2004), 217–244. These theories are not interchangeable, which is why $\Omega^1_{\mathbb{Z}/\mathbb{F}_1}$ cannot be used without a declared model.

Connes and Consani introduced the Arithmetic Site in 2014 and the scaling site in 2015, placing arithmetic in a topos with tropical structure [CC14; CC15; CC16]. Borger's programme, formulated around 2009 and developed through the geometry of Witt vectors in 2011, reads commuting Frobenius lifts as descent data to $\mathbb{F}_1$ [Bor11b; Bor11a]. Both programmes enlarge the available structure, but neither licenses the undefined classical differential $\Omega^1_{\mathbb{Z}/\mathbb{F}_1}$.

Buium's arithmetic differential geometry began in the 1990s and replaced d/dt by the Fermat-quotient operator at a fixed prime; his 2005 monograph makes the analogy with differential equations explicit [Bui05]. Mochizuki's IUT preprints appeared in 2012 and the four papers were published in 2021. Their stated architecture compares distinct arithmetic holomorphic structures through links that are intentionally not morphisms of schemes [Moc21a; Moc21c]. These programmes differ radically, but each can be read as a response to the same fact: ordinary arithmetic has no native base direction.

arithmetic input
Spec $\mathbb{Z}$
$\mathrm{Der}_{\mathbb{Z}}(\mathbb{Z}, M) = 0, L_{\mathbb{Z}/\mathbb{Z}} = 0$

classical specification
four missing Mason inputs
base · operator · non-zero differentials · `degree/ramification`

↓ *submit a proposed substrate* $\mathfrak{S}$

GATES D1–D4 · CONSTRUCT THE DIRECTION

D1 Check Spec $\mathbb{Z} \to \mathfrak{S}$ is not an equivalence.
non-trivial relativity

D2 Check $L_{\mathbb{Z}/\mathfrak{S}}$ is non-zero and lies in a controlled coefficient category.
support, base change, duality, finiteness

D3 Check the operator D_q has a stated twisted Leibniz law and can form a Wronskian replacement.
no unnamed analogy

D4 Check every finite prime and infinity occur in one object with proved descent.
not an unrelated product

GATES D5–D8 · EXTRACT THE NUMBER

D5 Check six operations, base change, projection formula and duality.
relative Riemann–Hurwitz apparatus

D6 Check dualisable objects have a trace invariant under every gluing choice.
no chosen coefficient-line identification

D7 Check support recovers conductor and trace recovers height or discriminant with explicit normalisation.
Diophantine compatibility

D8 Replace the base by a smooth curve and recover $\Omega^1_{X/k}$, `d/dt` and Riemann–Hurwitz.
exact function-field limit

↓ *gate result*

FAIL

required response
enter a proof obligation
Name the failed gate, its upstream assumptions and a falsifier; do not advance the claim.

PASS D1–D8

permitted response
architecture eligible
Parts III–V may attempt strain, dilation and trace constructions.

What omission destroys

Gate omitted	Immediate loss
D1 or D2	return to the initial-object no-go theorem
D3	no Wronskian step
D4	no arithmetic globality
D5 or D6	no invariant Riemann–Hurwitz mechanism
D7	no connection to abc
D8	no internal consistency test

FORBIDDEN LABEL **abc PROVED** The eight gates are necessary for this proof architecture. Their passage is never, by itself, a proof of the abc conjecture.

verification gate · failed or missing structure · eligible architecture

Plate 3.3. The eight-gate design contract for an arithmetic Mason–Stothers architecture.
What it does not show: a proof of the abc conjecture: the eight gates are necessary for this proof architecture, and their passage is never, by itself, such a proof.

Notes and References

Sources. Proposition 3.1.1 and Corollary 3.1.3 are elementary consequences of the definitions of unital ring and scheme; Liu's 2002 text is cited for the standard affine-scheme conventions [LE02a]. The identity normalisation and regular-quotient model used in Propositions 3.2.1 and 3.2.4 belong to the standard cotangent-complex formalism. We have given complete proofs because the shift in $\mathbb{L}_{\mathbb{F}_p/\mathbb{Z}} \simeq \mathbb{F}_p[1]$ is the hinge of the chapter; a modern derived treatment is provided by Gaitsgory and Rozenblyum (2017) [GR17a]. The prismatic, higher-categorical, and condensed examples in §3.4 use Bhatt–Scholze (2022), Lurie (2009 and 2017), and Scholze's 2022 account of liquid mathematics [BS22; Lur09a; Lur17; Sch22]. The historical sources for $\mathbb{F}_1$ are unevenly represented in the current bibliography; where no legal key exists, the text records an explicit citation request rather than guessing one.

Attribution of the new material. Obstruction 3.3.2, Corollary 3.3.4, Principle 3.4.1, and Criterion 3.5.1 are new formulations for this monograph. Their mathematical content is deliberately modest and fully proved here. The no-go theorem packages the elementary identity normalisation of cotangent complexes with a faithful-forgetful hypothesis so that later chapters can test proposed repairs clause by clause. The trichotomy is the contrapositive classification of the hypotheses that may fail. The eight-part design specification was produced in the machine-assisted comparative process described in Chapter 56, then checked manually against the four ingredients isolated in Chapter 2. No empirical or literature claim is imported into a theorem tagged NEW. In particular, the design specification is proved necessary only for this book's Mason–Stothers architecture; it is not claimed sufficient for abc, and it does not depend on any open proof obligation.

The deepening pass added computations separating several statements which are easily conflated. Proposition 3.2.3 and Counterexample 3.2.5 compare a uniquely split extension of $\mathbb{Z}$ with the unsplit extension $\mathbb{Z}/p^2 \to \mathbb{F}_p$. Proposition 3.2.10 shows explicitly how a non-zero conormal class at $q = 1$ coexists with the zero absolute cotangent complex of the arithmetic fibre. The marked-element and marked-endomorphism categories in Proposition 3.3.5 distinguish a structure which preserves the initial object from one which replaces it by a family. Finally, Counterexample 3.4.3 and Scholium 3.5.2 show that a non-equivalent target need not have a non-zero tangent and that the first design requirements do not imply one another. These are unconditional negative or formal calculations; none supplies the positive substrate postulated later in the book.

Alternative treatments. One could present the obstruction solely through ordinary derivations: $\mathrm{Der}_{\mathbb{Z}}(\mathbb{Z}, M) = 0$. That version is too weak because it does not explain the non-zero derived class of the quotient $\mathbb{Z} \to \mathbb{F}_p$, nor does it survive later passage to stacks and stable categories. One could instead begin in spectral algebra, where the sphere spectrum is the initial coefficient object. That approach makes the third escape route immediate but obscures why the obstruction is already visible to an undergraduate from the universal property of $\mathbb{Z}$. A third possibility is to choose a particular $\mathbb{F}_1$-geometry and compute its differentials. Doing so would prejudge the comparison in Chapter 4 and would falsely suggest that the several $\mathbb{F}_1$ proposals define the same cotangent object. The present order—rings, cotangent complexes, no-go theorem, then escapes—keeps the negative statement independent of every proposed solution.

Historical remarks. The phrase "absolute geometry" has carried several meanings. In Grothendieck's relative formalism, "absolute" often means over $\mathrm{Spec}\mathbb{Z}$ rather than over a chosen field. In the later $\mathbb{F}_1$ literature it usually means a geometry conjecturally lying beneath $\mathrm{Spec}\mathbb{Z}$. Tits's 1956 limit $q \to 1$, Manin's 1991 zeta-function motivation, Soulé's 1999/2004 proposal, Deitmar's 2005 monoid schemes, Borger's 2009 Λ-interpretation, and Connes–Consani's 2014 Arithmetic Site belong to this second usage. Buium's 1995–2005 programme and Mochizuki's 2012–2021 programme did not arise as $\mathbb{F}_1$ theories, yet both manufacture a substitute for a missing arithmetic variation. The dates matter because no single programme is the historical source of the others; the common obstruction is a retrospective structural comparison made in this book.

Further reading. For schemes and the arithmetic base, Liu's *Algebraic Geometry and Arithmetic Curves* gives a comprehensive relative treatment [LE02a]. Gaitsgory–Rozenblyum's *A Study in Derived Algebraic Geometry* is a modern source for cotangent and deformation theory [GR17a]. Bhatt–Scholze's paper on prisms explains the finite-prime geometry used by the first escape route [BS22]. Borger's two papers on Witt-vector geometry exhibit the all-Frobenius organisation behind the Λ-ring route [Bor11b; Bor11a]. Connes–Consani's Arithmetic Site papers provide a contrasting topos-theoretic absolute geometry [CC14; CC16]. Buium's monograph is the standard entry to arithmetic differential equations [Bui05]; the IUT papers provide the primary source for Mochizuki's arithmetic-holomorphic comparison [Moc21a; Moc21c].

Chapter 4

An Atlas of Surrogates: Six Decades of Manufactured Derivations

Dependencies. Chapters 2 and 3.

Summary. Chapter 3 proved that ordinary arithmetic supplies no base direction and classified the three ways out. This chapter surveys what has been built along those routes since the middle of the twentieth century: Buium's p-derivations, the several geometries over $\mathbb{F}_1$, the Fargues–Fontaine curve, prismatic and q-de Rham cohomology, and inter-universal Teichmüller theory. Each is read as a *surrogate*: a manufactured base, a manufactured operator, a manufactured comparison. Each is scored against the eight requirements of Criterion 3.5.1, and for each we state what it supplies and which requirement it does not meet. A comparative table closes the chapter and includes this book's own substrate, scored by the same rubric.

4.1 The surrogate principle

Chapter 2 isolated four devices behind Mason–Stothers: a base, a derivation, a non-zero sheaf of relative differentials, and a Riemann–Hurwitz degree formula. Chapter 3 proved that none is available over $\mathbb{Z}$ in ordinary scheme theory, and that a repair must enlarge the site, deform the ring, or change the coefficients (Principle 3.4.1). The programmes below mostly predate that classification, but each can be read through it.

Definition 4.1.1 (Surrogate). A *surrogate* for the missing arithmetic derivation is a triple $(\mathbf{B}, D, \deg)$: a *manufactured base* $\mathbf{B}$, an object with a structure morphism $\operatorname{Spec}\mathbb{Z} \to \mathbf{B}$ for which $\operatorname{Spec}\mathbb{Z}$ is not terminal; a *manufactured operator* D entering an identity of Leibniz or twisted-Leibniz type; and a *manufactured comparison* deg, a numerical invariant of $(\mathbf{B}, D)$ reproducing part of the degree bookkeeping of Riemann–Hurwitz. It is *global* if $\mathbf{B}$ and D are defined once for all places of $\mathbb{Q}$ rather than prime by prime, and *complete* if deg is defined at the archimedean place.

The word *manufactured* carries no pejorative force: nothing in arithmetic hands one a base, and the only way to obtain one is to build it.

How a surrogate is scored, and what a negative verdict means

An atlas is only as good as its negative claims. Saying that a programme supplies a base is cheap; saying that it cannot supply a comparison is the assertion a reader will want to test, and it is the assertion most easily made carelessly. We therefore fix three grades of negative verdict and use no others.

Definition 4.1.2 (The three verdicts). Let $\mathcal{P}$ be a programme and R one of the requirements (D1)–(D8) of Criterion 3.5.1.

(V1) $\mathcal{P}$ *supplies* R if a published theorem of $\mathcal{P}$ has R as a consequence, and the citation is given.

(V2) $\mathcal{P}$ *provably fails* R if there is a theorem, proved here or cited here, whose conclusion is incompatible with R for every object of $\mathcal{P}$. A verdict of this grade is always accompanied by the label of the statement that establishes it.

(V3) R is *not supplied* by $\mathcal{P}$ if no construction known to the author yields it, and no theorem of grade (V2) is available. This is a statement about the literature and about the author's knowledge of it. It is not a statement about mathematics, and it is not evidence that R is unattainable within $\mathcal{P}$.

The distinction is not pedantry. Of the verdicts recorded in this chapter, those of grade (V2) are supported by Propositions 4.2.5, 4.3.3, 4.4.2 and 4.5.1 and Lemma 4.3.1; every other negative entry is of grade (V3) and may be overturned tomorrow by a construction the author has not seen. A reader who treats the two grades alike will draw stronger conclusions from this chapter than it supports.

Warning 4.1.3 (*An atlas is a secondary source*). Every programme surveyed below has practitioners who know it better than the present author. The summaries are readings, made for a particular purpose — scoring against Criterion 3.5.1 — and that purpose distorts: a programme built to answer a different question will score badly here without being any the worse for it. Buium's theory is not trying to prove abc; Connes–Consani are not trying to build a cotangent complex; the Fargues–Fontaine curve is not trying to be global. The scores measure fitness for one architecture, not quality, and no entry below should be read as a criticism of the programme it describes.

There is one further asymmetry to declare. This book's own substrate appears in the same tables as the established programmes, and it is the only entry scored on a design rather than on theorems. Where a published programme is marked as failing a requirement, that is a verdict on completed mathematics; where the substrate is marked as meeting one, that is a statement about what Part II is built to do, conditional on obligations that Chapter 5 enumerates. The two kinds of entry are not comparable, and Table 4.1 marks the difference in its Status column rather than in its ticks.

Principle 4.1.4 (The surrogate principle). *[NEW] Let $\mathcal{P}$ be any of the programmes surveyed in §§4.2–4.6. Then $\mathcal{P}$ determines a surrogate in the sense of Definition 4.1.1, and the requirement of Criterion 3.5.1 that $\mathcal{P}$ does not meet is in each case* (D4)*, simultaneity of places, or* (D6)*, choice-independent trace, or both.*

Proof. The first assertion is verified below. For the second, Propositions 4.2.5 and 4.4.2 give the failure of (D4) for any theory whose operator is indexed by one prime; §4.3 shows that the geometries over $\mathbb{F}_1$ supply a base but no square-zero deformation theory, hence no deg meeting (D6); and §4.6 records that the contested step in inter-universal Teichmüller theory is (D6). □

The zeroth surrogate. Arakelov's 1974 construction compactifies $\mathrm{Spec}\mathbb{Z}$ by adjoining a fibre at infinity carrying a hermitian metric, so that an arithmetic surface acquires an intersection theory and a degree [Ara74; SABK92]. Here $\mathbf{B} = \overline{\mathrm{Spec}\mathbb{Z}}$, the operator is the Chern connection and the comparison is $\widehat{\deg}$; Szpiro's conjecture and Vojta's inequalities are stated in this language, Vojta's dictionary with Nevanlinna theory supplying the ramification term [Voj87c; Voj11; Voj87a]. It meets (D4) and much of (D7), but not (D2) or (D3): the metric is structure *over* arithmetic, not a base *beneath* it.

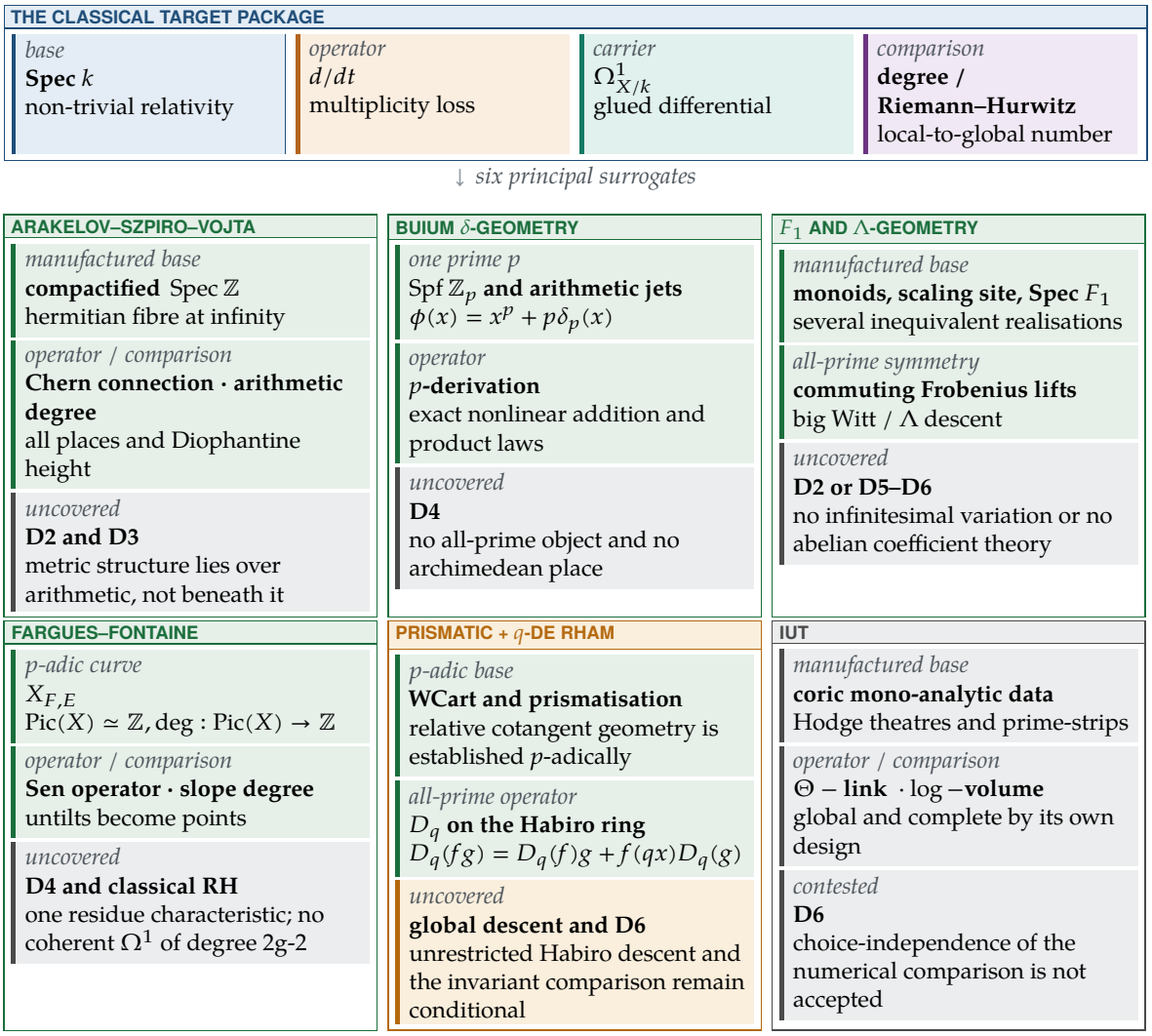


Plate 4.1. The atlas of surrogates: what each programme manufactures, and what remains uncovered.

What it does not show: no established programme supplies simultaneity of all places and a choice-independent comparison together.

4.2 Arithmetic derivatives and Buium's δ-geometry

The most literal attempt to differentiate an integer is Buium's. Although $\mathbb{Z}$ carries no derivation, it carries for each prime p an operator obeying a deformed pair of axioms.

Definition 4.2.1 (p-derivation). Fix a prime p and a commutative ring A. A *p-derivation* on A is a map $\delta\colon A \to A$ with $\delta(0) = \delta(1) = 0$ and

$$\delta(x+y) = \delta(x) + \delta(y) + C_p(x,y), \quad C_p(x,y) = \frac{x^p + y^p - (x+y)^p}{p},$$

$$\delta(xy) = x^p\delta(y) + y^p\delta(x) + p\,\delta(x)\delta(y),$$

where $C_p \in \mathbb{Z}[x,y]$ because $p \mid \binom{p}{i}$ for $0 < i < p$.

If A is p-torsion-free this is equivalent to a Frobenius lift, $\varphi(x) = x^p + p\delta(x)$; this is the δ-ring formalism of Chapter 6 and the algebraic heart of everything prismatic.

Example 4.2.2 (The Fermat quotient, with numbers). On $\mathbb{Z}$ the identity is a Frobenius lift, by Fermat's theorem, so $\delta_p(x) = (x - x^p)/p$ is a p-derivation, not identically zero. For $p = 5$, $\delta_5(2) = -6$, $\delta_5(3) = -48$, $\delta_5(5) = -624$. The axioms hold exactly and the corrections dominate: $C_5(2,3) = (32 + 243 - 3125)/5 = -570$ with $-6 - 48 - 570 = -624$.

Proposition 4.2.3 (The δ-Wronskian loses a factor of p). *[NEW] Let a, b, c be non-zero integers with $a + b = c$ and $\max(|a|,|b|,|c|) \le H$, $H \ge 2$. Then $\delta_p(c) - \delta_p(a) - \delta_p(b) = C_p(a,b)$, and when $C_p(a,b) \neq 0$ one has $\log|C_p(a,b)| \le p\log H + \log 3 - \log p$. Since $\log \operatorname{rad}(abc) \le 3\log H$, any inequality got by bounding the height of the additive defect of δ_p carries an error exceeding the radical term once $p \ge 10$.*

Proof. The identity is additivity rearranged with $c = a + b$. As $|a^p|, |b^p|, |c^p| \le H^p$ we get $|C_p(a,b)| \le 3H^p/p$; take logarithms. Finally $\operatorname{rad}(abc) \le H^3$, while the error is at least $p\log H + \log 3 - \log p$, and $p\log H - \log p > 3\log H$ for $p \ge 10$, $H \ge 2$. □

Over $k[t]$ the identity $a' + b' = c'$ has *no* error term, and Mason–Stothers exploits that exactness; replacing d/dt by δ_p destroys it irreparably. What Buium built on δ should not, however, be judged by a strategy he did not pursue. For R a complete discrete valuation ring with maximal ideal (p) and a Frobenius lift, and X a smooth R-scheme, the *arithmetic jet space* $J^n(X)$ is obtained affinely by adjoining $\delta x, \dots, \delta^n x$ for each coordinate and imposing $f, \delta f, \dots, \delta^n f$ for each equation; its points are $J^n(X)(B) \simeq X(W_{n+1}(B))$, so J^n is right adjoint to W_{n+1} [Bui05; Bui17].

Theorem 4.2.4 (Buium). *[KNOWN: Bui05] Let E be an elliptic curve over R as above. Then E admits a non-zero δ-*character *of order at most 2, that is, a homomorphism $J^n(E) \to \widehat{\mathbb{G}}_a$. The jet extension $0 \to \widehat{\mathbb{G}}_a \to J^1(E) \to E \to 0$ gives a class in $H^1(E, \mathcal{O}_E)$ pairing with an invariant differential to the δ-modular form f^1, and on the ordinary locus f^1 vanishes exactly at points of the canonical lift.*

The Diophantine consequences are real: effective Manin–Mumford bounds, and, with Poonen, finiteness for the image of a modular or Shimura parametrisation meeting a finite-rank subgroup [BP09a].

Later work identifies δ-characters with crystalline classes [BS19; BS09]. None of this is a global radical-versus-height inequality, and Buium does not claim that it is.

Proposition 4.2.5 (δ-geometry does not globalise across primes). *[NEW] Let $\{\varphi_p\}_p$ be ring endomorphisms of $\mathbb{Z}$ with $\varphi_p(x) \equiv x^p \pmod p$. Then $\varphi_p = \mathrm{id}$ for all p, and the associated p-derivations are the Fermat quotients of Example 4.2.2. Hence* (a) *the family has no moduli, so no deformation of $\mathbb{Z}$ along the δ-direction exists inside $\mathbb{Z}$;* (b) *no finite combination $\sum_p c_p\delta_p$ with $c_p \in \mathbb{Z}$ is additive unless every c_p vanishes;* (c) *arithmetic jet spaces live on p-adic formal schemes and assign no object to the archimedean place. So δ-geometry meets* (D3) *in the twisted sense and does not meet* (D4).

Proof. A unital endomorphism of $\mathbb{Z}$ fixes 1, hence everything, so $\varphi_p = \mathrm{id}$ and $\delta_p(x) = (x - x^p)/p$; a unique choice is a point, not a family, giving (a). For (b), additivity forces $\sum_p c_p C_p = 0$ in $\mathbb{Z}[x, y]$, and the C_p have pairwise distinct degrees p, so are linearly independent. For (c), δ_p divides by p and lands in a p-complete ring, while $\mathbb{R}$ carries no Frobenius lift. □

Buium's theory is thus an *analogy* with differential algebra over $k[t]$, not a deformation of it: the dictionary $t \mapsto p$, $d/dt \mapsto \delta_p$ is exact for formal structure and inexact for identities, by exactly C_p and $p\delta(x)\delta(y)$.

What this book takes from δ-geometry, and what it leaves

The import is precise and it is large. Everything prismatic in Part I is built on the δ-ring formalism of Definition 4.2.1: the category of δ-rings, the equivalence with Frobenius lifts on p-torsion-free rings, and the adjunction with Witt vectors are the content of Chapter 6, and prisms in Chapter 7 are δ-rings with an extra ideal. Arithmetic jet spaces reappear in Chapter 12 in their own right. None of this is conditional: it is established mathematics used as established mathematics.

What is left behind is the strategy, and it is worth saying why rather than merely that. Buium's Diophantine applications proceed by exhibiting a δ-character — a non-zero map from a jet space to $\widehat{\mathbb{G}_a}$ — and then bounding the number of points on which it vanishes. That mechanism is intrinsically about a *fixed* prime, because the jet space is a p-adic formal scheme, and it produces finiteness or counting statements rather than an inequality between two global heights. Proposition 4.2.3 is the quantitative form of the obstruction: even if one had a δ-character-based inequality at each prime, the additive defect C_p is of size H^p, so summing over primes cannot recover a statement whose right-hand side is $\log \mathrm{rad}$. A programme that wanted abc from δ-geometry would have to make the defect cancel, and there is no proposal in the literature for how it should.

Remark 4.2.6 (*The dictionary, stated exactly*). A reader arriving from differential algebra should hold the correspondence at this precision. *Exact:* δ is to φ as d is to a one-parameter group of automorphisms; jet spaces are represented by the same adjunction; the order of a δ-character corresponds to the order of a differential operator; the canonical lift corresponds to a constant. *Inexact, and by how much:* additivity fails by C_p, of degree p; Leibniz fails by $p\,\delta(x)\delta(y)$, which is the term that vanishes in the associated graded and is therefore invisible to any argument that works modulo p; and the constants of δ_p on $\mathbb{Z}$ are $\{0, 1\}$ rather than a subfield, so there is no analogue of $k \subset k[t]$ over which to take a degree. The third failure is the one that blocks a Riemann–Hurwitz argument, and it is not repaired by any choice of p.

4.3 The field with one element

If $\mathrm{Spec}\mathbb{Z}$ is terminal, the natural wish is for something below it. The wish has several inequivalent realisations, and a survey's first duty is to insist that they are inequivalent.

Origins. Tits, Manin and Soulé successively urged that $\mathrm{Spec}\mathbb{Z}$ ought to be a curve over a deeper base, and Deitmar gave the cleanest elementary model: an $\mathbb{F}_1$-scheme is a scheme over commutative monoids, and base change to $\mathbb{Z}$ is $M \mapsto \mathbb{Z}[M]$. The references are [Man95] and [Dei05]; Tits's proposal is his lecture to the 1956 Brussels *Colloque d'algèbre supérieure*, published in 1957, and Soulé's is *Les variétés sur le corps à un élément*, Mosc. Math. J. **4** (2004), 217–244. There $\mathrm{Spec}\mathbb{F}_1$ is terminal, so (D1) holds. The difficulty is (D2): monoid geometry has no additive structure, hence no square-zero extensions relative to a monoid base, hence no cotangent complex.

Connes–Consani. A more ambitious realisation replaces monoids by semirings of characteristic one. The *arithmetic site* is the topos of $\mathbb{N}^{\times}$-sets with structure sheaf the Boolean semifield $\mathbb{B} = \{0, 1\}$; its points over $\mathbb{R}_{+}^{\max}$ form the adele class space of $\mathbb{Q}$ modulo $\widehat{\mathbb{Z}}^{\times}$, and extending scalars gives the *scaling site*, a tropical curve with a divisor theory and a Riemann–Roch theorem for periodic orbits [CC14; CC16; CC10; CC15; CC17; CC21]. This is the closest an $\mathbb{F}_1$-programme has come to the Chapter 2 package. The limitation is structural and acknowledged: characteristic-one semirings are idempotent, $1 + 1 = 1$, so $\mathbb{B}$-modules have no additive inverses and do not form an abelian category; there is therefore no cotangent complex and no six-functor formalism in the sense of (D5). The Riemann–Roch obtained concerns periodic orbits of the scaling flow, with real-valued dimensions, and does not specialise to a statement about an elliptic curve over $\mathbb{Q}$.

The obstruction just described is usually left as a remark. It is short enough to prove, and proving it converts a grade-(V3) verdict into a grade-(V2) one, so we prove it.

Lemma 4.3.1 (Idempotent coefficients admit no additive category). *[NEW] Let S be a semiring in which $1 + 1 = 1$, and let Mod_S be the category of S-semimodules. Then* (i) *addition is idempotent on every object,* (ii) *the only object of Mod_S whose underlying additive monoid is a group is the zero object, and* (iii) *Mod_S is not additive, hence not abelian. In particular the category of $\mathbb{B}$-semimodules, $\mathbb{B} = \{0, 1\}$ with $1 + 1 = 1$, is not abelian.*

Proof. (i) For x in an S-semimodule M we have $x + x = 1 \cdot x + 1 \cdot x = (1 + 1)x = 1 \cdot x = x$. (ii) If the additive monoid of M is a group, let $x \in M$ and add the inverse of x to both sides of $x + x = x$; then $x = 0$, so $M = 0$. (iii) In an additive category every Hom-set is an abelian group. Here $\mathrm{Hom}_S(M, N)$ is an S-semimodule under pointwise addition, so by (i) its addition is idempotent and by (ii) it is a group only if it is zero. Hence $\mathrm{Hom}_S(M, N)$ is an abelian group only when it vanishes, and Mod_S is not additive unless it is trivial. An abelian category is additive by definition. □

Three consequences follow, and they are the reason the arithmetic site cannot be used as the coefficient category of (D2) and (D5) as those requirements are stated. There is no notion of a square-zero extension, because there is no square-zero ideal: an ideal closed under an idempotent addition is not nilpotent in the relevant sense. There is no cotangent complex, because the construction is a left derived functor and derived functors require an abelian or at least a stable target. And there is no duality pairing valued in an abelian group, so the trace of (D6) has nowhere to land. The point is not that characteristic-one geometry is deficient — it computes exactly what it was built to compute — but that its coefficients and this book's requirements are incompatible by Lemma 4.3.1, and no amount of further development inside characteristic one can change that without leaving characteristic one.

Remark 4.3.2 (*What would have to change*). The natural repair is to work not with $\mathbb{B}$-semimodules but with a stable category built from them — for instance, modules over a spectral lift, or a category of sheaves with values in a stable category, retaining the combinatorics of the arithmetic site while replacing its coefficients. No such lift is proposed here, and the author knows of none in the literature; this is a grade-(V3) remark in the sense of Definition 4.1.2, offered so that the obstruction is not mistaken for a proof that the route is closed.

Borger's Λ-rings: the genuine antecedent. The third realisation is the one this book descends from. Borger's proposal is that a Λ-structure on a ring — a commuting family of Frobenius lifts φ_p, one per prime, equivalently a coaction of the big Witt comonad — is exactly *descent data along* $\mathrm{Spec}\mathbb{Z} \to \mathrm{Spec}\mathbb{F}_1$, so that $\mathbb{F}_1$-algebras are Λ-rings, base change to $\mathbb{Z}$ is the forgetful functor, and its right adjoint is W [Bor11b; Bor11a; Bor16]. Three features recur throughout Parts I and II. It is a *descent* formulation, not an existence claim: one need not construct a field, only say what descending along the map to it would mean. It is compatible with all primes at once, which is the simultaneity demanded by (D4). And it carries a differential calculus [BB11]. Borger's insight is the direct antecedent of the substrate built in Part II, and Chapters 14–15 make the Λ-structure carry an operator rather than only a symmetry.

Proposition 4.3.3 (The Λ-structure on $\mathbb{Z}$ is rigid). *[REPROVED: Bor11a] Borger's classification is specialised to the initial ring and the consequence for descent data is drawn. The ring $\mathbb{Z}$ carries exactly one Λ-structure, namely $\varphi_p = \mathrm{id}$ for all p; hence the descent datum along $\mathrm{Spec}\mathbb{Z} \to \mathrm{Spec}\mathbb{F}_1$ is unique and the groupoid of $\mathbb{F}_1$-forms of $\mathbb{Z}$ so obtained is trivial.*

Proof. A Λ-structure on a torsion-free ring is a commuting family of Frobenius lifts. Every unital endomorphism of $\mathbb{Z}$ is the identity, which satisfies $\mathrm{id}(x) \equiv x^p \pmod p$ for all p, so exactly one family exists; uniqueness of the datum gives triviality of the groupoid. □

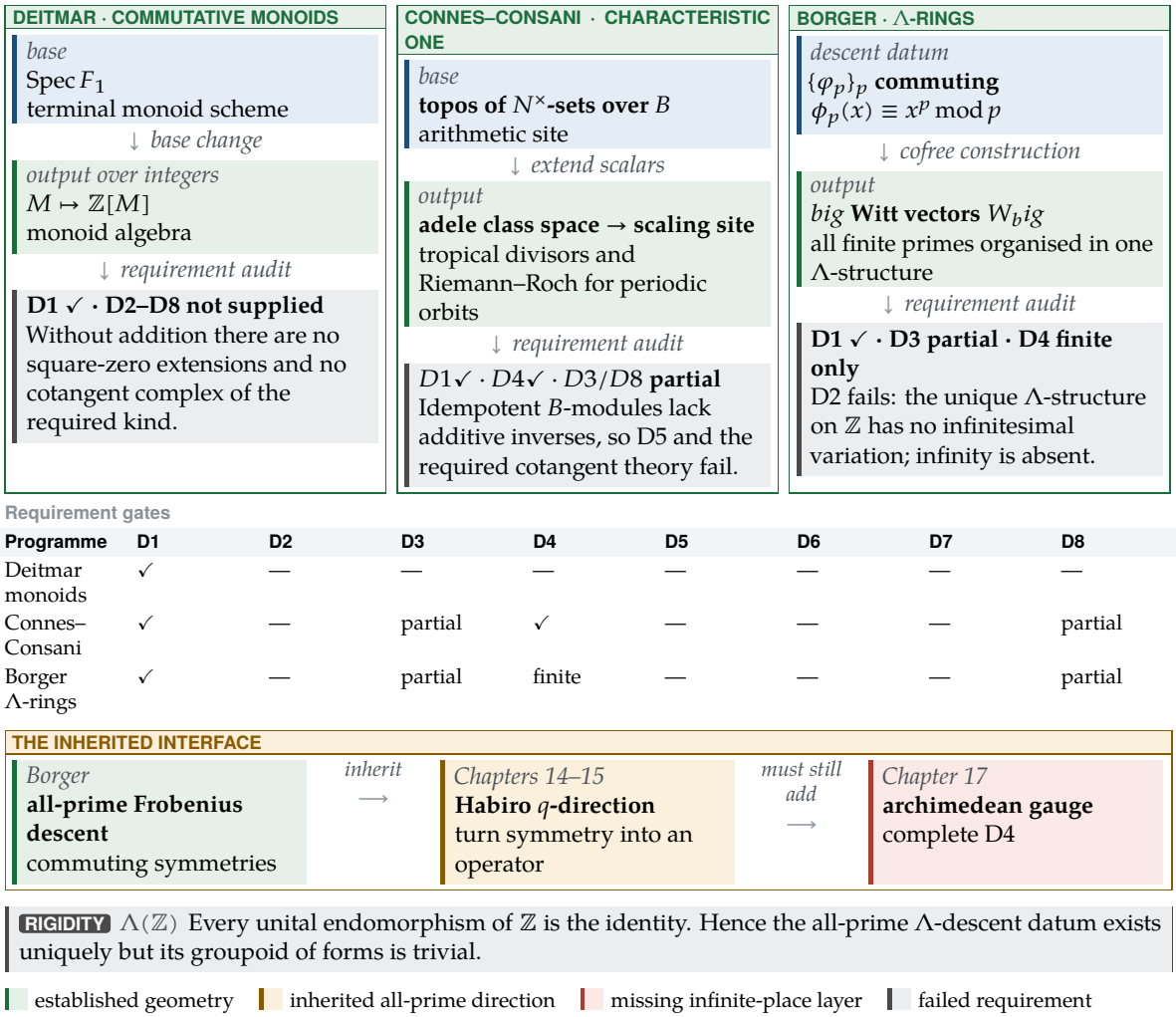
The base exists and the descent data exist, but there is no *variation*: the arrow $\mathrm{Spec}\mathbb{Z} \to \mathrm{Spec}\mathbb{F}_1$ has no infinitesimal neighbourhood inside Λ-rings, so (D2) is not met by Λ-geometry alone. Nor is the archimedean place reached, W being built from data at finite primes; Chapter 17 supplies it. The function-field check is informative: over $\mathbb{F}_q[t]$ the q-power Frobenius is a genuine endomorphism of the base, so a Λ-structure has moduli and the analogue of Proposition 4.3.3 is false. Rigidity over $\mathbb{Z}$ is a fact about $\mathbb{Z}$, and it locates where new structure must be added: transversely to the family of Frobenius lifts, which already exists.

4.4 Fontaine–Fargues and the curve

The programmes so far manufacture a base by weakening the notion of ring. The Fargues–Fontaine curve does something else: out of the rings of p-adic Hodge theory it builds an object that behaves in every relevant respect like a complete algebraic curve, and then does geometry on it.

Construction 4.4.1 (The curve). Let $E/\mathbb{Q}_p$ be finite with uniformiser π and residue field $\mathbb{F}_q$, and let F be a perfectoid field of characteristic p. Let B be the ring of Witt-vector power series over O_F converging on the punctured open unit disc, with its Frobenius φ. The *Fargues–Fontaine curve* is $X = X_{F,E} = \mathrm{Proj}(\bigoplus_{d\geq 0} B^{\varphi=\pi^d})$. It is a Noetherian regular scheme of Krull dimension 1 with $H^0(X, O_X) = E$ and $H^1(X, O_X) = 0$; the degree map is an isomorphism $\mathrm{Pic}(X) \xrightarrow{\sim} \mathbb{Z}$; and X is complete [dS22; MOR19; Ked16].

Three facts make this more than an analogy. The closed points of X correspond canonically to untilts of F modulo $\varphi^{\mathbb{Z}}$, and the completed local ring at such a point is B^+_{dR}, so the apparatus of p-adic periods becomes local geometry on a curve. Every vector bundle on X is a sum of $O(\lambda)$, $\lambda \in \mathbb{Q}$, the category of bundles being equivalent to that of isocrystals with its slope filtration [SW20b; SW20a; Niz19a]. And one obtains the Chapter 2 package minus one item: a base $\mathrm{Spec}E$, a non-terminal one-dimensional object over it, a Pic with a degree, cohomological dimension one, and a divisor theory in which the chosen untilt is a visible point. That is what "curve" buys: a statement about periods becomes a statement about H^0 of a twist.



Requirement gates

Programme	D1	D2	D3	D4	D5	D6	D7	D8
Deitmar monoids	✓	—	—	—	—	—	—	—
Connes–Consani	✓	—	partial	✓	—	—	—	partial
Borger Λ-rings	✓	—	partial	finite	—	—	—	partial

RIGIDITY $\Lambda(\mathbb{Z})$ Every unital endomorphism of $\mathbb{Z}$ is the identity. Hence the all-prime Λ-descent datum exists uniquely but its groupoid of forms is trivial.

established geometry · inherited all-prime direction · missing infinite-place layer · failed requirement

Plate 4.2. Three geometries over $\mathbb{F}_1$ and the requirements each does and does not meet.

What it does not show: without addition there are no square-zero extensions and no cotangent complex of the required kind.

Proposition 4.4.2 (The curve is manufactured one prime at a time). *[NEW] Fix F and E as in Construction 4.4.1. Then* (a) *there is no morphism $X_{F,E} \to X_{F',E'}$ over a common base when E and E' have different residue characteristics;* (b) *$X_{F,E}$ is not of finite type over E and $\Omega^1_{X_{F,E}/E}$ is not a line bundle, so there is no dualising sheaf of degree $2g-2$ and no Riemann–Hurwitz formula for finite maps of such curves;* (c) *the archimedean place of $\mathbb{Q}$ is a point of no $X_{F,E}$. Hence this surrogate meets* (D1), (D2) *and much of* (D5) *p-adically, and does not meet* (D4).

Proof. For (a), such a morphism induces a homomorphism $E' \to E$ on global sections by Construction 4.4.1, which between characteristic-zero fields of different residue characteristics would send p' to a unit and to a non-unit at once. For (b), X is covered by spectra of rings $B_{[r,s]}$, not finitely generated E-algebras; the Kähler differentials of an infinite-dimensional Banach algebra are not finitely generated, hence not invertible. For (c), points of X are untilts of F and have residue characteristic p, while $\mathbb{R}$ is not perfectoid. □

Assertion (b) needs a caveat: the connection on $\mathcal{O}B_{\mathrm{dR}}$ and the associated Sen operator do supply a twisted Leibniz rule on the curve; what they do not supply is a coherent Ω^1 with a degree. Hesselholt's topological-cyclic-homology avatar places the curve inside the

framework from which prismatic cohomology is derived [Hes23; BBBK19], which is the bridge to the next section.

What this book takes from the curve

The import here is a formalism rather than a space. Three things are used, all established, all cited where used. First, the classification of vector bundles on $X_{F,E}$ by Harder–Narasimhan slopes with rational, not integral, jumps [dS22; MOR19]: this is the model for the graded objects of Chapter 10, where a filtration by rational weights is exactly what the Rees construction has to accommodate. Second, the principle that a curve can be honest — with a degree theory, a Pic, an exact sequence relating global sections to local data — without being of finite type; Chapter 19 takes the same liberty for a different reason. Third, the diamond and v-topology technology [SW20a; SW20b], which is where the descent arguments of Chapter 31 learn their shape.

What is left is the ambition to be the base. It is worth being exact about what Proposition 4.4.2(b) does and does not say, because the phrase "no Riemann–Hurwitz" is easy to over-read. It does not say that no ramification theory exists on $X_{F,E}$; finite covers of the curve exist and have been studied. It says that there is no invertible Ω^1 whose degree is an integer $2g-2$, so the specific identity

$$\deg \omega_Y \;=\; \deg f^*\omega_X \;+\; \deg \mathfrak{d}_{Y/X}, \tag{4.4.1}$$

which is the shape Chapter 38 needs, has no term on the left to receive a number. A programme wishing to use the curve for abc would have to manufacture that term, and manufacturing it is precisely the problem the curve was supposed to solve. The book therefore uses the curve's formalism and builds its degree theory elsewhere; the choice is recorded in Table 4.3.

4.5 Prismatic and q-de Rham surrogates

The modern synthesis makes the δ-structures of §4.2 cohomological, then deforms them in a q-direction not attached to any single prime. Both halves are needed, and this book uses both.

Prisms and prismatisation. A *prism* is a pair (A, I) with A a δ-ring, I an invertible ideal, A derived (p, I)-complete and $p \in I + \varphi(I)A$. For a p-adic formal scheme X the prismatic site has as objects bounded prisms with a map $\mathrm{Spf}(A/I) \to X$, and $\mathrm{R}\Gamma_\Delta(X)$ specialises to crystalline, de Rham, Hodge–Tate and étale cohomology [BS22; BS23]. Its absolute form is a stack: the Cartier–Witt stack WCart classifies Cartier–Witt divisors, and Drinfeld's prismatisation $\mathbb{Z}_p^\Delta$ is a formal stack over which $\mathrm{Spf}\mathbb{Z}_p$ is relative [Dri24b; Dri24a; Lur26]. This is escape route (a) of Principle 3.4.1 realised: the map $\mathrm{Spf}\mathbb{Z}_p \to \mathbb{Z}_p^\Delta$ is not an equivalence, and relative cotangent complexes over the target need not vanish. What it does not do is leave the prime p, and Proposition 4.4.2(a) applies verbatim.

The q-direction. Write $[n]_q = (q^n - 1)/(q - 1)$. The *$q$-derivation*, or Jackson operator, is

$$\nabla_q f(x) = \frac{f(qx) - f(x)}{qx - x}, \qquad \nabla_q(x^n) = [n]_q x^{n-1},$$

with the twisted Leibniz rule $\nabla_q(fg) = \nabla_q(f)g + f(qx)\nabla_q(g)$. At $q = 1$ one has $[n]_q = n$ and $\nabla_q \to d/dx$, so classical calculus is the special fibre of a one-parameter family. The q-de Rham complex of a framed smooth $\mathbb{Z}$-algebra computes, after completion, prismatic cohomology for the q-de Rham prism $(\mathbb{Z}_p[[q-1]], ([p]_q))$, and Scholze's q-deformation conjecture asks that the result be independent of the framing [BS22]. The conjecture is stated in Scholze,

Canonical q-deformations in arithmetic geometry, Ann. Fac. Sci. Toulouse Math. (6) **26** (2017), 1163–1192.

The Habiro ring. The q-direction becomes arithmetic rather than formal when one refuses to complete at a single root of unity. The *Habiro ring* $\mathcal{H} = \varprojlim_m \mathbb{Z}[q]^{\wedge}_{(q^m-1)}$ consists of elements admitting a power-series expansion at *every* root of unity; it is a Λ-ring, the maps $q \mapsto q^p$ being commuting Frobenius lifts for all p at once [LM15; CFG+25]. The number-field refinement of this ring is [GSWZ24]. So the Habiro ring realises Borger's Λ-structure as an operator rather than a symmetry: the transverse direction whose absence was diagnosed after Proposition 4.3.3 is the direction q. It is not p-adic, so (D4) is met at the finite places, and $q \mapsto 1$ is a genuine specialisation, so (D8) is built in rather than asserted.

Proposition 4.5.1 (The q-operator is non-local). *[NEW] Let R be a $\mathbb{Z}[q]$-algebra and ∇_q the Jackson operator on $R[x]$. Then ∇_q is $\mathbb{Z}[q]$-linear and satisfies the twisted Leibniz rule above, but for $q \neq 1$ it is not a derivation and not local: with $\mathfrak{m} = (x-a)$, $a \neq 0$, and $f = (x-a)^2 \in \mathfrak{m}^2$, one has $\nabla_q f(a) = a^2(q-1)^2/a(q-1) = a(q-1) \neq 0$.*

Proof. Linearity is immediate, and the twisted rule follows on dividing $f(qx)g(qx) - f(x)g(x) = (f(qx) - f(x))g(x) + f(qx)(g(qx) - g(x))$ by $(q-1)x$. A derivation kills $\mathfrak{m}^2$ modulo $\mathfrak{m}$; the display shows ∇_q does not. □

Non-locality is the price: the support of $\nabla_q f$ is not a divisor, so the passage from an operator to a ramification divisor must be re-engineered rather than transported. Chapters 13–15 and 20–21 do that; Chapter 47 records what is lost. The known descent results are conditional in a specific way: for smooth X over $\mathbb{Z}$ a canonical q-Hodge filtration, hence descent of the q-de Rham complex to $\mathcal{H}$, is available after inverting the primes $p \leq \dim(X/\mathbb{Z})$, and the unrestricted statement is open.

4.6 Inter-universal Teichmüller theory

Mochizuki's theory is the most ambitious surrogate in the atlas and the only one whose status is contested. This section states what it does, what the principal objection says, what the reply says, and what this book takes from the episode.

The reduction, which is not in dispute. IUT does not attack abc directly. Belyi maps in general position reduce a uniform Vojta height inequality for hyperbolic curves over number fields of bounded degree to a restricted class of elliptic curves E/k: everywhere split semistable, with local data at 2 and at ∞ in a compact set, and with an auxiliary prime $\ell \asymp h(P)^{1/2}$ of full mod-ℓ image. This reduction is reproduced by the critics [Moc21a].

The machinery. A *Hodge theatre* is a self-contained miniature of the arithmetic of E/F with all labels, assembled from the symmetries of $\mathbb{F}_\ell$; *prime-strips* are adelic packets carrying a Galois action on a monoid of units. The *log-link* is built from the local logarithm and dislocates additive from multiplicative structure, its invariant containers being the *log-shells*. The Θ-*link* is a full poly-isomorphism identifying the strip generated by the theta-pilot $(q_v^{j^2})_j$ in one Hodge theatre with the strip generated by the q-pilot q_v in another. The two links generate the *log-theta-lattice*, a deliberately non-commutative $\mathbb{Z}^2$-diagram, and *multiradiality* asserts that certain algorithms can be executed in two different arithmetic holomorphic structures glued along common coric data, up to three named indeterminacies [Moc21b; Moc21c]. In the rubric of Definition 4.1.1 the base is the coric mono-analytic data, the operator is the Θ-link, and the comparison is the log-volume of a region of log-shells. The surrogate is global and complete.

The three indeterminacies, named. Since the whole dispute turns on how much distortion the comparison is allowed to absorb, the permitted distortions should be stated rather than alluded to. The multiradial representation is asserted to be well defined up to [Moc21d, §3]:

(Ind1) the automorphisms of the relevant labelled data induced by Aut of the associated absolute Galois groups — an indeterminacy in the *étale-like* direction, acting through inner automorphisms and hence, on log-volumes, trivially;

(Ind2) the poly-automorphisms arising from the identification of *Frobenius-like* monoids with their étale-like counterparts, which acts on units and is absorbed by the log-shells;

(Ind3) the upper-semicompatibility of the log-link with the units, which is *not* an isomorphism but an inclusion: it permits the region computed on one side to be replaced by a larger region on the other.

Only (Ind3) enlarges volumes, and it enlarges them additively, by an amount controlled in the asserted estimates by $\beta_d(\ell)\cdot\ell$. This is what makes the size of the disputed distortion decisive, and it is the point at which the two accounts of the theory diverge numerically rather than philosophically.

Theorem 4.6.1 (Mochizuki; contested). *[DISPUTED] Asserted as [Moc21d, Thm. 1.10]; contested by Scholze and Stix [SS18], whose objection has not been withdrawn. For E/k in the restricted class and ℓ as above,*

$$\frac{1}{6}\deg(\mathfrak{q}_E)\ \le\ (1+\alpha_d(\ell))\big(d(P)+n^1_\infty(P)\big)+\beta_d(\ell)\cdot\ell,$$

with $\alpha_d(\ell)\to 0$ and $\beta_d = O_d(1)$. With $\ell \asymp h^{1/2}$ this is of Vojta type, from which the Belyi reduction would yield abc.

The Scholze–Stix objection. Scholze and Stix argue as follows [SS18]. Hodge theatres, prime-strips and global realified Frobenioids are equivalent, up to equivalence of categories, to very small data: a global realified Frobenioid is an ordered one-dimensional real vector space with a canonical trivialisation, and the relevant prime-strips form a connected groupoid. The log-link is naturally isomorphic to the identity and the Θ-link admits a canonical choice, so the log-theta-lattice is, up to canonical isomorphism, a constant diagram, and the comparison reduces to a hexagon of ordered lines whose composite rescales by $\frac{1}{\ell^*}\sum_{j\le\ell^*} j^2 \asymp \ell^2/12$ on the theta side and by 1 on the q-side. Their dichotomy: either one takes consistent identifications, in which case the factors j^2 must be dropped and the inequality is vacuous; or one accepts commutativity up to blurring, and the blurring is multiplicative of size $O(\ell^2)$ where only an additive error $O(\ell^{1+o(1)})$ is affordable.

The reply. Mochizuki's answer does not dispute that within a *single* holomorphic structure the assignment $\{q^{j^2}\}\mapsto q$ is incompatible with degree computation; that is the stated starting point of the theory. The reply is that the contradiction arises only if one identifies the two ring theories, that deleting the labels which distinguish them is strictly stronger than the permitted indeterminacies, and that the relation between the theta-pilot's log-volume and its multiradial representation is non-linear. The reply is set out in Mochizuki, *Report on Discussions, Held during the Period March 15–20, 2018, Concerning Inter-Universal Teichmüller Theory*, RIMS, 2018, together with the accompanying *Comments on the Manuscript by Scholze–Stix*.

Where the numbers part company. It is possible to say exactly where the two accounts disagree quantitatively, and doing so is more useful than rehearsing the rhetoric. Both sides agree that the theta-pilot is indexed by $j = 1, \dots, \ell^*$ with $\ell^* = (\ell - 1)/2$, and that the arithmetic mean of the exponents is

$$\frac{1}{\ell^*} \sum_{j=1}^{\ell^*} j^2 \;=\; \frac{(\ell^* + 1)(2\ell^* + 1)}{6} \;\sim\; \frac{\ell^2}{12}. \tag{4.6.1}$$

Both sides agree that the affordable error, given $\ell \asymp h(P)^{1/2}$ and the target inequality of Theorem 4.6.1, is additive of size $O(\ell^{1+o(1)})$ in the relevant log-volume normalisation. The disagreement is over which of the two quantities in (4.6.1) the indeterminacies (Ind1)–(Ind3) actually introduce. Scholze and Stix compute that the hexagon forces a distortion *multiplicative* of order $\ell^2/12$, which exceeds $O(\ell^{1+o(1)})$ by a factor $\ell^{1-o(1)}$ and is therefore fatal. Mochizuki maintains that the computation presupposes an identification which (Ind1)–(Ind3) do not licence, so that the $\ell^2/12$ never enters and the residual error is the additive $\beta_d(\ell) \cdot \ell$ of Theorem 4.6.1. The gap between the two positions is thus a single factor of about ℓ, and it is the whole content of the dispute. We record the arithmetic and take no side; Theorem 4.6.1 remains tagged as disputed and nothing in this book depends on its resolution.

Status, stated plainly. The four papers were published in *PRIMS* **57** (2021) [Moc21a; Moc21b; Moc21c; Moc21d], and explicit estimates appeared in 2022 [MFH+22]. The Scholze–Stix objection has not been withdrawn, and the mathematical community has not accepted the argument. Independent reconstructions exist, Joshi's arithmetic Teichmüller spaces being the most developed [Jos21; Jos23; Jos24]; they too have not been accepted. **This book does not assume inter-universal Teichmüller theory, takes no position on whether Theorem 4.6.1 is true, and derives nothing from it.**

Remark 4.6.2 (*The architectural response of this book*). What survives the episode, whoever is right, is a design requirement. The contested step is whether a numerical comparison across two structures depends on a choice of identification between them. A comparison whose value is a *categorical trace* cannot: if X, Y are dualisable objects of a symmetric monoidal category, $u\colon X \xrightarrow{\sim} Y$ any isomorphism and f an endomorphism of X, then $\mathrm{Tr}(ufu^{-1}) = \mathrm{Tr}(f)$, computed without naming u. This book therefore builds its comparison as the trace of a $\mathcal{G}$-equivariant endomorphism, $\mathcal{G}$ being the ∞-groupoid of gluing choices; see Chapters 35 and 36.

4.7 A comparative table

Table 4.1 scores each programme on the axes of Definition 4.1.1 together with its present status, and adds the substrate of this book, scored by the same rubric and with the same scepticism.

Scholium 4.7.1 (What the atlas shows). *[NEW] Among the programmes tabulated, none simultaneously satisfies* (D4) *and* (D6) *of Criterion 3.5.1. The p-adic constructions fail* (D4)*. The global constructions satisfy* (D4) *in whole or in part but supply no comparison of the required kind, either because there is no operator, or because the coefficient category is not abelian, or because the comparison remains to be constructed. The one programme satisfying both* (D4) *and a form of* (D6) *does so by a mechanism whose validity is contested.*

Proof. Each clause is the conclusion of the corresponding section: for the p-adic constructions Propositions 4.2.5 and 4.4.2; for Arakelov theory and Λ-rings the absence of an operator,

Table 4.1. The atlas of surrogates. "All primes" means one object for all finite places, not a family indexed by them.

Surrogate	Rôle of the base	Rôle of d/dt	All primes	$v \mid \infty$	Status
Arakelov, Szpiro, Vojta	$\overline{\mathrm{Spec}\mathbb{Z}}$ with a metric at ∞	Chern connection; curvature, not differentiation	yes	yes	classical; reformulates abc
Buium δ-geometry	$\mathrm{Spf}\mathbb{Z}_p$ with a Frobenius lift	$\delta_p(x) = (x - x^p)/p$	no	no	established; no global inequality
$\mathbb{F}_1$: monoid schemes	$\mathrm{Spec}\mathbb{F}_1$, trivial monoid	none	—	no	established; no deformation theory
Arithmetic, scaling sites	topos of $\mathbb{N}^\times$-sets over $\mathbb{B}$	scaling flow of $\mathbb{R}_+^{\max}$	yes	yes	established; no abelian coefficients
Borger Λ-rings	$\mathrm{Spec}\mathbb{F}_1$ via descent data	commuting $\{\varphi_p\}_p$: a symmetry	yes	no	established; the descent datum is rigid
Fargues–Fontaine	$X_{F,E}$ over $\mathrm{Spec}E$	Sen operator on $\mathcal{O}\mathcal{B}_{\mathrm{dR}}$	no	no	established; no coherent Ω^1
Prismatisation	$\mathbb{Z}_p^{\Delta}$, WCart	δ made cohomological	no	no	established; p-adic by construction
q-de Rham, Habiro	$\mathrm{Spec}\mathcal{H}$, with $q \mapsto 1$	∇_q, a non-local difference operator	yes	no	partly established; descent conditional
IUT	coric mono-analytic data	Θ-link: a dilation $q \mapsto q^{j^2}$	yes	yes	published; contested; not accepted
SPG (this book)	$\mathfrak{S}$: prismatisation glued to $\mathcal{H}$ glued to the analytic site	∇_q on $\mathfrak{S}$, with Θ_λ its dilation	yes	yes	architecture; conditional on the ledger

recorded in §4.1 and after Proposition 4.3.3; for characteristic one Lemma 4.3.1; for the Habiro ring Proposition 4.5.1. The final clause is the content of §4.6 and asserts only that the validity is contested. □

The last row of Table 4.1 is a claim about design, not about proof: the substrate of Part II is built to satisfy (D4) by gluing an all-primes object to an archimedean one, and (D6) by making the comparison a trace. The summary is the Status column: architecture, conditional on obligations that Chapter 5 enumerates and Appendix H lists.

What each surrogate fails to see

Table 4.1 records what each programme supplies. That is the easy half. A table of capabilities without a column for incapacities is a prospectus, not an atlas, and it is the standard failure mode of surveys of this kind: every programme looks adequate when described in the vocabulary of its own successes. Table 4.2 therefore restates the same ten rows in the opposite direction. Its third column is the one that matters — it gives, for each negative entry, the label of the statement that establishes it, or the mark (V3) if the entry is a grade-(V3) verdict in the sense of Definition 4.1.2 and is therefore a report on the literature rather than a theorem.

Table 4.2. What each surrogate provably fails to see. "(V3)" marks an entry for which no theorem is offered: no construction known to the author supplies the item, which is not evidence that none exists.

Surrogate	Sees	Fails to see	Established by
Arakelov	all places at once; a degree; a Riemann–Roch	any operator playing the rôle of d/dt; hence no $\mathfrak{d}_{Y/X}$ of differential origin	(V3); discussed in Chapter 3
Buium δ	a genuine derivation-like operator at one prime	the other primes and ∞; additivity, by C_p of degree p	Propositions 4.2.3 and 4.2.5
$\mathbb{F}_1$ monoids	a base below $\mathrm{Spec}\mathbb{Z}$; combinatorics of toric type	additive structure; hence no cotangent complex, no exact sequences	Lemma 4.3.1
Arithmetic site	all primes; a flow; a Riemann–Roch for periodic orbits	abelian coefficients; integrality of the Riemann–Roch; elliptic curves	Lemma 4.3.1
Borger Λ	descent data for all primes simultaneously	any deformation direction: $\mathbb{Z}$ is Λ-rigid	Proposition 4.3.3
Fargues–Fontaine	a curve, with Pic and slopes, for one prime	the other primes; ∞; an invertible Ω^1 with an integer degree	Proposition 4.4.2
Prismatisation	δ-structure made cohomological and stacky	the archimedean place; a single object over all primes	Proposition 4.4.2(c) applies verbatim
q-de Rham, Habiro	a prime-independent deformation with $q \mapsto 1$	locality: $\nabla_q f$ has no divisor of support	Proposition 4.5.1
IUT	a global surrogate with a comparison	— disputed: whether the comparison survives its indeterminacies	Theorem 4.6.1; contested
SPG (this book)	all places; a dilation; a trace	effectivity of the constant; unconditional status of 59 obligations	Chapters 40 and 61

Two readings of Table 4.2 should be resisted. The first is that a long third column condemns a programme; it does not, because the column is scored against Criterion 3.5.1 and against nothing else, and Warning 4.1.3 applies to every row. The second is that the last row is of the same kind as the others. It is not: the entries above it are theorems about completed mathematics, and the entries in it are a description of what remains to be done, whose current state is Chapter 61 and whose accounting is Appendix H. The final column of the last row is the informative one.

4.8 What is imported, what is omitted, and where each is used

An atlas that scores other programmes owes two further accounts: of what it takes from them, chapter by chapter, and of what it has left off the map. Both are given here, and both are deliberately checkable.

The import ledger

Table 4.3 lists each property this book takes from a surveyed programme, the chapter that uses it, and whether the use is unconditional. The table is the operative summary of the chapter: it is what a reader verifying provenance should consult, and it is what a reader wondering whether the book has quietly assumed a disputed result should check. There is exactly one row whose source is contested, and its Use column reads *none*.

Table 4.3. What the book imports from each surrogate, and where. "Unconditional" means the import is established mathematics used as such; "design" means the programme supplied a requirement or a shape of argument rather than a theorem.

From	What is taken	Used in	Status of the import
Buium	δ-rings; Frobenius lifts; jet spaces	Chapters 6 and 12	unconditional
Borger	Λ-structures as all-primes descent data	Chapters 6 and 11	unconditional
Connes–Consani	the demand that a base carry a flow, not only a point	Chapters 11 and 27	design
Fargues–Fontaine	slope filtrations with rational jumps; v-descent	Chapters 10, 19 and 31	unconditional
Bhatt–Scholze	prisms; prismatic cohomology; the Nygaard filtration	Chapters 7 and 9	unconditional
Drinfeld	the prismatisation stack WCart	Chapters 8 and 18	unconditional
q-de Rham	∇_q; the q-Hodge filtration; Habiro descent	Chapters 13 to 15	partly conditional
Arakelov	metrised bundles; arithmetic degree; the archimedean gauge	Chapters 17 and 37	unconditional
Condensed	a category in which archimedean and p-adic analysis cohabit	Chapters 16 and 17	unconditional
IUT	the design requirement of Remark 4.6.2 only	Chapter 36 (as motive)	**none mathematically**

The last row deserves emphasis because it is the row most likely to be misremembered. This book takes from inter-universal Teichmüller theory a question — can a comparison across

AUDIT PROCEDURE

1. Identify the manufactured base and test non-trivial relativity and cotangent variation.
   D1–D2
2. Write the operator's exact law; distinguish derivation, twisted operator, flow and symmetry.
   D3
3. Test all finite places, infinity, six operations and invariant trace separately.
   D4–D6
4. Demand an explicit `height/conductor` comparison and the exact function-field limit.
   D7–D8
5. Record ✓, partial, absent, contested or conditional without interpolation.
   status is part of the datum

Ten programmes against the eight requirements

Surrogate	D1	D2	D3	D4	D5	D6	D7	D8	Status
Arakelov / Vojta	—	—	—	✓	partial	—	partial	partial	classical
Buium δ	partial	partial	✓	—	—	—	partial	—	established p-adically
F_1 monoids	✓	—	—	—	—	—	—	—	established
arithmetic / scaling site	✓	—	partial	✓	—	—	—	partial	established
Borger Λ	✓	—	partial	partial	—	—	—	partial	established; rigid
Fargues–Fontaine	✓	✓	partial	—	partial	—	—	—	established p-adically
prismatisation	✓	✓	✓	—	partial	—	—	—	established p-adically
q-de Rham / Habiro	partial	partial	✓	partial	partial	—	—	✓	descent conditional
IUT	✓	—	partial	✓	—	?	?	—	published; contested
SPG substrate	*C*	*C*	*C*	*C*	*C*	*C*	*C*	*C*	architecture; ledger

CONTESTED ROW

IUT: D6 and D7 remain contested
Publication is not represented as community acceptance, and this book derives nothing from the asserted inequality.

CONDITIONAL ROW

SPG: every global pass is obligation-guarded
The row records the design target, not an established construction.

EMPTY INTERSECTION D4 ∩ D6 Among accepted established programmes, the columns for all-place simultaneity and choice-independent trace are never both passed.

✓ established pass partial or — not supplied ? contested *C* conditional architecture comparative conclusion

Plate 4.3. Scorecard: ten surrogates against the eight requirements of Criterion 3.5.1.

What it does not show: among accepted established programmes, the columns for all-place simultaneity and choice-independent trace are never both passed.

two structures be made independent of the identification between them? — and answers it with a construction of its own, the categorical trace. It takes no statement, no lemma and no estimate. If Theorem 4.6.1 is false, nothing in this book changes; if it is true, nothing in this book is thereby proved.

What is not on the map

Six programmes bearing on the same circle of questions are omitted, and the omissions are listed rather than concealed. *Deninger's* programme of foliated dynamical systems, which seeks a flow whose periodic orbits are the primes and whose Lefschetz trace is an explicit formula, is close in spirit to the trace formalism of Part V and is not used here; the survey in [Man95] is the entry point taken. *Haran's* non-additive geometry and *Durov's* generalised rings offer bases below $\mathrm{Spec}\mathbb{Z}$ with better categorical properties than monoid schemes, and would be the natural place to look for the repair contemplated in Remark 4.3.2. *Blueprints* and the systematic $\mathbb{F}_1$-geometry of Lorscheid [Lor12; Lor13] subsume several of the earlier proposals and are cited but not scored, since the obstruction of Lemma 4.3.1 applies to the additively idempotent part of the theory and the non-idempotent part is not a characteristic-one object. *Relative schemes* over a symmetric monoidal category [TV07] give the general framework into which most $\mathbb{F}_1$-proposals embed. *Weil cohomology in mixed characteristic* beyond the prismatic case, and *anabelian* approaches other than Mochizuki's [Jos20; Jos22], complete the list.

Scholium 4.8.1 (The atlas is not a classification). [NEW] *The claim established by Scholium 4.7.1 is about the ten programmes tabulated and about no others. It is not a theorem that no surrogate can satisfy* (D4) *and* (D6) *together — Part II of this book is an attempt to build one — and it is not a survey of all attempts. Its force is this: the requirements of Criterion 3.5.1 are not idly chosen, since each of the most developed programmes in the field fails at least one of them, and the two that are hardest to satisfy simultaneously are the two this book spends Parts II–V on.*

Proof. Immediate from Scholium 4.7.1 together with the observation that the quantifier there ranges over the rows of Table 4.1. The list of omissions above shows that this range is not exhaustive. □

Warning 4.8.2 (*Do not read the atlas as a priority claim*). Several of the ideas the substrate uses appear in the surveyed programmes first, and Table 4.3 is the record of that. Where a construction of Part II resembles one in the literature, the resemblance is acknowledged at the point of use, and the reader should assume borrowing rather than coincidence. In particular the gluing of a p-adic object to an archimedean one along a comparison is a very old ambition, and the novelty claimed here is confined to the specific substrate of Chapter 18, the dilation of Chapter 28, and the trace argument of Chapter 36 — and even those are claimed as constructions, not as proofs of abc.

Notes and References

Sources. Definition 4.2.1 and the jet-space adjunction follow Buium's monograph [Bui05] and his later foundations [Bui17]; Theorem 4.2.4 is the theory of δ-characters of elliptic curves from the same sources, with the crystalline interpretation from [BS19]. The Λ-ring formulation of descent to $\mathbb{F}_1$ is Borger's [Bor11b; Bor11a; Bor16], the differential calculus on arithmetic jet spaces Borger–Buium [BB11]. The arithmetic and scaling sites are Connes–Consani [CC14; CC16; CC15; CC17; CC10; CC21]. Construction 4.4.1 and the classification of bundles are as presented by de Shalit and Morrow [dS22; MOR19], with Noetherianity from Kedlaya [Ked16] and the relative theory from Scholze–Weinstein [SW20b; SW20a]. Prisms are Bhatt–Scholze [BS22; BS23]; prismatisation is Drinfeld [Dri24b; Dri24a]. The account of inter-universal Teichmüller theory follows the four published papers [Moc21a; Moc21b; Moc21c; Moc21d]; the objection is quoted from [SS18]; the alternative reconstruction is Joshi's [Jos21; Jos23; Jos24]. Where a source is not in the bibliography, the text gives the reference in full rather than guessing a key.

Attribution of the new material. Definition 4.1.1, Principle 4.1.4, Proposition 4.2.3, Proposition 4.2.5, Proposition 4.4.2, Proposition 4.5.1 and Scholium 4.7.1 are formulations made for this monograph. Their content is elementary and each is proved here in full; none imports an unproved literature claim. Proposition 4.2.3 makes quantitative a remark that is folklore among workers in arithmetic differential equations, and the numerical values in Example 4.2.2 were checked against the routines of Chapter 48. The comparative rubric of Table 4.1 was drafted in the machine-assisted process described in Chapter 56, then checked clause by clause against the primary sources; the scoring of the last row is the author's design claim and is labelled as such. Remark 4.6.2 records a design decision and is deliberately not an assertive environment.

Alternative treatments. Three other organisations were considered. A chronological survey, from Tits to prismatisation, makes the intellectual debts vivid but hides the structural point, since programmes decades apart fail the same requirement. A survey by escape route, following Principle 3.4.1, was rejected because several programmes use two routes at once — prismatisation enlarges the site and deforms the ring — and forcing them into one column misrepresents them. Omitting inter-universal Teichmüller theory on the ground that its status is unsettled was rejected because the design requirement extracted in Remark 4.6.2 is the most important lesson the atlas teaches, and it is teachable only by stating the dispute exactly. The chosen organisation flatters programmes with clean statements and penalises programmes still under construction; the reader should discount accordingly.

Historical remarks. The dates show how long the same obstruction has been felt. Tits proposed a geometry of characteristic one in 1957; Arakelov compactified $\operatorname{Spec}\mathbb{Z}$ in 1974; Manin connected $\mathbb{F}_1$ to zeta functions in lectures of the early 1990s, and Soulé's functorial proposal followed. Buium's arithmetic differential equations began in 1995 and reached monograph form in 2005. Borger's Λ-ring interpretation of $\mathbb{F}_1$ dates from 2009. The Fargues–Fontaine curve was constructed around 2010 and published in 2018. Connes and Consani introduced the arithmetic site in 2014 and the scaling site in 2015. Mochizuki's preprints appeared in 2012 and were published in 2021; the Scholze–Stix report is dated 16 July 2018. Prisms appeared in 2019 and were published in 2022. Six decades separate the first and last of these, and in that time the four devices of Chapter 2 have never all been supplied at once.

Further reading. Buium's *Arithmetic Differential Equations* remains the entry point to δ-geometry and is unusually explicit about what its methods cannot do [Bui05]. Borger's two papers on the geometry of Witt vectors are the clearest statement of the Λ-ring philosophy and repay reading alongside Chapter 6 [Bor11a; Bor11b]. De Shalit's lectures are the most economical introduction to the Fargues–Fontaine curve for a reader who wants the geometry before the p-adic Hodge theory [dS22]. Bhatt–Scholze's *Prisms and prismatic cohomology* is the foundational reference for Part I [BS22]. For the contested material the reading list is the primary one: the four IUT papers, the Scholze–Stix report, and Joshi's reconstruction, read in that order and without intermediaries [Moc21c; SS18; Jos24].

Chapter 5

Design of the Framework: Main Theorems, Proof Architecture, and the Reading Contract

Dependencies. Chapters 1–4.

Summary. This chapter is the specification of the book. It restates the eight requirements extracted in Chapter 3, describes the substrate $\mathfrak{S}$ that Parts I–II construct to meet them, and states the three principal results: a master inequality, its abc specialisation over $\mathbb{Q}$, and a trace-invariance theorem. Theorems A and B are conditional, and their obligations are named here in full rather than abbreviated; Theorem C has an unconditional categorical core and a conditional substrate instance, and the two are stated separately. We then give the dependency graph between the parts, summarise the ledger of 98 obligations by part and difficulty, record the structural facts about the obligation graph that Chapter 61 audits, set out the epistemic tagging contract, and provide reading routes. The abc conjecture is not proved in this book.

Warning 5.0.1 (*This chapter is a promise, and it is audited*). A specification chapter placed at the front of a long book is the natural home of overstatement: it is written first, read first, quoted first, and revised last. This one is therefore written to be checked against Chapter 61, which states what has actually been proved, and against the machine-readable ledger from which Table 5.3 is generated. Section 5.8 tabulates the correspondence promise by promise. Where the present chapter and Chapter 61 could be read as disagreeing, Chapter 61 governs; where either could be read as disagreeing with the ledger, the ledger governs. No statement of this chapter is offered as an independent source of truth.

5.1 The eight requirements

Criterion 3.5.1 isolated eight requirements that any substrate must meet if the Mason–Stothers architecture of Chapter 2 is to have an arithmetic transposition. They are the specification against which everything that follows is to be judged, and we restate them in compressed form so that the present chapter is self-contained.

(D1) *Non-trivial relativity.* A morphism $i\colon \operatorname{Spec}\mathbb{Z} \to \mathfrak{S}$ that is not an equivalence.

(D2) *Controlled cotangent direction.* $\mathbb{L}_{\mathbb{Z}/\mathfrak{S}} \neq 0$, in a coefficient category with support, base change, duality and finiteness.

(D3) *Derivation-like operator.* An operator ∇_q with an explicit twisted Leibniz rule, functorial enough to build a Wronskian replacement.

(D4) *Simultaneity of places.* Every finite prime and the archimedean place in one object, glued by a proved descent datum.

(D5) *Six operations and duality.* The identities needed for a relative Riemann–Hurwitz argument.

(D6) *Choice-independent trace.* A categorical trace Tr on dualisable objects, invariant under the groupoid of gluing choices.

(D7) *Compatibility with Diophantine size.* Support recovers conductor, trace recovers height or discriminant, with normalisations stated.

(D8) *The function-field limit.* Over a smooth curve the substrate direction becomes $\Omega^1_{X/k}$, ∇_q becomes d/dt, and the trace identity becomes Riemann–Hurwitz.

Table 5.1 records, for each requirement, the construction that is intended to supply it and the obligation that guards it. The right column is the actual content of the table: a requirement is not met by an intention, and where the supplier is conditional the entry says so.

Table 5.1. The eight requirements, their suppliers, and their guards.

Req.	Supplier	Status of the supplier
D1	Absolute prismatisation, Ch. 8, 11	Established p-adically
D2	Strain complex, Ch. 20	Conditional (PO-037–PO-042)
D3	∇_q on the Habiro ring, Ch. 14–15	Conditional (PO-011–PO-016)
D4	Gluing of Ch. 18	Conditional (PO-023–PO-029)
D5	Six-functor formalism, Ch. 34	Conditional (PO-062–PO-064)
D6	Trace invariance, Ch. 36	Conditional (PO-065–PO-068)
D7	Heights as traces, Ch. 37	Conditional (PO-042, PO-072, PO-075)
D8	Function-field limit, Ch. 38	Required of every construction

Requirement (D8) is not an afterthought. A framework that produced an inequality of the right shape without recovering the Chapter 2 proof in the geometric limit would have manufactured a coincidence, not a transposition; we therefore impose the limit as a standing check on each construction rather than as a single theorem at the end.

What each requirement costs

The table above records suppliers. It does not record price, and the price is the interesting quantity: a requirement met by an established theorem costs nothing, a requirement met by a conjecture costs the conjecture, and a requirement met by an object whose existence is itself an obligation costs that object plus everything downstream of it. We therefore restate the eight with their prices attached.

(D1) *Free, at one prime at a time.* Non-trivial relativity is supplied by absolute prismatisation, which is established mathematics [BS22; Dri24b]. The price is not the existence of the relativity but its uniformity: nothing in the p-adic theory produces a single object over all primes, and (D1) as stated quantifies over $\mathrm{Spec}\mathbb{Z}$, not over $\mathrm{Spec}\mathbb{Z}_p$. The gap between the two is where (D4) begins.

(D2) *Priced at the substrate.* The strain complex of Chapter 20 is defined relative to $\mathfrak{S}$. Until $\mathfrak{S}$ exists, $\mathbb{L}_{\mathbb{Z}/\mathfrak{S}}$ is notation. The obligations are PO-037–PO-042, declared in Chapter 21 where the transgression that uses them is set up, and the falsifier is explicit: a proof that the cotangent complex of the glued object vanishes, or fails base change, would end the programme at Part III.

(D3) *Priced at PO-011–PO-016.* The Jackson operator exists and its Leibniz rule is classical; what is not classical is that a q-connection descends to a structure defined at all roots of unity simultaneously with an effective bound. PO-012 is the effectivity, and the distinction between existence and effectivity is the whole of the cost.

(D4) *Priced at PO-023–PO-029, and this is the largest single price in the book.* Simultaneity of places is not a property one verifies of a known object; it is the assertion that the object exists.

(D5) *Priced at PO-062–PO-064.* Six-functor formalisms over non-schematic bases are an active field, and the identities required here are of a shape that the literature supports in analogous settings [Gal25; DG22]; the obligation is that they hold over $\mathfrak{S}$ specifically.

(D6) *Priced at PO-065–PO-068, of which PO-065 is dualisability.* The categorical statement that a trace of a dualisable object is invariant under equivalences is unconditional and is proved in Chapter 36. What is conditional is that the objects in play are dualisable and that the dilation is equivariant.

(D7) *Priced at PO-042, PO-072 and PO-075.* Compatibility with Diophantine size is a normalisation problem before it is a theorem: the trace must be shown to recover a height with the constants fixed once, and a factor of $\log 2$ lost here is a factor of 2 in the final constant. PO-042 is the discriminant–conductor comparison, PO-072 the compatibility of the trace with the Arakelov degree, PO-075 the agreement of degree and trace on determinants. Chapter F fixes the Arakelov-side normalisations so that this comparison has a fixed target.

(D8) *Free to state, expensive to maintain.* The function-field limit is a check, not a supplier, and the cost is that every construction in Parts I–V must be arranged so that the check can be applied to it. Chapter 38 performs the check; Chapter 2 is the answer it must reproduce.

Three of the eight — (D2), (D4) and (D6) — are priced at objects rather than at theorems. That is the structural fact about this design, and it is the reason the ledger is dominated by class-H records rather than by class-R ones.

Part 0 **specification** D1–D8	*Part I* **absolute base** δ, prisms, gauges
Part II **global substrate** q, Habiro, infinity	*Part III* **strain** cotangent and KS
Part IV **dilation** theta and rigidity	*Part V* **trace** six functors and RH
Part VI **Diophantine output** Szpiro and abc	*Part VII* **computation** verification engine
Part VIII **formalisation** machine audit	*Part IX* **falsifiers** obstruction atlas

Requirement → supplier → epistemic status

Req.	Required structure	Supplier	Arrow
D1	Spec $\mathbb{Z} \to \mathfrak{S}$ not an equivalence	Ch. 8, 11 · absolute prismatisation	solid p-adically
D2	non-zero controlled $L_{\mathbb{Z}/\mathfrak{S}}$	Ch. 20 · strain complex	dashed · conditional on $\mathfrak{S}$
D3	operator D_q with twisted Leibniz law	Ch. 14–15 · Habiro descent	dashed · PO-011–PO-016
D4	all finite places and infinity in one object	Ch. 18 · global gluing	dashed · PO-023–PO-029
D5	six operations and duality	Ch. 34	dashed · PO-062–PO-064
D6	choice-independent categorical trace	Ch. 36	dashed · PO-065–PO-068
D7	support = conductor; trace = height	Ch. 37	dashed · conditional
D8	exact function-field limit	Ch. 20, 21, 38 and every construction	standing check

ESTABLISHED RANGE

one prime at a time
prismatic relativity
The p-adic base, Frobenius and gauge technology are established mathematics.

GLOBAL RANGE

ledger-guarded
gluing, trace and height
Intentions count as suppliers only after their named obligations are discharged.

D8 IS A BAND ACROSS THE WHOLE ARCHITECTURE

smooth curve over k
$L \to \Omega^1_{X/k}$, $D_q \to \mathtt{d/dt}$, trace identity → Riemann–Hurwitz

acceptance criterion
recover height $\leq N^{(1)} + 2g - 2$
A similar-looking inequality is not enough.

solid established supply · dashed conditional supply · standing function-field check

Plate 5.1. The specification and its suppliers: eight requirements, ten parts, one standing check.
What it does not show: solid supply is presently p-adic; every global bridge is explicitly conditional.

5.2 The substrate in outline

The object called for by (D1)–(D8) is written $\mathfrak{S}$ and called the *global substrate*. It is not a scheme, and it is not built in one piece. Three constructions supply three ranges of places, and a descent datum is asserted to glue them; the assertion is an obligation, not a theorem.

The p-adic base. For a fixed prime p, absolute prismatisation supplies a relative situation. The Cartier–Witt stack WCart and the absolute prismatisation Δ of Bhatt–Lurie and Drinfeld carry a Frobenius that is not the identity, so a δ-structure is extra data rather than a tautology, and the cotangent complex of $\mathrm{Spec}\mathbb{Z}_p$ over the prismatic base is non-zero. This is the one place where the obstruction of Chapter 3 is already broken by established mathematics. Parts I develops it: δ-rings and Witt vectors (Chapter 6), prisms (Chapter 7), the prismatisation (Chapter 8), F-gauges and the syntomic stack (Chapter 9), Rees constructions (Chapter 10), and the resulting absolute base (Chapter 11). What it does not give is a single object over all primes at once.

The all-primes operator. The q-de Rham complex replaces d by the Jackson operator ∇_q, whose Leibniz rule is twisted by the q-shift. Over a single prism this is prismatic cohomology in disguise. The Habiro ring $\mathcal{H}$ is the arithmetic of q-series convergent at every root of unity simultaneously, and it is the mechanism by which a q-connection can be asked to exist at all primes at once rather than prime by prime. Part II builds this: q-de Rham cohomology (Chapter 13), the Habiro ring (Chapter 14), and the descent statement (Chapter 15) whose effectivity is PO-012. What it does not give is the archimedean place.

The archimedean place. Condensed and analytic geometry supply a category in which $\mathbb{R}$ and $\mathbb{C}$ are treated by the same formalism as $\mathbb{Q}_p$, so that the archimedean contribution is an object rather than an estimate appended by hand. Part II continues with condensed mathematics (Chapter 16) and the archimedean gauge (Chapter 17), whose obligations are PO-017–PO-022, and Chapter 18 proposes the gluing $\mathfrak{S}$ itself, with adelic curves in Chapter 19 as the comparison to the existing arithmetic of places.

Warning 5.2.1 (*What is asserted and what is proved*). Each of the three ingredients is established mathematics within its own range. The object $\mathfrak{S}$ is not. Its existence, its symmetric monoidal structure and the effectivity of its descent datum are the obligations PO-023–PO-029, and every later statement in this book that quantifies over $\mathfrak{S}$ inherits them. No argument here treats $\mathfrak{S}$ as constructed.

The three ingredients are not interchangeable

It is worth saying plainly why three constructions are needed rather than one, because a reader who suspects that the plurality is a symptom of not knowing which one is right is entitled to an answer.

The p-adic ingredient supplies *relativity*: over the prismatic base, $\mathrm{Spec}\mathbb{Z}_p$ has a non-zero cotangent complex, which is exactly the failure of absoluteness that Chapter 3 shows to be indispensable. It supplies this completely and unconditionally, and it supplies nothing else; in particular it is silent about the interaction between two distinct primes, because WCart is a p-adic object and there is no morphism relating WCart_p to WCart_ℓ for $\ell \neq p$.

The Habiro ingredient supplies *simultaneity at the finite places*. A q-series that converges at every root of unity is an object that sees all primes at once, and the q-de Rham complex converts that into a cohomological statement. It supplies this conditionally — PO-011–PO-016 — and it supplies nothing at the archimedean place, because there is no root of unity there.

The condensed ingredient supplies *the archimedean place as an object*. The point is not that $\mathbb{R}$ can be studied analytically; it is that in a category of analytic rings, $\mathbb{R}$ and $\mathbb{Q}_p$ are objects of the same kind, so that a descent datum can be formulated over the collection of all of them. It supplies this conditionally — PO-017–PO-022 — and it supplies no derivation-like operator of its own.

Each therefore covers exactly what the others miss, and the arrangement is forced rather than chosen. What is *not* forced, and what is the substantive risk, is that the three admit a common refinement at all. That is PO-023, and there is no argument in this book that reduces it to the three ingredient theories. **[CONDITIONAL : PO-023]**

5.3 Statement of the main theorems

Three results carry the architecture. Each is stated here in the form used later, with its obligations named; the proofs, such as they are, occur in the chapters indicated. The reader who reads nothing else in this chapter should read the tags.

One editorial decision needs defending before the statements. The obligation lists below are printed in full, at length, rather than summarised as "the obligations of Part II" or abbreviated to the four or five that a reader is most likely to remember. A summarised list is a list one cannot check, and a list one cannot check is exactly the mechanism by which a conditional theorem becomes an unconditional one over the course of a long manuscript. The lists are therefore reproduced verbatim from the chapters that prove the statements, and any discrepancy between a list here and a list there is a defect in this chapter, not there.

Theorem 5.3.1 (Theorem A: the master inequality). ***[CONDITIONAL : PO-001, PO-002, PO-007, PO-011, PO-012, PO-013, PO-014, PO-015, PO-016, PO-017, PO-018, PO-019, PO-020, PO-021, PO-022, PO-023, PO-024, PO-025, PO-026, PO-027, PO-028, PO-029, PO-037, PO-038, PO-039, PO-040, PO-041, PO-042, PO-043, PO-044, PO-045, PO-046, PO-047, PO-048, PO-049, PO-050, PO-053, PO-054, PO-055, PO-056, PO-062, PO-063, PO-065, PO-066, PO-067, PO-068, PO-070, PO-071, PO-072, PO-073, PO-074, PO-075, PO-076, PO-077, PO-078, PO-079, PO-080, PO-081, PO-083]*** *Let $\mathfrak{S}$ be the global substrate. For every number field K and every real ε with $0 < \varepsilon \leq 1$ there exists a real constant $C_A(K, \varepsilon)$, depending only on K, on ε and on the substrate — and on no feature whatever of X — such that for every admissible semistable strain package X over $\mathfrak{S}$ with base field K,*

$$\vartheta(X) \;\leq\; (1+\varepsilon)\bigl(q_K(X) + r_K(X)\bigr) \;+\; C_A(K,\varepsilon). \tag{5.3.1}$$

Here $\vartheta(X) = \mathrm{Tr}(\Theta_\lambda \mid M_{X/\mathfrak{S}})$ is the trace height, $q_K(X)$ the normalised conductor term and $r_K(X)$ the different term, all normalised as in Chapter 40.

This is Theorem 39.1.6 verbatim, and it carries fifty-nine obligations. Three features of it are load-bearing and are easy to lose in paraphrase.

(A1) *The factor $(1+\varepsilon)$ multiplies both terms on the right.* The variant in which it multiplies q_K alone, leaving r_K unweighted, is strictly stronger and is *not* what the argument of Chapter 39 delivers; the different term is absorbed by the same archimedean estimate that produces the conductor term, and the estimate does not distinguish them. A base-rigid form in which r_K disappears into the constant is available and is Corollary 39.1.9, but it holds only for base-rigid packages.

(A2) *The range of ε is $0 < \varepsilon \leq 1$, not $\varepsilon > 0$.* The restriction is harmless for the applications, since a statement for small ε implies one for large ε with the same constant, but it is stated as proved rather than as one would like it.

(A3) *The quantifier order is $\forall K\, \forall \varepsilon\, \exists C\, \forall X$.* Moving $\exists C$ to the left of $\forall K$ would make the statement uniform over number fields; that is strictly stronger than anything proved

here and stronger than the classical abc conjecture over number fields. Moving it to the right of $\forall X$ would make it vacuous. Remark 39.1.8 discusses both.

The shape of (5.3.1) is the arithmetic transposition of Riemann–Hurwitz: a degree on the left, a support term and a ramification term on the right. It is proved, conditionally, in Chapter 39, by combining the dilation comparison of Chapter 30 with the trace formalism of Part V. The constant decomposes as an analytic part, a field part, and a contribution of PO-079; the first two are computable by the audit of Chapter 40 and the third is not.

Theorem 5.3.2 (Theorem B: abc over $\mathbb{Q}$). *[CONDITIONAL : PO-001, PO-002, PO-007, PO-011, PO-012, PO-013, PO-014, PO-015, PO-016, PO-017, PO-018, PO-019, PO-020, PO-021, PO-022, PO-023, PO-024, PO-025, PO-026, PO-027, PO-028, PO-029, PO-037, PO-038, PO-039, PO-040, PO-041, PO-042, PO-043, PO-044, PO-045, PO-046, PO-047, PO-048, PO-049, PO-050, PO-053, PO-054, PO-055, PO-056, PO-062, PO-063, PO-065, PO-066, PO-067, PO-068, PO-070, PO-071, PO-072, PO-073, PO-074, PO-075, PO-076, PO-077, PO-078, PO-079, PO-080, PO-081, PO-082, PO-083, PO-086] Assume the sixty-one obligations listed, together with the three unassigned named hypotheses of Warning 5.3.4. Then for every ε with $0 < \varepsilon \leq 1$ there is a constant $K_B(\varepsilon) < \infty$ such that, for all coprime non-zero integers $a + b = c$,*

$$\max(|a|, |b|, |c|) \;\leq\; K_B(\varepsilon)\ \mathrm{rad}(abc)^{1+\varepsilon}. \tag{5.3.2}$$

Warning 5.3.3 ($K_B(\varepsilon)$ is finite, not computable). Theorem 5.3.2 asserts that a finite constant exists. It does not assert that one can be computed, and no reading of it may supply the word "effective". Equation (43.1.3) exhibits $K_B(\varepsilon)$ in closed form in terms of $C_A^{\mathrm{mod}}(\mathbb{Q}, \varepsilon)$, and every factor of that expression except one is effectively bounded; the exception is the archimedean contribution controlled by PO-079, whose effectivity is the separate and *open* obligation PO-084. Theorem 43.3.4 states precisely what fails. A version of this chapter that promised an effectively computable constant would be promising PO-084 as well, without listing it, and would therefore be overstating the theorem in the one way most likely to escape notice.

Warning 5.3.4 (Three hypotheses carry no identifier). The route to Theorem 5.3.2 passes through three named hypotheses which have no ledger identifier: the classifying-package obligation (CM), the discriminant metric interface (MI_Δ) of Chapter 41, and the semistable-reduction passage (SR). They are recorded in the *Upstream* fields of PO-082 and PO-086 and used in the text of Chapters 41–43, but they are not `proofobligation` environments, so no count in this volume, no graph, and no automated gate sees them. The accurate description of the antecedent of Theorem 5.3.2 is therefore *sixty-one ledger obligations together with three unassigned named hypotheses*, and the second half of that description is invisible to the tooling. Warning 61.1.3 states the same fact where the audit is performed.

Warning 5.3.5 (How Theorem 5.3.2 is to be read). Theorem 5.3.2 is an implication. Its hypotheses are unproved, and this book does not prove them; what is offered is a derivation of (5.3.2) from a named finite list, together with the means to check whether that list is being discharged. The abc conjecture is not proved in this book. A reader who removes the conditional tag from Theorem 5.3.2 has misquoted it.

Theorem C is stated twice, because it has an unconditional half and a conditional half and conflating them would misdescribe both. The first statement is a theorem about symmetric monoidal categories and owes nothing to this book's substrate; the second is its instance over $\mathfrak{S}$, and owes everything.

Theorem 5.3.6 (Theorem C: trace invariance, categorical form). *[NEW] Let $\mathcal{C}$ be symmetric monoidal, let $\mathcal{G}$ be a pointed connected anima, let $\rho\colon \mathcal{G} \to \mathcal{C}^{\simeq}$ be an action with $X := \rho(g_0)$ dualisable, and let $\tilde{f}$ be an equivariant lift of an endomorphism f of X. Then $\mathrm{Tr}(f_g \mid X_g) = \mathrm{Tr}(f \mid X)$ in $\pi_0\mathrm{End}_{\mathcal{C}}(\mathbf{1})$ for every g and every path from g_0 to g, and $g \mapsto \mathrm{Tr}(f_g \mid X_g)$ is a canonical point of $\mathrm{End}_{\mathcal{C}}(\mathbf{1})^{h\mathcal{G}}$. Its component $[\mathrm{Tr}(f)] \in \pi_0\mathrm{End}_{\mathcal{C}}(\mathbf{1})$ is independent of every choice in the connected groupoid $\mathcal{G}$.*

Scholium 5.3.7 (Theorem C: substrate instance). *[CONDITIONAL : PO-065, PO-066, PO-067, PO-068] Let $\mathcal{C}_{\mathfrak{S}}$ be the symmetric monoidal category of nuclear modules over $\mathfrak{S}$, let $M_{X/\mathfrak{S}}$ be the arithmetic avatar of X, let $\mathcal{G}$ be the ∞-groupoid of gluing choices of Chapter 25, and let Θ_λ be the theta dilation of Chapter 28. Assume PO-065 (dualisability of $M_{X/\mathfrak{S}}$), PO-066 (coherence of the gluing action), PO-067 (coherent equivariance of Θ_λ), and the canonical scalar-local-system trivialisation PO-068 as stated in Proof Obligation 36.7.4. PO-065–PO-067 give the fixed-category conclusion of Theorem 5.3.6; if substrate descent instead yields a local system of ambient categories, PO-068 identifies its trace target with the common coefficient object. Granting all four, $\mathrm{Tr}(\Theta_\lambda \mid M_{X/\mathfrak{S}})$ is a well-defined element of $\pi_0\mathrm{R}\Gamma(\mathfrak{S}, \mathcal{O}_{\mathfrak{S}})$, independent of every gluing choice.*

Remark 5.3.8 (What the split is for). The categorical statement is proved in Chapter 36 from the standard manipulations of duality data, and its difficulty is not the point; what is new there is the arrangement in which it is applied. Separating it from the substrate instance costs a page and buys the ability to say exactly where the difficulty is. It is not in the invariance. It is in the four hypotheses of Scholium 5.3.7, each of which asserts something about objects that do not yet exist, and of which PO-067 is class O and is described in the ledger as the load-bearing obligation of the framework. A presentation that stated only the substrate instance, tagged [CONDITIONAL], would leave a reader unable to tell whether the conditionality lay in the category theory or in the arithmetic; it lies entirely in the arithmetic. In one fixed ambient category the unit target is constant automatically. PO-068 is a separate arithmetic requirement only for the local-system implementation: it canonically trivialises the varying scalar target, exactly as stated in Proof Obligation 36.7.4.

Remark 5.3.9 (Why Theorem C is the architectural claim). The comparison of Chapter 30 relates two descriptions of the same deformation. In the surrogate reviewed in Section 4.6 such a comparison is effected by transporting numerical data along an identification of two copies of a coefficient line; the objection of [SS18] is that the identification is not compatible with the structures on either side. Theorem 5.3.6 is this book's structural response: it replaces the identification by a trace, and a trace of a dualisable object is invariant under the groupoid of choices by construction, so there is no identification left to be incompatible. This does not vindicate the surrogate, and it is not offered as a repair of it; it is a different design with a different, and explicitly listed, set of unproved inputs. Note also what the response does not do: it removes an objection to a mechanism, and removing an objection is not the same as supplying a proof. The four obligations of Scholium 5.3.7 remain open.

Theorem C is proved in Chapter 36 as Theorem 36.3.1 in its categorical form; the audit of the four substrate hypotheses is Scholium 36.4.1. Of those four, PO-067 rather than PO-065 is the one the ledger records as load-bearing, and in the dependency graph of the whole book neither is the deepest: PO-017, the archimedean gauge, reaches sixty-eight later identifiers, against ten for PO-065 (Lemma 61.3.2). The obligation a reader should attack first is therefore not the one this book's own machinery makes most conspicuous.

5.4 The proof architecture

The book is a chain of four movements. Parts I–II construct a candidate base. Parts III–IV construct, over that base, the deformation-theoretic data of a Frey-type object and the dilation that compares two of its descriptions. Part V supplies the categorical apparatus that turns the comparison into a number. Part VI converts that number into Diophantine statements. Parts VII–IX are computational, formal and reflective, and the mathematical spine of the book does not route through them; the one qualification to that statement is recorded in Warning 5.4.1 and is not cosmetic.

```
Candidate base:
  Part I:  prisms, gauges
           absolute base (Ch. 11)
  Part II: q-de Rham, Habiro, condensed
           all-primes operator
           archimedean input
                    |
  Part II, Ch. 18: substrate S
                    |
Deformation data:
  Part III: strain, Kodaira-Spencer
            log shells, theta line
  Part IV:  dilations, theta
            cocycle, rigidity
                    |
Categorical and arithmetic output:
  Part V:   six functors, dualisability
            trace invariance (Thm C)
            heights as traces
            Riemann-Hurwitz
                    |
  Ch. 39:   Theorem A
                    |
  Part VI:  Szpiro (Ch. 41), Frey (Ch. 42)
            Theorem B (Ch. 43), number fields
            consequences, limits
```

Read the diagram as a supply chain. Table 5.2 names, for each part, what it must deliver and which later part consumes it; a part that delivers nothing consumed downstream is expository, and is marked as such.

Two features of the chain deserve emphasis. First, it has a single point of failure at Chapter 18: if the gluing of the three ranges of places is not effective, nothing downstream survives, and no amount of work in Parts III–VI compensates. Second, the conversion of Part V is where the trace becomes a number, and it is therefore where (D6) and (D7) must both hold at once; the book's principal novelty and its principal risk are located in the same place.

Warning 5.4.1 (*The chain is not quite a chain*). The reading of Table 5.2 as a supply chain running left to right and stopping at Part VI is a simplification, and one place it is false should be recorded here rather than discovered later. PO-084, the effectivity of the constant in Theorem B, is declared in Chapter 43, which is in Part VI; its *Upstream* field names PO-087, PO-088 and PO-089, which are declared in Chapter 50, which is in Part VII. There is therefore at least one edge of the obligation graph running from Part VII back into Part VI, and the sentence "nothing in Parts I–VI depends on Parts VII–IX" — which earlier drafts of this chapter contained —

Table 5.2. What each part supplies and where it is consumed.

Part	Supplies	Consumed by
0	The problem, the obstruction, the specification	All
I	Absolute base; δ-structures; F-gauges	II, III
II	∇_q at all primes; archimedean gauge; $\mathfrak{S}$	III–V
III	Strain complex; log shells; theta line	IV, V
IV	Theta dilation; cocycle; rigidity	V
V	Six functors; Tr; heights; Riemann–Hurwitz	VI
VI	Theorems A and B and their consequences	—
VII	Algorithms; verification engine	Expository
VIII	Formalisation; the ledger as an artefact	Expository
IX	Obstruction atlas; falsifiers; problems	Expository

is false as stated. What is true is the weaker and still useful statement that no obligation in the closure of Theorem A lies in Parts VII–IX, so the *inequality* does not depend on the computational part of the book; what depends on it is the claim that the constant can be computed, which Warning 5.3.3 does not make.

The obligation graph, and what its shape says

The supply chain above is drawn between parts. The ledger supports a finer graph, between individual obligations, and the facts recorded here are the ones this chapter promises and Chapter 61 audits.

Proposition 5.4.2 (Structural facts about the declared graph). *[NEW] Let G be the directed graph whose vertices are the* 98 *ledger records and whose edges are the declared* Upstream *relations. Then G has* 291 *edges, is acyclic, and has eleven vertices of indegree zero. The declared hypothesis set of Theorem 5.3.1 has* 59 *elements and is closed upstream, so it equals its own transitive closure; the closure of the route to Theorem 5.3.2 adds PO-082 and PO-086 and has* 61 *elements; and* 37 *of the* 98 *records lie outside that closure.*

The proposition is a finite computation on a machine-readable file, and it is performed in Proposition 61.1.2, which is where the proof belongs. Four consequences are worth stating in the specification, because each of them is a promise a reader may hold this book to.

(G1) *The hypothesis set is closed, so the list is the whole list.* Because the 59 obligations of Theorem 5.3.1 contain every upstream of every one of their members, a reader who checks the fifty-nine has checked everything the theorem declares. There is no second, hidden layer.

(G2) *One of the eleven roots is not a root.* Ten of the eleven indegree-zero records declare an upstream lying outside the ledger — a representability theorem, a descent package, a pinned library interface — and are genuine roots. The eleventh, PO-082, declares "all obligations of Theorem A", which is an *internal* dependence written in prose rather than in identifiers, so the tooling computes its upstream closure as empty when the intended closure has 59 elements. Proposition 61.3.1 states this; it is a defect of the ledger's own encoding, and it is disclosed rather than repaired, because repairing it silently would change a published count.

(G3) *Risk is concentrated, and not where the chapter titles suggest.* PO-017, the existence and uniqueness of the archimedean gauge, reaches sixty-eight later identifiers; PO-011, the Habiro-side all-primes operator, reaches thirty-four; PO-023, the substrate itself, reaches twenty-six. By contrast PO-065, which the statement of Theorem C makes most conspicuous, reaches ten. A referee allocating effort should follow the reach numbers of Lemma 61.3.2 and not the prominence of a statement.

(G4) *Small cuts exist.* Three families of five to eight obligations each disconnect the declared route, and are recorded as C_{base}, C_{inv} and C_{arith} in Proposition 61.3.3. That small cuts exist is the useful fact: refuting any one member of any one family ends the programme, so the programme is falsifiable at a cost far below the cost of reading it.

The reader should notice what (G2) implies about this chapter's own reliability. A specification is a set of promises; a ledger is the mechanism by which the promises are counted; and (G2) exhibits a case in which the mechanism miscounts because a dependency was written in words. The remedy adopted here is not to claim that the tooling is complete but to state, at each point where it is not, what it misses — Warning 5.3.4 and (G2) being the two such points known at the time of writing.

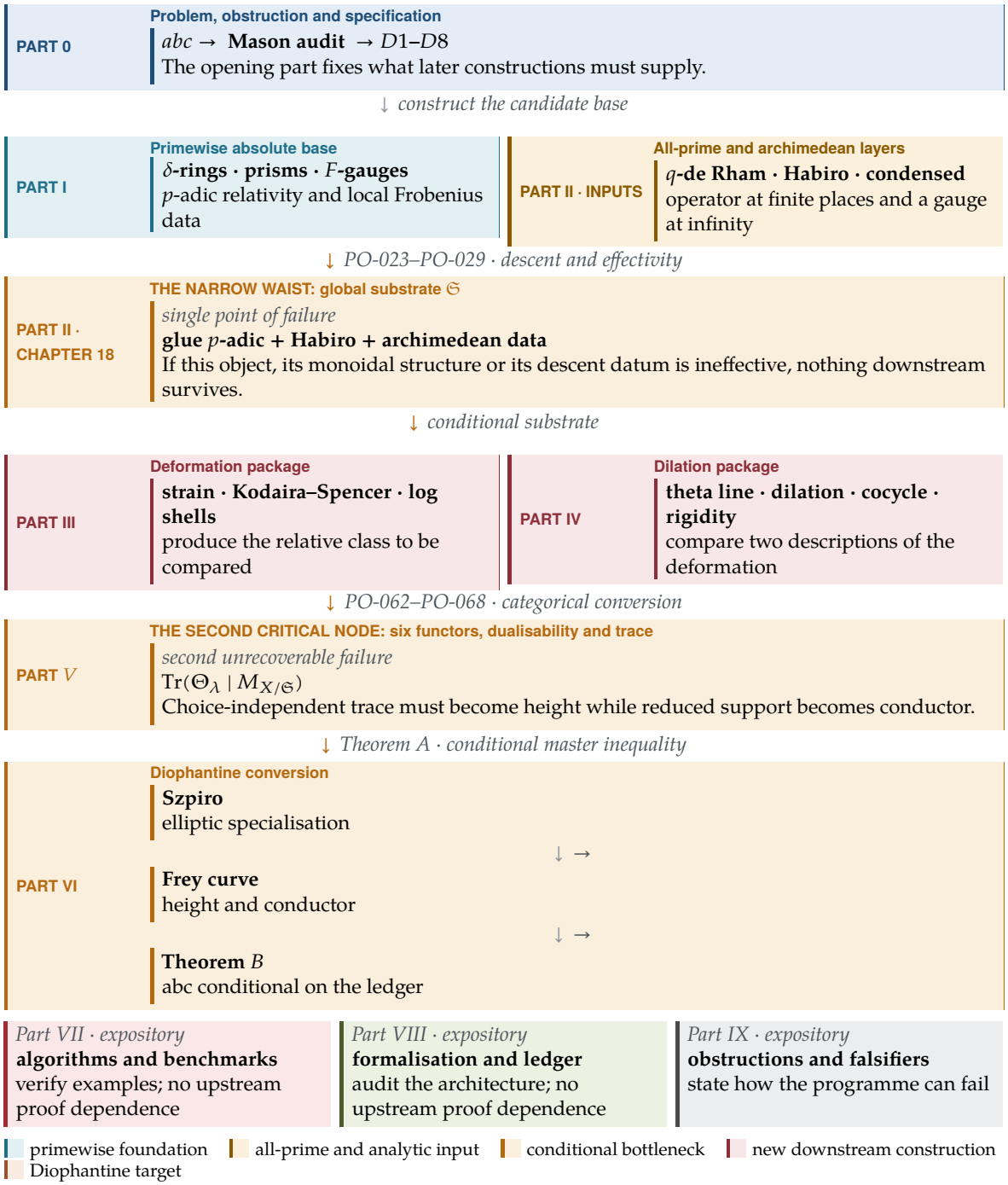


Plate 5.2. The dependency graph of the ten parts, with the single point of failure at Chapter 18.

What it does not show: if this object, its monoidal structure or its descent datum is ineffective, nothing downstream survives.

Table 5.3. The ledger by part and difficulty class.

Part	R	S	H	O	Total
0	0	0	0	0	0
I	0	5	2	0	7
II	0	8	13	8	29
III	0	0	2	4	6
IV	0	4	10	5	19
V	6	9	4	1	20
VI	1	3	1	0	5
VII	0	0	1	2	3
VIII	1	0	2	2	5
IX	0	0	1	1	2
App.	1	1	0	0	2
Total	9	30	36	23	98

5.5 The proof-obligation ledger

A framework whose unproved inputs are scattered through sixty chapters cannot be audited. We therefore maintain a source-generated numbered ledger, declared once each in the chapter where an obligation is first needed, in a quarantined environment that cannot be mistaken for a theorem. Every declaration carries seven fields: statement, status, risk, owner, upstream, downstream, and falsifier. The last is the operative one: an obligation with no stated falsifier is a wish, and the ledger admits none. Appendix H reproduces the ledger in full, sorted and cross-indexed.

Each obligation carries a difficulty class, one of four letters. Class R collects the obligations that are formal or technical within existing theory: a diagram chase, a compatibility, a convergence estimate of a kind already in the literature. Class S collects those requiring a substantial new comparison between two established theories. Class H collects the foundational ones, on which a whole part rests. Class O collects the programme-critical ones, for which no method of proof is currently known to the author. The letters map onto the five risk grades used inside chapters as R = I–II, S = III, H = IV, O = V. The classification is a claim about difficulty, and it may be wrong in either direction; it is recorded so that it can be disputed.

The shape of Table 5.3 is itself informative. Twenty-three obligations lie in the hardest class, concentrated principally in Parts II–IV and in the certification and formalisation layers. The thirty-six class-H records show that the problem is not confined to a few spectacular conjectures: many comparison and effectivity interfaces remain open. A reader forming a judgement about the framework should begin with the high-reach chain PO-017–PO-021, then inspect PO-023–PO-026, PO-037, PO-046–PO-049, and PO-067. These are load-bearing interfaces; the remaining records are not thereby reduced to bookkeeping.

Remark 5.5.1 (*Why the ledger is numbered rather than narrated*). Unproved inputs described in prose migrate. A hypothesis introduced as an assumption in one chapter reappears two hundred pages later as a known fact, not through dishonesty but through the ordinary erosion of memory. Numbering prevents this: a numbered obligation cannot be silently promoted, because the number travels with the statement into every [CONDITIONAL] tag that depends on it, and the audit tooling of Chapter 57 checks the correspondence mechanically.

What the numbering does not prevent

The remark above is a claim about a mechanism, and mechanisms have edges. Three failure modes survive numbering, and a specification that did not name them would be advertising a guarantee it cannot give.

A hypothesis that is never numbered is never checked. The three unassigned hypotheses of Warning 5.3.4 are used in the text of Chapters 41–43 and appear in the *Upstream* prose of two records, but because they are not `proofobligation` environments they contribute nothing to any count in this volume. The discipline catches migration; it does not catch omission.

An upstream written in prose is invisible to the graph. This is (G2) above. The field is free text by design, because most upstreams are external theorems that have no identifier; the price is that an internal upstream written the same way is silently dropped.

A difficulty class is an opinion. The letters R, S, H, O are the author's estimate of how hard each obligation is, recorded so that the estimate can be contradicted. Nothing in the tooling validates them, and the history of this subject offers several examples of statements that moved two classes in one direction or the other once someone competent looked at them. A reader who disputes a class should say so; a reader who treats the class as a measurement has misread the table.

Against these three, the numbering buys one thing that nothing else in the book buys: it makes the antecedent of every conditional statement a finite, printed, checkable object. Whether the antecedent is plausible is a separate question, on which this chapter takes no position and Chapter 58 takes several.

5.6 The reading contract

Every non-trivial assertion in this book carries exactly one epistemic tag, printed with the statement and coloured in the electronic version. There are five, and their meanings are fixed.

Table 5.4. The epistemic tags.

Tag	Meaning
[KNOWN]	In the literature; a citation is mandatory and the proof, if given, is a convenience
[REPROVED]	Known, but proved here by a different route or in greater generality; the delta is stated in one sentence
[NEW]	First proved here, unconditionally, from tagged inputs
[CONDITIONAL]	Proved modulo the listed obligations, which are always named
[OPEN]	Asserted, not proved here, and not proved elsewhere

A sixth tag, [DISPUTED], is used only for statements claimed in the literature and contested there; where it appears, the contesting party is named. The tags obey one rule, which is the hinge of the whole contract:

> **The cardinal rule.** No statement tagged [NEW] may depend, directly or transitively, on a statement tagged [OPEN], or on a [CONDITIONAL] statement whose obligations it does not itself declare.

There is no third option between [NEW] and [CONDITIONAL] and no exception. A result whose proof invokes an obligation is conditional on that obligation, however plausible the obligation may be, and the transitive closure is computed mechanically rather than by judgement. The promise this makes to the reader is narrow and precise: *no untagged assertion appears in this book*, so that the epistemic status of any statement can be determined by looking at it, without reconstructing its provenance.

Four things the contract does not promise

The tags are worth exactly what they say and no more, and it is easier to keep them honest if the reader knows in advance where their authority stops.

(T1) *[KNOWN] is a claim about the literature, not about correctness.* It asserts that the statement appears in a cited source in essentially this form. It does not assert that the source is correct, that the present formulation is faithful in every detail, or that the normalisations agree; the last of these is a real risk in Arakelov-theoretic statements and is why Chapter F fixes normalisations once and states which convention each cited source uses.

(T2) *[CONDITIONAL] is a claim about the declared list, not about the proof.* Tagging a statement conditional on a list asserts that the proof given here uses those obligations. It does not assert that they are sufficient in any absolute sense, and it certainly does not assert that they are necessary. An error in a proof is not detected by its tag.

(T3) *[OPEN] does not mean hard, and [NEW] does not mean deep.* [OPEN] means the statement is not proved here and the author knows no proof elsewhere. [NEW] means first proved here, unconditionally, from tagged inputs; several [NEW] statements in this book are finite computations on a data file, and they are tagged [NEW] because that is what the tag means, not because they are difficult. Proposition 5.4.2 is one such.

(T4) *The cardinal rule is enforced only where it can be seen.* The closure check reads `\COND` tags and cross-references. A dependence that is present in the mathematics but absent from both — a lemma used but not cited, a hypothesis carried in prose — is not visible to it. This is the same limitation as (G2) and Warning 5.3.4, in a third guise, and it is the reason the audit of Chapter 61 is performed by reading as well as by running the tools.

We therefore state the position of the book without hedging. A framework is constructed. Within it, an inequality of Riemann–Hurwitz shape is derived from a named finite list of unproved inputs, and the abc inequality over $\mathbb{Q}$ is derived from that inequality together with the classical Frey–Hellegouarch construction. The abc conjecture is not proved in this book, and nothing in it should be read as claiming otherwise; abc follows from the framework if, and only to the extent that, the obligations of the ledger are discharged. What the book offers instead is a design whose failure modes are enumerated, whose unproved inputs are counted, and whose falsifiers are written down.

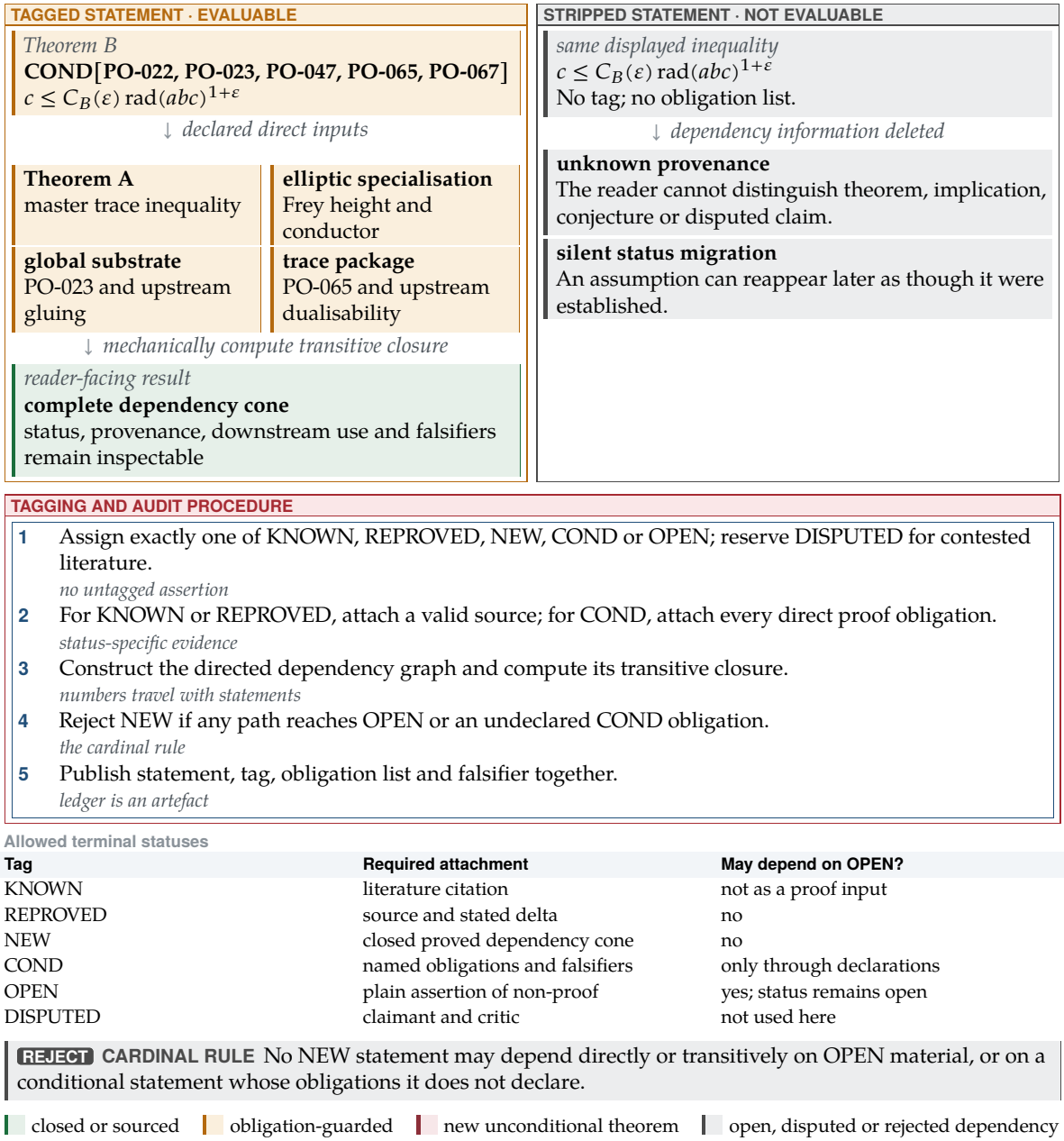


Plate 5.3. The tagging contract: a statement, its tag, and the dependency cone the tag makes visible.

What it does not show: no NEW statement may depend directly or transitively on OPEN material, or on a conditional statement whose obligations it does not declare.

5.7 How to read this book

Sixty-two chapters is more than most readers want. Four routes are provided; each is self-contained in the sense that it states its own imports.

The arithmetic geometer. Chapters 1–5, then Part I entire (6–12) for the absolute base, then 13–15 and 18 for the all-primes operator and the gluing, then Part III (20–26) for the deformation theory. Chapters 28, 30 and 36 give the dilation, the comparison and the invariance. Part VI may be read for the payoff. This is the longest route and the one for which the book is primarily written.

The analytic number theorist. Chapter 1 for the statement and its equivalents, Chapter 2 for the model argument, then Chapters 37–41 and 43 for heights, the master inequality, the constants and the abc specialisation. Chapter 40 is the one to read adversarially: it is where the uniformity of every constant is fixed, and a defect there would be visible without any prismatic machinery. Chapter 47 states what the framework cannot reach.

The category theorist. Chapters 3–5, then Part V entire (34–40). Chapters 34–36 are the technical core: six-functor formalisms, dualisability, and the invariance of the trace under a groupoid of choices. Chapter 25 supplies the groupoid, and Chapter 26 the indeterminacy calculus it replaces. The substrate may be treated as a black box satisfying the axioms listed in Chapter 34; a reader who does so should note that the black box is PO-023–PO-029.

The reader who wants only Theorem B. Theorem 5.3.1 and Theorem 5.3.2 above, then Chapter 42 for the Frey–Hellegouarch specialisation and Chapter 43 for the derivation, then Appendix H for the obligations on which both depend. That is four chapters and an appendix, and it is enough to evaluate the claim; it is not enough to evaluate the framework.

The referee. The shortest route that permits a judgement is this chapter, Chapter 61, and Chapter H, read in that order and then this chapter again. Chapter 61 is written to be read against the present one and states, statement by statement, what survives when the obligations are withheld; Chapter H supplies the falsifiers. A referee who then wishes to attack rather than to assess should go to PO-017 in Chapter 17 and to PO-023 in Chapter 18, in that order, on the reach evidence of (G3); attacking the trace machinery of Part V is a larger investment for a smaller return.

Remark 5.7.1 (*On the Leitfaden*). The front matter contains a chapter-level dependency diagram. It is drawn from the same source as Table 5.2 and should agree with it; where they disagree, the table is authoritative, because it is the one the ledger indexes.

5.8 The specification checked against the audit

Chapter 61 is the chapter that states what has been proved. It is written after the mathematics and against the ledger; the present chapter is written before the mathematics and against an intention. The two are therefore the natural place for a discrepancy to hide, and this section closes that gap explicitly, promise by promise.

Two entries of Table 5.5 deserve comment, because they are the places where an earlier form of this chapter did not agree with the audit and where a reader comparing editions may notice a change.

The obligation counts. A specification chapter is under continuous pressure to abbreviate. Four obligations are memorable, fifty-nine are not, and a list of fifty-nine consumes half a page. The abbreviation is nevertheless illegitimate: a reader who checks four and concludes that the antecedent is small has been misled by the presentation, not by the mathematics. Both theorems above therefore print their lists in full, and the lists are copied from Theorem 39.1.6 and from Setup 43.1.1 rather than reconstructed.

Table 5.5. Each promise of this chapter and the statement that audits it.

Promise made here	Audited by	Verdict
Theorem A holds under 59 named obligations	Proposition 61.1.2(i), (ii)	Agrees; the set is upstream-closed
Theorem B holds under 61 obligations	Proposition 61.1.2 (iii)	Agrees
Three further hypotheses carry no identifier	Warning 61.1.3	Agrees
$K_B(\varepsilon)$ is finite, not computable	Theorem 43.3.4, PO-084	Agrees; effectivity is open
Theorem C has an unconditional core	Theorem 36.3.1	Agrees
The substrate instance is conditional	Scholium 36.4.1	Agrees; four obligations
37 records lie outside the closure	Proposition 61.1.2(iv)	Agrees
Eleven roots, of which one is false	Proposition 61.3.1	Agrees; PO-082
Risk concentrates on PO-017	Lemma 61.3.2	Agrees; reach 68
Small cuts exist	Proposition 61.3.3	Agrees; three families
Something unconditional is proved	Proposition 61.1.9	Agrees; classical inputs

The shape of the master inequality. The right-hand side of (5.3.1) is $(1+\varepsilon)(q_K+r_K)+C_A$, with the factor on both terms. The variant $(1+\varepsilon)q_K + r_K + C_A$ is a different and strictly stronger statement, and it is not what Chapter 39 proves; the difference is $\varepsilon\, r_K(X)$, which is unbounded over the class of packages and therefore not absorbable into $C_A(K,\varepsilon)$. A specification that stated the stronger form would be promising a theorem the book does not contain, and the error would be invisible to every count in the volume, since both forms carry the same obligations. This is the reason (5.3.1) is transcribed rather than paraphrased.

Principle 5.8.1 (The specification may not exceed the audit). *[NEW] Where the present chapter and Chapter 61 can be read as making different claims about the same statement, the reading on which the present chapter claims less is the correct one, and the difference is a defect here. Nothing in this chapter is a source of mathematical content: every statement in it is either a restatement of something proved elsewhere in the volume, with its label given, or an organisational claim about the volume itself.*

The purpose of Principle 5.8.1 is not rhetorical. It fixes the direction in which an inconsistency is to be resolved, so that a discrepancy found by a reader has a determinate remedy rather than becoming an occasion for adjudication between two chapters of equal standing.

Notes and References

Sources. The eight requirements restated in Section 5.1 are those of Criterion 3.5.1, which was in turn extracted from the analysis of Mason–Stothers in Chapter 2; for the classical statement of that theorem see [GAL20]. The abc conjecture in the form (5.3.2) is due to Masser and Oesterlé [Mas85]; the survey [Mue93] remains a convenient entry point, and [Elk91] is the standard reference for the implication to Mordell. The three ingredients of Section 5.2 are, respectively, [BS22; BS23; Lur26; Dri24b] for prisms and absolute prismatisation, [CFG+25; GO26] for the Habiro ring in its analytic guise, and [Sch22; Com22; ABK+24] for condensed and analytic foundations. The six-functor language of Part V follows [Gal25; DG22; AGV22], and the higher-categorical conventions are those of [Lur09a; GR17a]. The normalisations $q_K(X)$, $r_K(X)$ and $C_A(K,\varepsilon)$ used in (5.3.1) are fixed in Chapter 40 and are not re-derived here.

Attribution of the new material. Nothing in this chapter is a mathematical result. It is a specification, and what is new to it is organisational: the ledger of 98 obligations with its seven mandatory fields and its four difficulty classes; the tagging discipline of Section 5.6 together with the cardinal rule and its mechanical enforcement; and the statement of Theorems A, B and C in a form where each carries its obligations in its head rather than in its preamble, together with the promise-by-promise reconciliation of Section 5.8 against Chapter 61. The substantive claims to novelty are made in the chapters that prove something: Chapters 15, 18, 28, 30, 36 and 39. The architectural response to the identification problem recorded in Remark 5.3.9 — replacing a transported identification by a categorical trace — is the one design decision of this chapter that is a mathematical choice rather than a presentational one, and it is defended in Chapter 36 rather than here.

Alternative treatments. Books that construct machinery towards an open problem have handled the epistemic question in three ways. The commonest is silence: hypotheses are stated as hypotheses and the reader is trusted to track them. The second is the conditional-throughout style of the analytic literature, where a Riemann-hypothesis-conditional theorem announces its hypothesis in its statement and nowhere else. The third, closest to what is done here, is the explicit assumption register of large formalisation projects, where every ‘sorry’ is a tracked object with an owner. We have taken the third route because the number of unproved inputs here is too large for the first and too heterogeneous for the second. Readers who find the tagging obtrusive may consult the untagged summary in the front matter, at the cost of the guarantee.

Historical remarks. The practice of publishing a proof-obligation ledger alongside a mathematical programme has no long history. Its nearest ancestors are the problem lists that accompanied major programmes in the twentieth century — Grothendieck’s standard conjectures, stated as what would be needed rather than as what was known, are the clearest case — and the audit culture of the formalisation community, where the distinction between a proved lemma and an admitted axiom is enforced by a machine. The specific pressure that produced the present ledger is more recent and is described in Section 4.6: when a long and difficult argument for a Diophantine conjecture reaches an impasse over whether a particular identification is legitimate [Moc21a; SS18], the cost of not having stated the load-bearing hypotheses separately becomes very high. The ledger is a response to that cost. Whether it is an adequate response is not for the author to say.

Further reading. For the requirements themselves and the no-go theorem that motivates them, Chapter 3. For the surrogates against which this design is to be compared, Chapter 4 and in particular Table 4.1. For the ledger in full, with statements, owners and falsifiers, Appendix H; for the tooling that checks the tags, Chapter 57; for the falsification experiments attached to the hardest obligations, Chapter 59. Readers interested in the function-field limit as a standing check should read Chapter 38 alongside Chapter 2. For the wider Diophantine context into which Theorem B would fit, if its hypotheses were discharged, see [Voj87c; Voj11].

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